MBA Thesis

# Bridging the GenAI Execution Gap:

# Barriers and Solutions for Scaling Generative AI in the IT Services Industry

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The author(s) declare(s) that they have completed the thesis work independently. All external sources are cited and listed under the References section. The thesis work has not been submitted in the same or similar form to any other institution(s) as part of another examination or degree.

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# Abstract

*Background.* Despite substantial strategic investment and widespread experimentation, a significant proportion of Generative AI (GenAI) initiatives within the IT (Information Technology) services industry remain constrained at the pilot stage, unable to transition into sustained, enterprise-scale, and governed operational deployment. This persistent divergence between experimental success and production execution, referred to throughout this thesis as the GenAI execution gap, represents a critical strategic challenge for IT service organisations whose competitive differentiation, client trust, and contractual accountability depend on responsible AI (Artificial Intelligence) deployment at scale.

*Purpose.* This thesis investigates the barriers that prevent IT service organisations from scaling GenAI beyond pilots and the solutions through which organisations may bridge the resulting execution gap. Using the Technology-Organisation-Environment (TOE) framework as the theoretical lens, the study produces an expert-validated, empirically grounded hierarchy of barriers and solutions specific to the IT services context and identifies cross-dimensional interaction effects that conventional single-method approaches cannot surface.

*Method.* The study employs a sequential four-phase mixed-methods design. Phase 1 comprised a survey questionnaire administered to 20 IT services professionals, measuring the perceived severity of barriers and effectiveness of solutions across seventeen TOE-derived constructs spanning 122 items. Phase 2 applied the Fuzzy Delphi Method (FDM) to these 20 survey responses using Triangular Fuzzy Numbers to validate and prioritise the barrier-solution taxonomy through expert consensus analysis. Phase 3 comprised 6 semi-structured individual expert interviews, selected from the 20 survey/FDM respondents, each of approximately 45 to 60 minutes in duration, conducted to explore the contextual mechanisms and organisational dynamics behind the quantitative findings. Phase 4 applied Braun and Clarke's (2006) thematic analysis framework with a hybrid deductive-inductive coding approach using the TOE framework as the initial theoretical coding template, synthesising the qualitative findings across eleven themes.

*Results and Analysis.* The FDM phase validated eleven consensus-supported barriers and forty-four consensus-supported solutions across the three TOE dimensions. The most critical barriers were concentrated in the Technology dimension: fragmented data governance across client environments (B2.2, crisp = 0.681) ranked first, followed by a cluster of integration complexity and technical debt items (B3.1 to B3.4), which achieved 100 per cent consensus coverage, the strongest consensus signal in the dataset. Organisational barriers, notably change resistance, low AI readiness, and prompt evaluation capability deficits, ranked below Technology in item-level consensus, yet expert holistic weight allocation placed the Organisation dimension as the leading contributor to the execution gap at 38.0 per cent, revealing a systemic rather than item-specific character. The Environment dimension produced no retained barriers due to jurisdictional variability yet yielded the highest mean retained solution effectiveness (0.710), confirming that environmental solutions are universally valuable despite contextual divergence in environmental barrier experience. Thematic analysis enriched and extended these findings, identifying three significant constructs not represented in the structured FDM taxonomy: a risk-tiered AI governance architecture principle; the client-side readiness bottleneck as an external scaling ceiling; and an incentive misalignment dynamic, termed the frozen middle, rooted in the billable-hour commercial model of IT service delivery. Cross-dimensional interaction analysis revealed that security and compliance gatekeeping processes systematically amplify technical integration barriers, and that iterative scaling frameworks produce sustained value only when anchored by explicit governance decision gates.

*Conclusions.* The GenAI execution gap in IT services is a multi-dimensional, socio-technical, and ecosystem-level challenge. It cannot be resolved through technical intervention alone; it requires simultaneous action on data and integration architecture, organisational capability and incentive structures, and external client and vendor relationship management. The findings challenge single-

dimensional explanations of the execution gap and support a structured, cross-dimensional model of barrier co-production and solution co-design. The commercial model of IT service delivery itself emerges as a structural barrier that conventional technology adoption frameworks do not capture.

*Recommendations for Future Research*. Future research should pursue longitudinal study designs to track how barrier severity evolves as organisations progress from pilot to production stages. Cross-industry comparative studies would assess the degree to which the identified barrier-solution hierarchy is IT services-specific or broadly generalisable. Governance maturity research should examine how organisations progress through AI governance development stages over time. Post-deployment operational studies should investigate LLMOps (Large Language Model Operations), GenAI operational resilience, and value attribution following production launch. Research into commercial model transformation, from headcount-based to value-based or outcome-based IT service pricing, is identified as a priority gap warranting dedicated empirical investigation.

*Keywords:* Generative AI, execution gap, IT services, TOE framework, Fuzzy Delphi Method, enterprise AI scaling, AI governance, mixed-methods research.

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# List of Abbreviations

| | |
|---|---|
| AI | Artificial Intelligence |
| AI COE | Artificial Intelligence Centre of Excellence |
| ADKAR | Awareness, Desire, Knowledge, Ability, Reinforcement (change management model) |
| APAC | Asia Pacific |
| API | Application Programming Interface |
| AHP | Analytic Hierarchy Process |
| ARB | Architecture Review Board |
| AWS | Amazon Web Services |
| BPO | Business Process Outsourcing |
| BSS | Barrier Severity Scale |
| CAIO | Chief Artificial Intelligence Officer |
| CDO | Chief Data Officer |
| CEO | Chief Executive Officer |
| ChatGPT | Chat Generative Pre-trained Transformer |
| CIO | Chief Information Officer |
| CISO | Chief Information Security Officer |
| CoC | Centre of Competency |
| CoE | Centre of Excellence |
| CTO | Chief Technology Officer |
| DOI | Diffusion of Innovations |
| DORA | Digital Operational Resilience Act |
| EEA | European Economic Area |
| EC2 | Elastic Compute Cloud |
| ERP | Enterprise Resource Planning |
| EU | European Union |
| FDM | Fuzzy Delphi Method |
| FOMO | Fear of Missing Out |
| FTE | Full-Time Equivalent |
| FinOps | Financial Operations |
| GDPR | General Data Protection Regulation |
| GenAI | Generative Artificial Intelligence |
| GPU | Graphics Processing Unit |
| HITL | Human-in-the-Loop |
| IEC | International Electrotechnical Commission |
| ISO | International Organisation for Standardization |
| IP | Intellectual Property |
| IT | Information Technology |
| ITSM | IT Service Management |
| JDBC | Java Database Connectivity |
| KPI | Key Performance Indicator |
| LLM | Large Language Model |
| LLMOps | Large Language Model Operations |
| LSA | Latent Semantic Analysis |
| MBA | Master of Business Administration |
| ML | Machine Learning |
| MLOps | Machine Learning Operations |
| MVA | Minimum Viable AI |
| MVP | Minimum Viable Product |
| NIST | National Institute of Standards and Technology |

| | |
|---|---|
| OKR | Objective and Key Result |
| PoC | Proof of Concept |
| PROSCI | Professional Science (change management methodology brand) |
| RAG | Retrieval-Augmented Generation |
| RFP | Request for Proposal |
| ROI | Return on Investment |
| RQ1 | Research Question 1 |
| RQ2 | Research Question 2 |
| RMF | Risk Management Framework |
| SaaS | Software as a Service |
| SAP | Systems, Applications, and Products (enterprise software) |
| SEM | Structural Equation Modelling |
| SES | Solution Effectiveness Scale |
| SLA | Service Level Agreement |
| SME | Small and Medium-Sized Enterprise |
| TAM | Technology Acceptance Model |
| TFN | Triangular Fuzzy Number |
| TOE | Technology–Organisation–Environment Framework |
| UK | United Kingdom |
| US | United States |
| VP | Vice President |

# Chapter 1: Introduction

## 1.1 Background and Motivation

Generative artificial intelligence has entered enterprise technology contexts with a speed and breadth that few prior innovations match. Recent studies indicate that GenAI experimentation has expanded rapidly across enterprise contexts and GenAI tools have progressed from academic research environments and early-adopter experiments into the core strategic agendas of organisations across virtually every industry sector. The IT services industry has been among the earliest and most substantial adopters of GenAI experimentation: system integrators, managed service providers, IT consulting firms, cloud and platform providers, and software-as-a-service organisations have collectively invested significantly in GenAI pilots directed at internal productivity improvement, knowledge management, automated service desk functions, security monitoring, and software development acceleration.

An increasingly well-documented challenge has emerged alongside this investment: most organisations that successfully pilot GenAI encounter significant barriers when attempting to scale those pilots into sustained, enterprise-wide, and operationally reliable deployments. Pilots demonstrate promise; production deployment exposes complexity. The gap between these two states, referred to throughout this thesis as the GenAI execution gap, is not primarily a technology capability problem. GenAI models have advanced substantially, and access to high-capability foundation models is no longer a meaningful barrier for large organisations. The execution gap is therefore understood as a multi-dimensional organisational, governance, and environmental challenge, shaped by deficiencies in data quality and management practices, skills and capability shortfalls, leadership and governance immaturity, regulatory complexity, and the institutional dynamics of the IT services delivery context itself.

Prior research confirms that while GenAI experimentation is widespread, organisations consistently encounter significant challenges when attempting to operationalise these capabilities in ways that satisfy requirements for reliability, governance, and measurable business value (Hughes et al., 2026; Kemell et al., 2025; Neumann et al., 2026). Organisations frequently demonstrate early promise in controlled pilots yet encounter substantial barriers when embedding GenAI into stable, governed, and production-grade service processes (Clear et al., 2025; Kasireddy & Sreenivasu, 2025; Voorneveld, 2024). These execution challenges are widely attributed to the interaction of technical, organisational, and environmental constraints, including data governance fragmentation, model reliability limitations, skills shortages, and regulatory uncertainty (Cheung, 2025; Smith, 2025; Taeihagh, 2025).

Understanding this execution gap, its constituent barriers, its organisational dynamics, and the strategies through which organisations successfully bridge it, is both a theoretically significant research problem and an urgent practical challenge for the industry. This thesis addresses that challenge directly, employing a mixed-methods design anchored to the Technology-Organisation-Environment (TOE) framework to produce a validated, expert-prioritised, and qualitatively enriched account of the barriers and solutions that define GenAI execution in the IT services sector.

## 1.2 Research Context: Generative AI in IT Services

Generative artificial intelligence refers to a class of AI systems designed to generate new content, including text, images, audio, video, and code, by learning the underlying patterns and structures of large training datasets and producing novel outputs in response to user inputs or prompts. Unlike traditional AI systems that focus primarily on classification or prediction tasks, GenAI systems create original outputs that resemble but are not identical to their training data. This generative capability makes GenAI particularly valuable for knowledge-intensive, creative, and communication-oriented tasks that are central

to IT service delivery, including documentation, incident diagnosis, code generation, client communication, and knowledge base management.

The IT services industry, for the purposes of this study, encompasses multiple organisational forms: system integrators that design and implement complex IT solutions for enterprise clients, managed service providers that operate and maintain IT infrastructure and services on clients' behalf, IT consulting firms that advise on technology strategy and transformation, cloud and platform service providers that offer infrastructure and application services, and SaaS (Software as a Service) organisations that deliver software applications as managed services. These organisations share a common set of structural characteristics that make the GenAI execution challenge particularly acute: service continuity obligations, client contractual accountability, auditability requirements, regulatory compliance demands, and the need to maintain consistent quality across distributed, multi-client delivery environments. Accordingly, IT services organisations operate under strict requirements for service continuity, auditability, compliance, and consistent service quality.

In this context, the execution gap is not merely an internal inefficiency. When GenAI outputs are embedded in client-facing service processes, the stakes of unreliable, unexplainable, or non-compliant outputs are contractual and reputational rather than merely operational. Scaling GenAI responsibly in IT services therefore demands a level of technical governance, organisational maturity, and environmental alignment that pilot programmes rarely require or expose.

## 1.3 Purpose and Scope of the Study

The purpose of this thesis is to investigate why GenAI initiatives in IT services organisations so frequently remain at the pilot stage, and to identify the barriers and corresponding solutions that determine whether organisations can bridge the execution gap. The study focuses on GenAI scaling in globally operating and multi-market IT services contexts, where service delivery is shaped by client accountability, distributed data environments, governance requirements, and operational execution demands.

The study is grounded in the Technology–Organisation–Environment (TOE) framework as its theoretical lens. In addition to its empirical purpose, the thesis aims to contribute to technology adoption theory by applying and refining the TOE framework in the context of GenAI scaling in IT services. The scope of the study is bound to IT services organisations operating in global or multi-market delivery contexts, with a focus on organisations of sufficient scale, exceeding 100 employees, to have encountered the execution gap as a meaningful strategic and operational challenge.

## 1.4 Structure of the Thesis

The thesis is structured as follows. Chapter 2 establishes the problem statement, articulates the research purpose, and states the two primary research questions. Chapter 3 presents a systematic review of the literature on GenAI adoption barriers and solutions, with particular attention to the IT services context, culminating in an analytical synthesis that maps barriers to solutions. Chapter 4 develops the theoretical framework, justifying the adoption of the TOE framework as the primary lens, operationalising its constructs for the GenAI execution context, developing the conceptual model with its ten research propositions, and demonstrating the methodological integration between the theoretical framework and the four-phase research design. Chapter 5 presents the full methodology, covering research philosophy, research design, each of the four phases of data collection and analysis, triangulation, validity, ethical considerations, and limitations. Chapter 6 presents the quantitative findings from the survey analysis and Fuzzy Delphi Method (FDM), including the prioritisation and validation of the most significant barriers and solutions affecting GenAI execution and scaling in IT services organisations. Chapter 7 presents the qualitative findings and thematic analysis derived from the semi-structured interviews, providing deeper

interpretive insight into the organisational, technological, and environmental factors influencing GenAI execution outcomes. Chapter 8 provides the overall discussion and conclusion of the study by integrating the findings from all research phases, answering the research questions, discussing theoretical and managerial implications, outlining the study's contributions and limitations, and identifying opportunities for future research. The References section provides a complete bibliographic record of all sources cited throughout the thesis.

This chapter has introduced the background, industry context, purpose, and scope of the study. It has positioned the GenAI execution gap as a strategic and operational challenge for IT services organisations seeking to move from experimentation to governed, enterprise-scale deployment. Building on this foundation, Chapter 2 develops the specific problem statement, research purpose, and research questions that guide the study.

# Chapter 2: Purpose and Research Question

Building on the background and scope established in Chapter 1, this chapter defines the specific research problem addressed by the thesis. While Chapter 1 introduced the broader GenAI execution gap in IT services, this chapter formulates the gap as a researchable problem by specifying its sectoral relevance, theoretical significance, and practical implications. The chapter then presents the research purpose and the research questions that guide the empirical investigation.

## 2.1 Problem Statement

IT services organisations occupy a structurally distinctive and demanding position in the AI adoption landscape. Unlike standalone enterprises that deploy AI primarily for internal value creation, IT service providers, including system integrators, managed service providers, IT consulting firms, cloud and platform providers, and SaaS organisations, must deploy AI in ways that are simultaneously productive for internal operations and sufficiently reliable, auditable, and compliant to be embedded in client-facing service delivery. This dual accountability structure amplifies both the strategic importance and the execution difficulty of moving GenAI from controlled experimentation to governed, enterprise-scale operations.

A persistent and empirically observable challenge characterises this sector: despite strong pilot outcomes and substantial investment in GenAI experimentation, IT service organisations consistently encounter barriers that prevent the transition from pilot-stage activity to sustained, value-generating, and enterprise-scale deployment. This thesis refers to this persistent challenge as the GenAI execution gap. It is not a novel observation, the gap between AI experimentation and operational execution has been documented across multiple research traditions, but it has acquired strategic urgency in the GenAI context because of the scale, speed, and commercial expectations of current GenAI investment cycles.

The empirical evidence for the execution gap is substantial. Enterprise AI adoption research consistently identifies that a significant proportion of AI initiatives encounter barriers at the scaling threshold, unable to satisfy the integration, governance, and capability requirements of production-grade deployment (Hughes et al., 2026; Kasireddy & Sreenivasu, 2025; Smith, 2025). Kemell et al. (2025), in a multiple-case study of GenAI adoption in software organisations, finds that GenAI use remains primarily individualised and disconnected from organisational workflows, with institutionalised best practices still emergent, a signal of structural immaturity that persists even where individual tool adoption is widespread. Neumann et al. (2026) document a compliance gap in agile development teams: a systematic misalignment between formal AI governance policies and day-to-day usage practices that results directly from governance frameworks being designed without reference to operational realities. Together, these studies confirm that the execution gap is not an artefact of individual organisations' inadequacy but a structural challenge that emerges at the boundary between experimentation and institutionalisation.

The barriers that generate the execution gap are multi-dimensional and mutually reinforcing. Technical limitations, including model hallucination risk, explainability deficiencies, data governance fragmentation, and integration complexity with legacy infrastructure, increase the governance and assurance demands of production deployment (Taeihagh, 2025). These technical demands in turn expose organisational capability shortfalls: skills deficits, change capacity limitations, and governance immaturity that prevent organisations from meeting the higher assurance requirements that production deployment imposes (Voorneveld, 2024). External environmental conditions, including regulatory uncertainty under frameworks such as the EU (European Union) AI Act and GDPR (General Data Protection Regulation), legal liability ambiguity in client-facing deployment, and competitive pressure that creates urgency to deploy before readiness conditions are adequately established, further shape organisational risk tolerance and constrain the sequencing of scaling decisions (Cheung, 2025; Fernández & Uriarte, 2026; Taeihagh, 2025).

Critically, these barriers do not operate in isolation. Hughes et al. (2026) empirically confirm that organisational change capacity moderates the relationship between technological complexity and GenAI adoption outcomes: organisations with lower change capacity find technological barriers more severe precisely because they lack the organisational machinery to navigate them. This interaction effect means that the execution gap is more than the sum of its individual barriers, it is a systemic condition produced by the co-occurrence and mutual reinforcement of technical, organisational, and environmental constraints. Taeihagh (2025) argues that GenAI's diffusion has outpaced institutional capacity to stabilise governance norms and accountability structures, creating a regulatory and organisational readiness vacuum that magnifies execution risk for organisations deploying GenAI in client-facing contexts where accountability is contractual and reputational rather than merely operational.

For IT services specifically, these dynamics are intensified by structural features of the sector. Service continuity obligations, client contractual accountability, auditability requirements, and multi-client data governance complexity create execution conditions that pilot programmes rarely expose (Clear et al., 2025; Kasireddy & Sreenivasu, 2025). When GenAI outputs are embedded in client-facing service processes, the consequences of unreliable, unexplainable, or non-compliant outputs are contractual and reputational rather than merely operational. The cost-benefit calculus of scaling therefore differs structurally from that of a single-enterprise adopter. This asymmetry helps explain why the execution gap is particularly pronounced in IT services contexts, and why it warrants dedicated empirical investigation rather than generalisation from enterprise AI adoption studies conducted in other sectors.

Despite a growing body of general AI adoption literature, there is limited consolidated empirical work that maps which specific barriers are most influential in IT services contexts and which strategic responses are most effective in addressing them. Existing empirical evidence is strongest in software and agile team contexts (Kemell et al., 2025; Neumann et al., 2026) and broader organisational and SME contexts (Fernández & Uriarte, 2026; Kittipanya-Ngam et al., 2025), but few studies explicitly theorise and empirically investigate the execution mechanisms in IT services, where pilot success must translate into repeatable service components, governed processes, and defensible client outcomes. Furthermore, existing studies tend to analyse barriers or solutions in isolation rather than as paired empirical constructs, and expert-validated prioritisation specific to the IT services sector is largely absent from the literature, making it difficult for practitioners to sequence their interventions strategically.

This thesis addresses these gaps through a theory-grounded, mixed-methods empirical investigation designed to produce actionable insights for both research and practice. It adopts the Technology-Organisation-Environment (TOE) framework as the organising theoretical lens, enabling systematic investigation of how technical, organisational, and environmental factors interact to produce the execution gap and shape the effectiveness of responses to it. The study is positioned to contribute a sector-specific empirical account of the execution gap that complements and extends the broader enterprise AI adoption literature while equipping IT service leadership with an evidence-based framework for scaling GenAI responsibly.

## 2.2 Research Purpose

The purpose of this thesis is threefold. First, it aims to systematically identify and categorise the barriers that prevent IT service organisations from scaling GenAI beyond pilots, structured within the TOE framework. Second, it seeks to identify and evaluate the solutions that organisations use, and the literature recommends bridging the execution gap, assessing their perceived effectiveness in the specific IT services context. Third, the study aims to develop a structured knowledge model that clarifies the relationships and sequencing between GenAI scaling barriers and solution mechanisms within the IT services industry, thereby supporting informed decision-making on how GenAI initiatives can progress beyond the pilot stage.

## 2.3 Research Questions

This study is bounded to the global IT services industry and focuses on large-scale organisations, defined in this study as organisations with more than 100 employees. Within this scope, the thesis is guided by two primary research questions:

*RQ1 (Research Question 1):* What barriers prevent IT service organisations from scaling GenAI beyond pilot stages?

*RQ2 (Research Question 2):* What solutions are used to overcome these barriers and bridge the GenAI execution gap?

These two research questions are intentionally separated but analytically connected. RQ1 focuses on diagnosing the technical, organisational, and environmental barriers that constrain the scaling of GenAI in IT service organisations. RQ2 shifts the analysis toward the solution mechanisms used to address these barriers, including their perceived feasibility, effectiveness, and contextual appropriateness. This separation strengthens analytical clarity by distinguishing problem identification from solution evaluation, while still allowing the study to develop an integrated account of the GenAI execution gap and how it may be addressed in practice.

## 2.4 Overarching Research Expectation

Although this study does not test formal statistical hypotheses, it is guided by an overarching research expectation derived from the literature and the Technology–Organisation–Environment framework. The study expects that the GenAI execution gap in IT services is not caused by technological limitations alone. Rather, it is expected to emerge from the interaction between technology-related barriers, organisational readiness constraints, governance maturity, client accountability requirements, and external environmental pressures.

In particular, the study expects that technical barriers, such as data governance fragmentation and integration complexity, become more severe when organisational capabilities, incentive structures, and governance mechanisms are insufficiently developed to absorb them. This expectation guides the empirical investigation while allowing the mixed-methods design to refine, extend, or challenge it through expert prioritisation and qualitative thematic analysis.

This chapter has defined the research problem, purpose, research questions, and overarching expectation guiding the study. It has argued that the GenAI execution gap in IT services is not only a technical implementation problem, but a multi-dimensional challenge involving organisational capabilities, governance structures, commercial incentives, client accountability, and environmental pressures. Chapter 3 now reviews the existing literature on GenAI adoption, scaling barriers, and solution approaches to position the study within current academic knowledge and identify the specific research gaps addressed by this thesis.

# Chapter 3: Previous Research

## 3.1 Overview and Search Strategy

This chapter reviews the academic literature relevant to understanding the barriers that prevent IT service organisations from scaling Generative AI (GenAI) beyond pilot stages, and the solutions used to bridge the resulting execution gap. The review is structured to serve two purposes: first, to position the study within existing knowledge and identify the specific gaps that the research addresses; and second, to derive the barrier–solution taxonomy that directly informs the empirical instruments used in Phases 1 through 4 of the research design.

Search activities were conducted using peer-reviewed academic databases including Summon, ScienceDirect, and Google Scholar, supplemented by targeted searches via ACM Digital Library and IEEE Xplore for conference proceedings. The search was conducted iteratively between January 2024 and March 2026. Initial keyword terms covered the core concepts of the study: 'Generative AI', 'GenAI', 'Generative Artificial Intelligence', 'barriers', 'execution gap', 'solutions', 'adoption', 'IT services', and 'enterprise AI'. These were progressively supplemented by more contextually specific terms including 'scaling', 'production deployment', 'organisational readiness', 'AI governance', 'Technology-Organisation-Environment', 'TOE framework', 'hallucination', 'explainability', 'data privacy', 'regulatory uncertainty', 'human-AI collaboration', and 'change capacity'. Boolean operators (AND, OR) and wildcard expansions were applied to maximise retrieval breadth within the thematic boundaries of the study.

The target organisational context was globally operating and multi-market IT services organisations, consistent with the empirical scope of the study. The literature search was not restricted to a single country or regulatory jurisdiction, reflecting the multi-market service delivery reality of the study's participant organisations. Instead, sources were selected for their conceptual or empirical relevance to enterprise GenAI adoption, IT services delivery, AI governance, scaling, and production deployment across diverse organisational and regulatory environments.

Inclusion criteria required that sources be: (i) peer-reviewed journal articles, conference proceedings, or scholarly working papers; (ii) published or accepted between 2021 and 2026, with selective inclusion of foundational technology adoption theory sources; (iii) relevant to enterprise or IT services contexts, or to the adoption, scaling, governance, or accountability dimensions of AI and GenAI; and (iv) of sufficient scholarly rigour to support theoretical or empirical claims. Sources were excluded where they were: non-scholarly, meaning they lacked peer-review or equivalent academic quality control, such as vendor white papers, trade press articles, or consultancy reports without empirical grounding; consumer-oriented studies focused on individual end-user experience in personal rather than enterprise contexts; or what this review terms purely technical in focus. In the context of this study, 'purely technical' refers to sources whose primary contribution is algorithmic, architectural, or model-performance focused, such as papers proposing new language model architectures, benchmarking model outputs on accuracy tasks, or optimising inference efficiency without addressing the organisational, governance, or deployment challenges that constitute the execution gap. Such sources were excluded because, while technically relevant to GenAI broadly, they do not address the multi-dimensional barriers and solutions that are the analytical focus of this thesis. This distinction is important in the IT services context, where the execution gap is a socio-technical problem requiring organisational, governance, and environmental analysis rather than purely engineering solutions.

The initial pool of potentially relevant publications exceeded 120 items. After applying the inclusion criteria and conducting abstract-level screening, 68 sources were retained for full-text review. Following full-text assessment against alignment with the study's research questions and TOE constructs, a final working corpus of 34 sources was selected for active use in the literature synthesis. This expanded corpus

reflects two phases of expansion from the initial 20-source draft: a first phase incorporating Fernández and Uriarte (2026), Hughes et al. (2026), Kemell et al. (2025) and Neumann et al. (2026), which substantially strengthen the empirical grounding of the barrier and solution analysis; and a subsequent targeted addition of Pushpakumara and Ahsan (2025), Smith (2025), Teng et al. (2025) and Wu et al. (2025), which strengthen the human-AI collaboration, IT services value pathway, output assurance, and strategic governance dimensions of the review.

The literature review is structured around four thematic areas: the GenAI execution challenge and the IT services context; the barrier landscape; solution approaches; and an integrated analytical synthesis. Critically, and in response to the complexity of the academic discourse on GenAI adoption, each thematic section explicitly acknowledges competing or contradictory findings where they exist. The final section identifies the specific research gaps that this study addresses. The review does not present the academic literature as uniformly convergent; rather, it treats tensions, contradictions, and boundary conditions as analytically informative, particularly for understanding why the GenAI execution gap is not a uniform phenomenon and why its solutions are not universally applicable.

## 3.2 GenAI and the IT Services Industry Context

The recent GenAI literature consistently distinguishes initial uptake from operational execution: organisations may experiment with GenAI tools, yet struggle to convert pilots into stable, governed, and value-producing capabilities embedded in everyday work. Empirical studies in software organisations and agile teams show GenAI being used primarily as an individual productivity aid, with limited evidence of durable, organisation-wide integration into standard processes and controls (Kemell et al., 2025; Neumann et al., 2026). Case study evidence corroborates this pattern: across seven European software organisations, none had systematically quantified any organisational benefit from GenAI adoption, and in several cases, the technology remained confined to individual tool use driven by bottom-up interest rather than strategic deployment (Kemell et al., 2025). This pattern is echoed in workplace-oriented research stressing that near-term impact is uneven, strongly conditioned by trust, privacy, and security concerns, and dependent on the absence of shared routines for responsible use (Clear et al., 2025).

Empirical research conducted within the IT industry itself adds a demand-side perspective to this execution gap analysis. Drawing on 32 semi-structured interviews with IT companies across the UK, EU, and US, Teng et al. (2025) identify five approaches through which GenAI creates new value propositions in IT services: AI agents, content creation, knowledge querying via cloud solutions, foundation models, and upstream infrastructure. These pathways range from incremental business model innovation, such as cloud providers adding GenAI knowledge querying as an add-on service, to radical innovation, such as autonomous AI agents that decompose complex tasks without human intervention. This empirical mapping is significant for the execution gap analysis: it shows that GenAI's value creation potential in IT services is not uniform but spans a spectrum from low-customisation infrastructure plays to high-customisation agentic deployments, each carrying different governance and readiness requirements. Organisations that conflate these pathways, treating them as interchangeable, risk misaligning their scaling strategies with the actual governance complexity their chosen use case demands.

At the governance level, policy scholarship argues that GenAI's diffusion has outpaced institutions' capacity to stabilise norms, responsibilities, and regulatory instruments, thereby magnifying uncertainty for organisations attempting production deployment (Cheung, 2025; Taeihagh, 2025). Taeihagh (2025) catalogues the specific risks this creates for organisations: hallucinations, output variability, sensitive information leakage, opacity, and novel deployment challenges that traditional IT governance frameworks, built around deterministic systems with predictable outputs, are poorly equipped to address. Critically, this is not merely a regulatory problem in the abstract; it is an operational cost problem, because the burden of compensating for governance immaturity falls on the organisations themselves,

typically through over-engineering assurance processes or restricting use cases to reduce liability exposure.

For the IT services industry specifically, these dynamics acquire particular structural significance. Service continuity, client trust, auditability, and regulatory compliance obligations are not peripheral concerns but are central to competitive differentiation and contractual accountability. Scaling requires aligning technical performance with organisational controls and stakeholder expectations, not merely increasing access to tools (Hughes et al., 2026). This framing positions the execution gap as a socio-technical production problem rather than a technology readiness problem, a distinction that has direct implications for where organisations should invest their resources when attempting to move beyond pilots. Large-scale empirical evidence from 34 European countries reinforces this point: even in highly innovative national contexts, firm-level AI adoption has been uneven and slow, with organisational barriers such as bureaucracy, legacy systems, and risk aversion often outweighing the advantage of access to financial and technical resources (Fernández & Uriarte, 2026).

However, the IT services framing itself introduces a contextual nuance that the broader AI adoption literature does not always capture. Most of the empirical evidence on GenAI barriers comes from software development teams (Kemell et al., 2025; Neumann et al., 2026), general enterprise surveys (Hughes et al., 2026), and cross-national studies (Fernández & Uriarte, 2026) rather than from IT services organisations operating under client-contracted, multi-vertical service delivery arrangements. The IT services context adds specific dimensions, including the need to obtain client consent for GenAI tool use, the presence of multi-party contractual liability structures, the requirement to demonstrate auditability and reproducibility to external clients, and the commercial pressure to absorb AI productivity gains within existing pricing models, that are not fully theorised in the existing literature. This contextual distinction motivates the sector-specific empirical focus of this study and explains why the execution gap in IT services cannot simply be resolved by applying solutions validated in software product organisations.

## 3.3 Barriers to GenAI Execution

Building on the barrier categories outlined in Chapter 2, this section organises the barrier landscape across five thematic constraint domains that anticipate the TOE framework operationalisation developed in Chapter 4. Critically, the literature does not present a uniformly convergent picture of which barriers are most severe or most fundamental. Competing interpretations, contradictory empirical findings, and context-dependent rankings are noted throughout, as these tensions inform the design and analytical scope of the empirical investigation.

*Model Reliability, Hallucination Risk, and Explainability Limitations*

A consistent finding across empirical and conceptual studies is that GenAI systems' output uncertainty produces a validation burden that grows substantially as organisations move from pilot demonstrations to client-facing or operationally consequential uses (Pushpakumara & Ahsan, 2025). In practitioner-focused evidence, inaccurate responses and hallucinations are explicitly identified as adoption inhibitors, requiring users to verify and refine outputs to avoid technical debt and misapplication (Simaremare & Edison, 2024). In agile team contexts, barriers are frequently framed as validation effort and trust constraints, with organisations requiring human review, and sometimes multi-person review policies, that become impractical at scale, thereby creating an execution bottleneck (Neumann et al., 2026). Public discourse analysis among software practitioners similarly shows widespread user awareness of hallucinations as a barrier to professional adoption, with developers recognising but frequently ignoring the need for output validation because the verification overhead undermines the productivity gains the tools are supposed to deliver (Basha & Rodríguez-Pérez, 2025). More general organisational adoption research highlights concerns about explainability and the necessity of human verification before outputs reach clients, a

constraint that is especially salient for IT services where accountability and service quality are audited (Hughes et al., 2026; Voorneveld, 2024).

Hughes et al. (2026) introduce a theoretically important reframing of this barrier. Rather than treating GenAI complexity as a generic integration or learning-curve problem (the classical TAM and DOI framing), they identify *explainability complexity* as a specific cognitive and epistemic challenge: the difficulty of making sense of outputs that may be simultaneously useful, biased, or unpredictable. This shift transforms the technological barrier from a purely technical assessment to a cognitive and interpretive challenge, suggesting that organisations face not only the engineering problem of controlling outputs but also the epistemic problem of knowing when and how much to trust them. The literature deepens this point by showing why the validation burden is structurally difficult to eliminate through engineering alone: GenAI's opacity and emergent behaviours make it difficult to anticipate failure modes, raising the need for robust guardrails, monitoring, and verification practices that scale poorly relative to the productivity gains pilots promise (Taeihagh, 2025).

It is worth noting that not all researchers frame explainability limitations as a primary scaling barrier. Voorneveld (2024) observes that organisations differ substantially in their tolerance for output variability depending on the use case and client exposure level, suggesting that explainability requirements are not uniform but are conditioned by context. In use cases with lower decision criticality, organisations may accept higher output variability and lower verification overhead, enabling a partial path around the validation bottleneck. This boundary condition is important for the IT services context, where not all GenAI applications carry equal accountability risk, and where staged deployment strategies can, in principle, manage explainability constraints by sequencing use cases from low-risk to high-risk.

*Data Privacy, Security Vulnerability, and Legal and Regulatory Ambiguity*

Across the reviewed corpus, privacy and security are among the most consistently cited adoption barriers, intensifying when organisations attempt to embed GenAI in workflows involving sensitive client or operational data. Practitioner evidence shows that privacy and confidentiality represent the most reported challenges, with explicit concerns about data leakage and training-data misuse (Simaremare & Edison, 2024). In agile organisations, data protection is presented as a core barrier shaping both tool licensing decisions and permitted use cases, with shadow IT practices emerging when approved tools are unable to match practical needs (Neumann et al., 2026). In IT services organisations specifically, clients have in several documented cases explicitly prohibited the use of GenAI tools for tasks involving their proprietary data or code, creating an additional external constraint on use-case scope that does not arise in product organisations (Kemell et al., 2025).

Public discourse analysis further indicates that adoption is constrained by intellectual property disputes and demands for transparency and legal clarity, suggesting that legitimacy concerns are not confined to regulators but are broadly salient across stakeholders (Basha & Rodríguez-Pérez, 2025). Regulatory uncertainty is reinforced by policy framing research: evolving regulation, cross-jurisdictional inconsistency, and uncertainty over intellectual property ownership and liability all increase the perceived downside of scaling beyond pilots, particularly in regulated client environments (Fernández & Uriarte, 2026; Taeihagh, 2025).

However, the empirical evidence on the actual behavioural impact of regulatory barriers is more ambiguous than the descriptive literature suggests. In one of the few quantitative studies directly testing regulatory pressure as a predictor of GenAI adoption, Hughes et al. (2026) found that regulatory pressure did not significantly influence adoption behaviour in their global sample of 304 decision-makers ($\beta = -0.123$, $p$ = n.s.). The authors interpret this as evidence of *regulatory flux*: GenAI adoption unfolds amid shifting, fragmented, and ambiguous rules, creating a regulatory void in which organisations cannot

anchor adoption decisions in clear external mandates. In such environments, internal factors, particularly perceived relative advantage and staff skills, become more decisive than external regulatory signals. This finding is in tension with governance scholarship that frames regulatory uncertainty as a primary scaling inhibitor (Cheung, 2025; Taeihagh, 2025), and it complicates the intuitive assumption that clearer regulation would straightforwardly accelerate responsible scaling. One plausible interpretation is that regulatory barriers have a threshold effect: below a critical level of enforcement maturity, organisations manage regulatory risk informally and proceed with adoption; above that threshold, compliance obligations begin to impose binding constraints. Hughes et al.'s (2026) data likely captures organisations operating below this threshold, while IT services organisations serving regulated sectors may operate above it. This contextual distinction is an important gap that the empirical investigation of this study is well positioned to address.

*Integration Complexity, Data Quality, and Policy-Practice Gaps*

A recurrent barrier to bridging the execution gap is integration complexity, both technical and procedural. Case evidence in agile environments shows that GenAI value depends strongly on whether tools are embedded into existing workflows and whether governance is practical to follow; when these conditions are absent, usage becomes fragmented and informal (Neumann et al., 2026). Enterprise-focused adoption research similarly identifies legacy systems, interoperability constraints, and data silos as structural bottlenecks when moving from experimentation to enterprise-wide integration (Kasireddy & Sreenivasu, 2025). Voorneveld (2024) highlights problems with data quality, including bias, incompleteness, and labelling errors, demonstrating that poor training data can lead to unreliable or ethically problematic model outputs, even when the model itself is technically sound. In software organisations, challenges include not only adoption decisions but also how tools are integrated into everyday work, with institutionalised best practices remaining emergent: a signal of institutional immaturity rather than mere tool immaturity (Kemell et al., 2025).

For IT service organisations, this integration problem is intensified because service delivery relies on standardised processes, repeatability, and predictable escalation paths. When GenAI usage remains individualised and disconnected from shared workflows, it cannot be reliably incorporated into service operations without amplifying operational variance and compliance risk (Hughes et al., 2026; Neumann et al., 2026). This explains why the execution gap persists even where organisational enthusiasm for GenAI is high: the barrier is not tool availability but the mismatch between what is easy to pilot, specifically ad hoc prompting, and what is required to operate, namely standardisation, monitoring, and governance.

A particularly important finding is the *compliance gap* identified by Neumann et al. (2026): a systematic divergence between formal governance policies (rooted in the environment context), organisational policies and procedures (reflecting the organisation context), and actual day-to-day technology usage patterns (technology context). This cross-dimensional misalignment arises when regulatory requirements are translated into governance policies without adequate consideration of technological characteristics or operational constraints. The result is that formally compliant organisations may still exhibit widespread informal non-compliance at the practitioner level, as in cases where developers reframe unapproved GenAI tool use as 'private time' to reconcile productivity needs with policy restrictions. This finding is theoretically significant because it suggests that integration barriers are not solely technical but are partly produced by governance frameworks that are insufficiently calibrated to technological realities and practitioner needs. Critically, it introduces a tension into the solutions literature: tightening governance policies to address integration risks may inadvertently worsen the policy-practice gap by pushing usage further underground, increasing rather than reducing organisational risk.

The cross-national study by Fernández and Uriarte (2026) adds an institutional dimension to this barrier, showing that large organisations in highly innovative country contexts, despite having greater access to

financial resources, specialised human capital, and data infrastructure, often adopt AI more slowly than expected due to internal bureaucracy, hierarchical decision-making, and risk aversion. This finding complicates the common assumption that resource availability is the primary enabler of scaling: even well-resourced organisations may be constrained by structural inertia that the TOE framework captures under the organisation context dimension.

*Skills Shortages, Cultural Resistance, and Human-AI Collaboration Risks*

The literature broadly converges on skills shortages as a persistent and fundamental barrier to scaling GenAI. Studies emphasise inadequate training and limited AI literacy as constraining factors, with change capacity moderating how organisations cope with skills gaps (Hughes et al., 2026). In software organisations, prompting competence and learning-to-use challenges are prominent at the individual level, while organisational barriers include change resistance and the difficulty of measuring tool impact (Kemell et al., 2025). Kasireddy and Sreenivasu (2025) identify a shortage of machine learning, data engineering, and AI governance skills as a major impediment, arguing that without focused upskilling, enterprises will struggle to scale AI use effectively. The large-scale European study by Fernández and Uriarte (2026) confirms this empirically: lack of expertise is the most statistically significant barrier across their 34-country sample, with a one standard deviation increase in national human capital specialisation reducing the probability of non-adoption by 1.6 percentage points and increasing awareness of AI-related challenges, suggesting that more skilled organisations not only adopt more but also engage more critically with the associated risks.

However, the relationship between skills and scaling is more complex than a simple deficit model implies. Hughes et al. (2026) report a theoretically unexpected finding: organisational change capacity *negatively moderates* the relationship between staff skills and GenAI adoption ($\beta = -0.147$, $p < 0.05$), meaning that in organisations with high change capacity, the positive effect of individual staff competencies on adoption diminishes. The authors interpret this as evidence of *capability complementarity*: when organisations have robust systemic enablers, including cross-functional learning routines, adaptive change infrastructure, and decentralised decision-making, they become less reliant on specific individuals' technical expertise to drive adoption. This finding challenges the additive-driver assumption that dominates the skills literature, which generally treats skills and change capacity as independently additive barriers. For IT services, this has a practical implication: organisations that invest exclusively in individual upskilling without simultaneously building organisational change infrastructure may see diminishing returns compared to organisations that develop both in parallel. It also introduces a caution against treating skills gaps as uniformly primary barriers across all organisational contexts.

Public discourse analysis adds a further dimension by highlighting *over-reliance* as a distinct human-AI collaboration risk: not only are skills deficits a barrier, but the misuse of AI capabilities by insufficiently critical users can introduce new quality and accountability risks into service delivery (Basha & Rodríguez-Pérez, 2025). Concerns about over-reliance extend beyond novice users to experienced practitioners, who may also gradually reduce their independent critical evaluation of AI outputs, a dynamic that is especially consequential in IT services where diagnostic reasoning and service quality judgement are core delivery competencies. This is not merely a training problem but a work design problem: organisations need to explicitly define task boundaries between human-led and AI-assisted activities rather than allowing them to emerge informally.

Experimental evidence provides a more precise account of the human-AI collaboration risk. Wu et al. (2025) conducted four pre-registered online studies ($N = 3,562$) examining the immediate and downstream performance effects of GenAI collaboration across professional task settings. Collaboration with GenAI consistently improved immediate task quality across all study conditions. However, this performance benefit did not persist in subsequent tasks completed independently. More significantly,

transitioning from AI-assisted to solo work produced a consistent psychological deprivation effect: workers reported reduced intrinsic motivation and increased boredom when moving from AI-supported tasks back to independent work, even as their sense of control partially recovered. Prolonged AI collaboration did not improve motivation outcomes; it sustained a stable but low motivational baseline. In IT services contexts, where diagnostic reasoning, client communication, and escalation judgement require sustained independent cognitive engagement, this motivational erosion carries direct service quality implications. Practitioners who regularly complete high-stakes tasks with GenAI assistance may find it progressively harder to sustain the focused independent critical engagement those tasks require when AI assistance is unavailable or inappropriate. The implication is that work design, specifically how organisations structure the balance and sequencing of AI-assisted and human-led activities, matters as much as tool access in sustaining long-term service performance quality.

*Governance Immaturity, Value Ambiguity, and External Pressures*

A key insight across studies is that organisations may observe micro-level productivity gains while still struggling to establish a robust scaling rationale, particularly when benefits are difficult to measure, unevenly distributed, or poorly connected to commercial incentives. Organisational studies explicitly identify the difficulty of measuring positive impact as a major adoption challenge (Kemell et al., 2025; Simaremare & Edison, 2024). Simaremare and Edison (2024) identify two previously undocumented barriers in this category: the absence of compatible use cases, where practitioners cannot identify any workflow in which GenAI offers meaningful improvement over existing automation; and the challenge of unforeseen benefits, where organisations decline to invest in exploration because the potential return is invisible until use cases are actively mapped. This creates a self-reinforcing pattern: without exploration investment, applicable use cases remain undiscovered; without applicable use cases, management lacks an investment rationale. In IT services, where investment decisions are constrained by margins, contractual KPIs, and client demand signals, GenAI that does not map cleanly to billable value, SLA improvement, or risk reduction is likely to remain at the pilot stage as the rational equilibrium. Al-Khatib (2023) further demonstrates in a quantitative study that relative advantage is a primary predictor of GenAI adoption: when the value proposition is ambiguous or unmeasurable, adoption stalls regardless of technical readiness.

Governance-oriented research argues that GenAI-specific properties require adapted governance mechanisms linking data, model, people, and organisational targets to concrete accountability levers, precisely because routine operational reliability cannot be assumed (Taeihagh, 2025). Traditional IT governance frameworks, built around deterministic systems, are inadequate for governing the complex adaptive behaviours of large language models (Taeihagh, 2025). The cost of ongoing model maintenance and continuous improvement is highlighted as a distinct barrier: even if a model performs well in controlled settings, performance degradation and operational upkeep create recurring costs that organisations frequently underestimate (Kasireddy & Sreenivasu, 2025).

External conditions, including regulatory uncertainty, evolving industry standards, vendor ecosystem immaturity, and competitive pressure, further shape GenAI adoption trajectories. However, a critical empirical tension exists here. Competitive pressure is widely cited as a dual-edged force: it simultaneously drives urgency to adopt GenAI and creates the risk of premature deployment before readiness conditions are met (Cheung, 2025; Taeihagh, 2025). Yet Al-Khatib's (2023) quantitative study of the Jordanian retail sector found that competitive pressures had no statistically significant effect on GenAI adoption, with customer pressures proving more influential. This divergence may reflect sectoral or cultural differences, but it cautions against treating competitive urgency as a universally powerful driver. In the IT services context, competitive pressure takes a specific form: clients may demand AI-enabled service improvements before the service provider has developed the governance infrastructure to deliver them safely, creating a demand-pull dynamic that can accelerate deployment beyond

organisational readiness. This dynamic is distinct from the competitive dynamics studied in product-market or retail contexts and underscores the importance of sector-specific empirical investigation.

## 3.4 Solutions for Bridging the Execution Gap

Although the barriers described above dominate the empirical literature, the reviewed corpus also converges on solution directions that are compatible with IT services realities. These can be grouped into five interrelated solution domains, each addressing a distinct set of execution barriers. Importantly, the solutions literature is not uniformly prescriptive: several studies identify tensions between solution approaches, and some solutions intended to address one barrier can exacerbate another. These tensions are noted explicitly, as they are relevant to the empirical investigation's objective of assessing perceived solution effectiveness in context.

*Foundational and Technical Readiness Solutions*

The first solution category focuses on strengthening the technical foundations required for GenAI to move beyond pilots into stable service operations. Enterprise-focused studies argue that GenAI adoption should follow phased and maturity-based deployment pathways, allowing organisations to stabilise data quality, system interoperability, and infrastructure capacity before expanding scope (Kasireddy & Sreenivasu, 2025; Smith, 2025). To address integration challenges, researchers emphasise modular and API-first architectures that enable GenAI capabilities to be embedded into existing IT service management workflows without disrupting standardised service processes (Neumann et al., 2026; Voorneveld, 2024). In parallel, security-by-design approaches, including prompt-level controls, data anonymisation, and vendor data sovereignty agreements, are proposed to mitigate novel attack surfaces and intellectual property risks (Taeihagh, 2025).

Kemell et al. (2025) observe that locally hosted language models represent an emerging technical readiness solution that allows organisations to manage data sovereignty and vendor continuity risks simultaneously, though the computational and infrastructure investment required means this option is currently accessible mainly to larger or more technically mature organisations. This represents a potential stratification in solution availability that is particularly relevant in the IT services context, where firms vary considerably in infrastructure maturity across their global delivery operations.

*Operational Assurance and Trust-Building Solutions*

A second solution category targets the output trustworthiness and assurance gap that emerges as GenAI moves into operationally critical contexts. The literature converges on the need for human-in-the-loop validation, structured review routines, and explainability mechanisms to manage hallucination risk and accountability requirements (Kemell et al., 2025; Neumann et al., 2026). Several studies propose grounding and validation techniques, such as Retrieval-Augmented Generation (RAG) and standardised evaluation pipelines, to reduce output variability and increase reproducibility across runs (Kasireddy & Sreenivasu, 2025; Pushpakumara & Ahsan, 2025). Basha and Rodríguez-Pérez (2025) identify RAG as a promising direction for reducing validation anxiety among practitioners, noting that when outputs can be traced to verifiable enterprise knowledge sources, trust in the system increases.

However, the literature introduces an important caveat: overly burdensome assurance controls can widen rather than narrow the policy-practice gap. Neumann et al. (2026) document a case where a mandatory two-person review requirement for all AI-generated content, while formally sound from a compliance perspective, was unworkable in practice and drove users toward informal workarounds that increased rather than decreased organisational risk. This finding implies that assurance mechanisms must be operationally feasible and aligned with service delivery realities, not merely theoretically defensible from

a governance perspective. This tension between compliance rigour and operational feasibility is one of the most consequential for IT services organisations, which face both strong client accountability demands and high operational velocity requirements. The solution, emerging from the literature, is not to choose between compliance and productivity but to co-design governance mechanisms with practitioners so that formal and practical compliance align.

*Organisational Capability and Change Enablement Solutions*

A third category focuses on organisational capability building, recognising that GenAI execution depends as much on human and organisational readiness as on technical performance. Across the literature, AI skills shortages and insufficient literacy are identified as persistent barriers, leading to fragmented use and over-reliance risks (Hughes et al., 2026; Kemell et al., 2025). Proposed solutions emphasise structured upskilling programmes, role-based training, and AI literacy initiatives that enable employees not only to use GenAI tools but also to critically evaluate outputs (Clear et al., 2025; Kasireddy & Sreenivasu, 2025). Fernández and Uriarte (2026) confirm empirically that countries and organisations with greater investment in specialised human capital exhibit both higher adoption rates and stronger ethical awareness, suggesting that capability building is not only a performance enabler but also a governance enabler that equips organisations to manage adoption responsibly.

Studies also highlight the importance of explicit human-AI collaboration design, including work redesign frameworks that clarify which tasks should remain human-led and which can be safely AI-assisted, particularly in diagnostic and escalation-critical IT service roles (Basha & Rodríguez-Pérez, 2025; Kemell et al., 2025). However, there is a notable tension between the capability-building and change management literature. Hughes et al. (2026) find that investing in change capacity can, under certain conditions, reduce the marginal benefit of individual skills investment, as systemic change infrastructure can substitute for, rather than complement, individual expertise. This complicates the common recommendation to invest simultaneously and equally in individual upskilling and organisational change management: the evidence suggests the relationship is substitutive rather than simply additive in high-change-capacity contexts, meaning that the optimal balance of these investments is likely context-dependent and organisations should assess their current change capacity maturity before deciding how to allocate capability-building resources.

*Governance, Leadership, and Value Realisation Solutions*

A fourth solution domain addresses institutional and strategic execution gaps, particularly those related to unclear accountability, weak governance structures, and ambiguous value capture. Governance-oriented research argues that GenAI requires adapted governance frameworks that explicitly account for GenAI-specific risks such as hallucinations, emergent bottom-up deployment, and model opacity (Taeihagh, 2025). Taeihagh (2025) advocates for a Complex Adaptive System (CAS) lens for AI governance, arguing that governance should move from 'predictive accountability,' which assigns fixed roles and responsibilities, toward 'joint accountability,' a collaborative approach using feedback loops that shares responsibility among all stakeholders. This represents a fundamental reconceptualisation of IT governance that has direct implications for how IT services organisations structure their AI oversight bodies, accountability frameworks, and client assurance mechanisms.

Leadership alignment is identified across multiple studies as a decisive enabler: senior sponsorship, clear ownership, and strategic integration of GenAI into enterprise objectives are necessary to move initiatives beyond experimentation (Kasireddy & Sreenivasu, 2025; Kumar et al., 2025). To address the business case gap, researchers propose ROI measurement frameworks and value tracking mechanisms that link GenAI outcomes to service quality, SLA performance, and risk reduction rather than isolated productivity gains (Hughes et al., 2026; Smith, 2025). Smith (2025) argues specifically that effective governance

requires embedding internal control principles from initial deployment rather than retrofitting them after scaling, and proposes a layered assurance model that distributes oversight responsibilities across technical, operational, and executive levels. Al-Khatib (2023) demonstrates quantitatively that relative advantage clarity, meaning the ability to demonstrate a clear performance improvement over existing approaches, is a primary positive predictor of GenAI adoption intention (confirmed in Hughes et al. (2026) with $\beta = 0.340$, $p < 0.001$), underscoring that the value articulation problem is not merely a marketing challenge but a fundamental adoption enabler.

*External Accountability and Legitimacy Safeguard Solutions*

The final solution category responds to external and environmental constraints that shape organisational risk tolerance. Regulatory and legal uncertainty is widely cited as a reason why pilots remain commercially safer than production deployments. The literature advocates compliance-by-design approaches, embedding regulatory interpretation, auditability, and documentation into GenAI systems from the outset (Cheung, 2025; Taeihagh, 2025). Fernández and Uriarte (2026) propose contractual frameworks that clarify liability allocation between providers, clients, and vendors to reduce ambiguity and enable confident scaling. Ethical leadership and transparency mechanisms are also highlighted as legitimacy enablers, helping organisations respond to growing stakeholder expectations around fairness, accountability, and responsible AI use (Kittipanya-Ngam et al., 2025; Kumar et al., 2025).

An important complication arises from Fernández and Uriarte's (2026) empirical finding that the structural conditions which accelerate AI adoption, such as higher national innovation capacity and greater enterprise size, simultaneously increase awareness of ethical concerns rather than resolving them. This suggests a paradox: the solutions that help organisations scale GenAI also heighten their exposure to ethical, legal, and reputational risks. For IT services organisations, this paradox is particularly acute: scaling GenAI creates client-facing accountability demands that require more sophisticated governance, even as it delivers the productivity and innovation gains that justify the investment. The implication is that governance investment is not a prerequisite to scaling that can be deferred; it must co-evolve with scaling ambition. Organisations that treat governance as a compliance activity to be addressed after scaling tend to encounter compounding governance deficits as deployment scope expands.

## 3.5 Analytical Synthesis: Barriers and Solutions Mapping

Table A.1 in Appendix A synthesises the barrier and solution landscape documented in the reviewed literature, mapping each identified execution barrier to its corresponding solution approach and indicating the key supporting literature. Rather than presenting barriers in isolation, the mapping highlights the extent to which these challenges and their responses persist across diverse organisational and industry settings and provides the direct input to the empirical phases of this study.

The literature indicates that the GenAI execution gap in IT services arises from interacting and mutually reinforcing constraints rather than isolated barriers. Technical limitations related to data, integration, and model reliability increase the need for governance oversight and human validation, thereby amplifying organisational capacity demands. Where skills, change capability, or governance maturity are insufficient, technically viable GenAI initiatives remain confined to pilots because they cannot be embedded into standardised, auditable service operations. As use cases move closer to client-facing or decision-critical contexts, assurance and governance requirements intensify simultaneously, often exposing misalignments between experimental controls and production expectations. External pressures, including regulatory uncertainty, legal liability, and competitive urgency, further shape organisational risk tolerance and deployment sequencing, frequently reinforcing pilot stasis when readiness conditions are unmet.

The synthesis also reveals that several solution pairings carry embedded tensions. Tightening governance to improve compliance can widen the policy-practice gap (Neumann et al., 2026). Investing heavily in individual skills may yield diminishing returns in high-change-capacity organisations (Hughes et al., 2026). Scaling adoption to capture competitive advantage may accelerate the rate at which ethical and governance challenges emerge (Fernández & Uriarte, 2026). These tensions do not invalidate the solution recommendations but they do mean that solution effectiveness is context-dependent, and that solutions must be sequenced, balanced, and calibrated to organisational maturity rather than applied uniformly. This insight motivates the empirical investigation's use of expert-validated prioritisation (FDM) and qualitative thematic analysis to assess which solutions are most feasible and effective in the specific IT services context.

## 3.6 Research Gaps

While existing literature richly catalogues risks, governance debates, and adoption barriers, four specific research gaps motivate this study. First, limited consolidated evidence examines how barriers and corresponding solutions interact specifically within IT service delivery contexts, where service governance, client contracting, regulated environments, and operational tooling shape what is feasible. The strongest existing empirical bases come from software and agile teams (Kemell et al., 2025; Neumann et al., 2026), general enterprise surveys (Hughes et al., 2026), and cross-national European studies (Fernández & Uriarte, 2026), but fewer studies explicitly theorise and evidence the execution mechanisms in IT services, where pilot success must be translated into repeatable service components, governed processes, and defensible client outcomes.

Second, existing studies tend to analyse barriers or solutions in isolation rather than as paired empirical constructs. The systematic relationship between specific barrier categories and their most effective corresponding solutions, in terms of contextual applicability and practical effectiveness, remains underdeveloped. The compliance gap identified by Neumann et al. (2026) and the change capacity moderation effect documented by Hughes et al. (2026) are suggestive of important cross-dimensional interactions, but neither study examines these dynamics within the structured IT services context this thesis investigates.

Third, quantitative or expert-validated prioritisation of barriers and solutions specific to the IT services sector is absent from the literature, making it difficult for practitioners to sequence their interventions strategically. Most existing studies report barriers as lists of equal salience; none apply a prioritisation method that enables differential weighting of barriers by severity or solutions by feasibility within a specific service delivery context.

Fourth, the empirically documented tensions in the literature, including the non-significant effect of regulatory pressure found by Hughes et al. (2026), the substitutive relationship between change capacity and skills found in the same study, the compliance gap documented by Neumann et al. (2026), and the paradox between scaling ambition and ethical exposure identified by Fernández and Uriarte (2026), have not been examined within a unified empirical model of the GenAI execution gap in IT services. These tensions are not merely theoretical puzzles; they have direct practical implications for how IT services organisations should prioritise and sequence their GenAI scaling strategies, and the absence of IT-services-specific empirical evidence leaves these implications unresolved.

This study addresses all four gaps through its four-phase sequential mixed-methods design, grounded in the TOE framework and operationalised through the barrier–solution taxonomy developed from this literature review. The empirical investigation is thus positioned to produce sector-specific, expert-validated, and contextually nuanced evidence about the GenAI execution gap that complements and extends the existing literature while directly addressing the practical needs of IT services leadership. The

TOE-grounded conceptual model developed in Chapter 4 explicitly integrates the cross-dimensional interaction effects identified in the literature review and links execution barriers to context-appropriate solution pathways, building on the insights documented here.

# Chapter 4: Theory

## 4.1 Theoretical Framework Selection

This research is grounded in a theoretical framework that provides the conceptual architecture for analysing how organisational, technological, and environmental factors interact to produce barriers to GenAI execution and enable the responses that overcome them. Given the multi-dimensional nature of the GenAI execution challenge, simultaneously involving technical infrastructure, organisational capabilities, governance structures, and external regulatory and competitive forces, this study requires a framework capable of capturing the full complexity of the adoption and scaling context. After evaluating competing theoretical options, the Technology-Organisation-Environment (TOE) framework is adopted as the primary theoretical lens.

## 4.2 The Technology-Organisation-Environment (TOE) Framework

The TOE framework, originally developed by Tornatzky and Fleischer (1990), provides a comprehensive model for understanding how organisations make decisions about adopting and implementing new technologies. The framework posits that technology adoption and implementation are shaped by three distinct but interrelated contextual dimensions: the Technology Context, which encompasses the characteristics of the technologies available to the organisation and their fit with existing infrastructure; the Organisation Context, which captures internal organisational characteristics including structure, culture, resources, governance capacity, and human capital; and the Environment Context, which reflects the external arena in which the organisation operates, including industry dynamics, regulatory conditions, competitive pressures, stakeholder expectations, and the broader technology ecosystem. These three contexts simultaneously influence each other and collectively determine the organisation's adoption trajectory, implementation depth, and scaling capability.

The TOE framework has been extensively validated across a wide variety of technology adoption contexts, including enterprise resource planning systems, cloud computing, e-commerce, social commerce, and blockchain. More recently, scholars have applied TOE specifically to AI and GenAI adoption contexts, consistently confirming its explanatory utility. Hughes et al. (2026) apply the TOE framework in a mixed-methods study of GenAI adoption among organisational decision-makers, identifying technological complexity, staff skills, relative advantage, and regulatory environment as key TOE-aligned constructs shaping adoption outcomes, and empirically confirming that change capacity operates as a significant positive moderator. Neumann et al. (2026) employ the TOE lens to interpret GenAI adoption patterns in agile software development teams across three German organisations, revealing systematic cross-dimensional misalignments that produce the compliance gap, a systematic divergence between formal policy and day-to-day practice. Voorneveld (2024) structures an investigation of GenAI implementation parameters around the TOE triad, demonstrating that data quality, infrastructure, organisational culture, leadership support, and regulatory conditions collectively determine implementation success or failure outcomes. Quantitative empirical validation of the TOE framework in an AI adoption context confirms that technological, organisational, and environmental constructs significantly influence adoption intention and behaviour.

## 4.3 Alternative Frameworks Considered

The TOE framework's theoretical lineage situates it alongside two other influential models: the Technology Acceptance Model (TAM), proposed by Davis (1989), and the Diffusion of Innovations (DOI) theory, formulated by Rogers (1995). TAM focuses primarily on individual-level psychological determinants of technology acceptance, specifically perceived usefulness and perceived ease of use. It has been widely applied to explain end-user adoption decisions. However, TAM offers limited insight into organisational-level dynamics, particularly for complex, multi-stakeholder technologies such as GenAI

that require coordination across departments, governance structures, and strategic decision-making hierarchies. While TAM constructs provide valuable explanatory power at the user level, they do not adequately account for the role of top management support, financial resource availability, or environmental pressures in shaping enterprise-wide technology decisions.

DOI theory, by contrast, conceptualises technology adoption as a social process through which innovations diffuse through organisational and social systems, emphasising characteristics such as relative advantage, compatibility, complexity, trialability, and observability as predictors of adoption rate and breadth (Rogers, 1995). While highly relevant for understanding innovation spread within organisations, DOI theory treats organisational and environmental factors as secondary rather than co-equal influences on adoption outcomes. The TOE framework subsumes and extends both models by treating technological characteristics, organisational conditions, and environmental forces as analytically equivalent and mutually constitutive drivers of adoption behaviour, while adding the critical environment dimension that substantially increases its capacity to predict and explain enterprise technology adoption decisions across diverse organisational contexts.

## 4.4 TOE Operationalisation in the GenAI Execution Context

### 4.4.1 Overview

For the specific purposes of this study, investigating barriers and solutions for bridging the GenAI execution gap in the IT services industry, the TOE framework is particularly appropriate for five reasons. First, its organisational focus is well-suited to the enterprise-level research question, which is concerned not merely with individual user acceptance but with the strategic, structural, and resource-based factors that determine whether GenAI initiatives successfully move beyond pilots to enterprise-scale execution. Second, the inclusion of the environmental dimension makes the TOE framework uniquely equipped to capture the regulatory, competitive, stakeholder, and ecosystem-level forces that profoundly shape IT services organisations' GenAI strategies, a dimension absent from TAM and underdeveloped in DOI. Third, the framework is sufficiently flexible to accommodate the full range of specific constructs, hallucination risks, explainability limitations, governance structures, leadership commitment, regulatory compliance, human-AI collaboration dynamics, legal liability, and vendor ecosystems, that the literature synthesis identifies as critical to the execution challenge. Fourth, the framework has demonstrated applicability across mixed-methods GenAI adoption studies, providing methodological precedent that directly supports the research design adopted in this thesis. Fifth, the TOE framework's structured three-dimensional architecture naturally translates into measurable constructs for survey instrument design, expert evaluation criteria for the FDM, and thematic coding categories for qualitative analysis, providing theoretical scaffolding for methodological integration across all four research phases.

### 4.4.2 TOE Constructs Applied to the GenAI Execution Context

This study operationalises seventeen specific constructs within the three TOE dimensions, providing full coverage of the twenty barrier categories documented in Table B.1 (Appendix B) and ensuring conceptual completeness across the identified execution constraints.

Within the *Technology Context*, six constructs are identified. The first, Model Reliability, Hallucination Risk, and Explainability and Reproducibility Limitations, encompasses the propensity of large language models to generate confident but factually inaccurate outputs, the inability to explain or audit model reasoning, and the monitoring and reproducibility challenges that arise when outputs vary unpredictably across runs (Basha & Rodríguez-Pérez, 2025; Hughes et al., 2026; Neumann et al., 2026; Pushpakumara & Ahsan, 2025; Taeihagh, 2025; Voorneveld, 2024). The second, Data Quality, Availability, and Privacy, encompasses both the adequacy of organisational data assets for safe GenAI use and the sensitivity constraints that restrict which data can be shared with GenAI systems (Neumann et al., 2026;

Pushpakumara & Ahsan, 2025; Smith, 2025; Taeihagh, 2025; Voorneveld, 2024). The third, Integration Complexity and Technical Debt, captures the technical effort and risk associated with connecting GenAI systems to existing IT infrastructure, legacy applications, ITSM workflows, and multi-client data architectures (Kasireddy & Sreenivasu, 2025; Kemell et al., 2025; Neumann et al., 2026; Voorneveld, 2024). The fourth, Security Vulnerabilities and Intellectual Property Risks, encompasses data leakage vulnerabilities, prompt injection attack surfaces, intellectual property disputes, and compliance obligations when processing sensitive client data (Kittipanya-Ngam et al., 2025; Neumann et al., 2026). The fifth, AI Infrastructure Readiness and Scalability, refers to the organisation's computational infrastructure, cloud architecture maturity, and API management capabilities required to deploy and scale GenAI workloads beyond controlled experimental environments (Kasireddy & Sreenivasu, 2025; Kemell et al., 2025; Smith, 2025; Voorneveld, 2024). The sixth, Relative Advantage Clarity and Value Demonstration, captures the difficulty of establishing a clear, measurable business case for GenAI investment beyond pilot-phase enthusiasm and linking GenAI use cases to specific IT services value pathways (Al-Khatib, 2023; Hughes et al., 2026; Kemell et al., 2025; Simaremare & Edison, 2024; Smith, 2025; Teng et al., 2025).

Within the *Organisation Context*, six constructs are identified. The first, Leadership Commitment and Strategic Alignment, refers to the degree to which senior executives actively champion GenAI initiatives, allocate resources, and embed AI objectives within organisational strategy (Kasireddy & Sreenivasu, 2025; Kumar et al., 2025; Voorneveld, 2024). The second, Organisational Readiness and Change Capacity, captures the structural, cultural, and process-level preparedness for the transformation GenAI implementation entails; Hughes et al. (2026) confirm empirically that change capacity acts as a positive moderator of the relationship between technological complexity and adoption outcomes. The third, AI Skills, Literacy, and Capability Gaps, denotes deficiencies in the internal availability of AI engineers, data scientists, prompt engineers, and AI-literate business professionals, the single most consistently documented barrier across the reviewed corpus (Clear et al., 2025; Fernández & Uriarte, 2026; Hughes et al., 2026; Kasireddy & Sreenivasu, 2025; Kemell et al., 2025). The fourth, Governance Structures, Accountability, and AI Policy Maturity, encompasses the formal institutional arrangements, AI policies, oversight bodies, ethical guidelines, accountability frameworks, and risk protocols that govern AI usage decisions at scale (Hughes et al., 2026; Neumann et al., 2026; Smith, 2025; Taeihagh, 2025). The fifth, Resource Availability and Investment Capacity, refers to the financial, human, and technological resources the organisation can commit to GenAI initiatives. Financial resource availability is a significant positive predictor of AI adoption intention. The sixth, Human-AI Collaboration Design and Over-Reliance Risk Management, captures the challenge of designing effective work routines that balance AI assistance with human judgement, particularly in diagnostic, escalation, and quality assurance roles in IT services (Basha & Rodríguez-Pérez, 2025; Clear et al., 2025; Kemell et al., 2025; Neumann et al., 2026; Wu et al., 2025).

Within the *Environment Context*, five constructs are identified. The first, Regulatory Frameworks, Compliance Requirements, and Legal Liability, captures both the external legal and regulatory environment, including the EU AI Act, GDPR, and sector-specific mandates, and the contractual liability ambiguity that arises when GenAI outputs cause harm or produce incorrect decisions in client-facing deployments (Kittipanya-Ngam et al., 2025; Neumann et al., 2026; Taeihagh, 2025). The second, Ethical Concerns, Bias, and Legitimacy Expectations, encompasses societal and stakeholder expectations regarding fairness, transparency, and accountability in AI decision-making (Kittipanya-Ngam et al., 2025; Kumar et al., 2025). The third, Industry Competition and Adoption Urgency, refers to the competitive dynamics that simultaneously accelerate adoption intent and create risks of premature deployment before readiness conditions are met (Cheung, 2025; Fernández & Uriarte, 2026). The fourth, Customer and Stakeholder Pressure, captures client demands for AI-enabled service improvement, auditability requirements, and broader market expectations (Hughes et al., 2026; Taeihagh, 2025). The fifth, Vendor Ecosystems and AI Platform Maturity, encompasses the availability, reliability, and compliance-readiness

of commercial GenAI platforms and implementation partners (Kasireddy & Sreenivasu, 2025; Neumann et al., 2026).

### 4.4.3 Methodological Rationale: How the TOE Framework Supports Research Design

A critical, and often under-articulated, strength of the TOE framework in this study is its direct methodological utility: the structured three-dimensional architecture of TOE translates naturally into the four components of the research design, ensuring coherence between theoretical constructs and empirical instruments.

*Survey questionnaire design.* Each TOE construct operationalised in Section 4.4.2 corresponds to a distinct thematic block in the survey instrument. The survey measures construct-level indicators derived from the barriers-solutions taxonomy in Table B.1 (Appendix B), using Likert-scale items to assess the perceived severity of each identified barrier and the perceived effectiveness of each corresponding solution. The Technology Context construct of Model Reliability and Hallucination Risk, for example, generates survey items assessing output trustworthiness, validation burden, and explainability requirements; the Organisation Context construct of AI Skills and Capability Gaps generates items on training adequacy, expertise availability, and literacy sufficiency. This approach is methodologically consistent with established TOE-based survey instruments (Hughes et al., 2026) and ensures that survey responses can be systematically compared across all three TOE dimensions.

*Fuzzy Delphi Method for expert validation and prioritisation.* The conceptual model variables presented in Table B.1 (Appendix B) constitute the direct input to the FDM instrument. Each TOE construct and its associated barrier–solution pairs form an item to be evaluated by IT services industry experts using triangular fuzzy number-based judgement. Experts individually rate each barrier item according to its severity to GenAI execution and each solution item according to its effectiveness and feasibility. Aggregated fuzzy scores and consensus levels are subsequently used as input for the Phase 3 semi-structured interviews to enable deeper qualitative exploration and validation of the identified barriers and solutions. Items meeting the defined threshold criteria are selected for further review and discussion during the interview phase. This process produces a prioritised and expert-validated barrier–solution hierarchy specific to the IT services context. Critically, the FDM outcomes directly inform the interview protocol by identifying the highest-priority barriers and solutions warranting in-depth qualitative exploration.

*Semi-structured interview guide design.* The interview guide is structured around the three TOE dimensions, with questions designed to explore how practitioners experience and make sense of the barriers and solutions identified in the conceptual model. The FDM prioritisation results are used to focus interview probing the most critically rated items, ensuring that qualitative exploration is directed towards the areas of highest practical significance. Interview questions operationalise TOE constructs through concrete practice-proximate prompts, "what happened," "how did you respond," "what governance mechanisms are in place", rather than abstract belief assessments, enhancing construct validity and enabling systematic cross-case comparison (Hughes et al., 2026; Neumann et al., 2026).

*Thematic analysis as a validation and enrichment mechanism.* The TOE framework serves as the theoretical codebook for the first-level theoretical template coding of interview transcripts, following established practice in TOE-grounded qualitative research (Hughes et al., 2026; Neumann et al., 2026). Initial codes map to TOE dimension categories (Technology, Organisation, Environment), enabling systematic comparison of barrier experiences and solution approaches across participants. Axial and selective coding then proceeds inductively to identify patterns, relationships, and mechanisms, including cross-dimensional interaction effects such as the compliance gap and change capacity moderation, that extend beyond or refine the a priori TOE categories. This abductive analytical approach ensures that the theoretical framework remains empirically responsive while maintaining analytical structure.

## 4.5 Conceptual Model and Research Framework

### 4.5.1 Model Description

Drawing on the TOE framework and the literature synthesis presented in the preceding sections, this study develops a conceptual model that maps the causal pathway from TOE contextual factors through GenAI execution barriers to organisational solutions and, ultimately, to improved GenAI adoption and scaling outcomes. The model is structured as: TOE Contextual Factors → GenAI Execution Barriers → Organisational Solutions → Bridging the GenAI Execution Gap. This logic reflects the theoretical position that execution barriers are not random occurrences but are systematically produced by identifiable deficiencies or tensions within the technology, organisation, and environment contexts; and that effective solutions must therefore be conceptually matched to the barrier-generating context in which they are deployed.

In the first layer, the Technology Context acts through its six constructs to create technological barriers to GenAI execution. These barriers manifest as concerns about output trustworthiness and explainability, breakdowns or persistent constraints in system integration, inadequate or insecure data management, insufficient computational infrastructure, and an inability to articulate measurable business value beyond pilot demonstrations. Technology barriers are primarily addressed through technical solutions: hallucination mitigation techniques such as retrieval-augmented generation and human-in-the-loop validation, enterprise data governance programmes, modular API-first integration architectures, security-by-design practices, cloud infrastructure investment, and ROI measurement frameworks (Pushpakumara & Ahsan, 2025; Kasireddy & Sreenivasu, 2025; Neumann et al., 2026; Voorneveld, 2024; Smith, 2025).

In the second layer, the Organisation Context acts through its six constructs to create organisational barriers. These barriers manifest as resistance to change, capability deficits, governance vacuums and policy-practice gaps, resource insufficiency, and inappropriate human-AI task allocation that creates over-reliance or deskilling risks. Organisational barriers are primarily addressed through strategic and managerial solutions: leadership-driven AI strategies, comprehensive upskilling and AI literacy programmes, AI governance framework development, structured change management initiatives, investment prioritisation roadmaps, and work redesign frameworks (Hughes et al., 2026; Kasireddy & Sreenivasu, 2025; Kittipanya-Ngam et al., 2025).

In the third layer, the Environment Context acts through its five constructs to create environmental barriers. These barriers manifest as regulatory compliance complexity, legal liability ambiguity, reputational and ethical exposure, premature deployment pressure, and vendor dependency risks. Environmental barriers are primarily addressed through strategic positioning solutions: compliance-by-design governance integration, contractual framework development for liability allocation, ethical leadership and AI ethics board establishment, deliberate adoption sequencing, and strategic vendor assessment and diversification (Cheung, 2025; Fernández & Uriarte, 2026; Kittipanya-Ngam et al., 2025; Kumar et al., 2025; Taeihagh, 2025; Teng et al., 2025; Smith, 2025).

The conceptual model also captures three documented cross-dimensional interaction effects. The first is the compliance gap (Neumann et al., 2026): a cross-dimensional misalignment between the Environment Context (regulatory requirements), Organisation Context (policy decisions), and Technology Context (practical AI usage patterns) that arises when governance frameworks are not calibrated to technological realities and practitioner constraints. The second is the change capacity moderation effect (Hughes et al., 2026): organisational change capacity (Organisation Context) moderates the negative relationship between technological complexity (Technology Context) and adoption outcomes, such that organisations with higher change capacity can absorb and navigate technological barriers more effectively. The third is resource-barrier amplification: resource constraints (Organisation Context) systematically amplify the severity of both technological barriers and environmental barriers, creating a compounding disadvantage

for resource-constrained organisations attempting to scale GenAI (Fernández & Uriarte, 2026; Kasireddy & Sreenivasu, 2025; Smith, 2025).

### 4.5.2 Research Framework: Research Propositions Rather Than Formal Hypotheses

A methodologically significant decision in the design of this study concerns the form in which the conceptual model is translated into empirical guidance: formal statistical hypotheses, research propositions, or a purely exploratory conceptual framework. The study adopts an interpretivist, mixed-methods research design combining a survey instrument, the Fuzzy Delphi Method, semi-structured interviews, and thematic analysis. This combination has important implications for the format of research guidance.

Formal statistical hypotheses, structured as directional null and alternative pairs tested through significance testing, are appropriate when the study's primary goal is to confirm or disconfirm pre-specified relationships through statistical analysis. However, FDM is not a hypothesis-testing technique: it is a structured expert consensus and prioritisation method that produces ranked and validated item lists rather than statistical relationship tests. Similarly, thematic analysis is not designed to confirm directional hypotheses but to inductively develop contextual understanding of how and why barriers manifest and how solutions are enacted in practice.

Research propositions, stated as directional theoretical relationships derived from the literature and subject to empirical exploration and refinement, are therefore the most appropriate form of empirical guidance for this study. Propositions structure the survey instrument and FDM evaluation criteria, direct interview probing, and provide a theoretical reference against which thematic analysis findings can be compared, confirming, extending, or challenging prior expectations. This approach is consistent with established practice in interpretivist and mixed-methods organisational research (Ghauri et al., 2020; Patton, 2015) and with the abductive analytical logic adopted in this study.

### 4.5.3 Research Propositions

Based on the conceptual model and literature synthesis, this study formulates ten research propositions covering the full range of TOE constructs. These propositions guide the selection of survey measurement items, the construction of FDM evaluation criteria, the design of the semi-structured interview guide, and the first-level theoretical template coding for thematic analysis.

*Proposition 1 (P1), Technological Complexity as a Barrier:* High technological complexity in GenAI systems, including model reliability risks, hallucination and explainability limitations, data quality deficiencies, and integration challenges with legacy IT infrastructure, constitutes a primary barrier to successful GenAI execution and scaling in IT services organisations. This proposition is grounded in extensive evidence that perceived complexity and technical readiness gaps are among the strongest inhibitors of technology adoption (Hughes et al., 2026; Voorneveld, 2024).

*Proposition 2 (P2), Relative Advantage Clarity as an Adoption Enabler:* The ability to clearly articulate and measure the relative advantage of GenAI over existing approaches, in terms of service quality improvement, SLA performance, knowledge management, or cost efficiency, positively influences an organisation's willingness to invest in moving from pilot to production deployment. This proposition is supported by the finding of Hughes et al. (2026) that relative advantage is the strongest positive predictor of GenAI use ($\beta = 0.340$, $p < 0.001$).

*Proposition 3 (P3), Leadership Commitment as an Organisational Enabler:* Strong leadership commitment and a clearly articulated strategic vision for GenAI positively influence an organisation's ability to overcome execution barriers by mobilising resources, reducing organisational resistance, and

sustaining implementation momentum (Kasireddy & Sreenivasu, 2025; Kumar et al., 2025; Voorneveld, 2024).

*Proposition 4 (P4), Skills Gaps as an Organisational Barrier:* Deficiencies in AI-related skills and capabilities within the organisation, including technical skills, prompt engineering competence, AI literacy, and the epistemic capacity to safely evaluate GenAI outputs, negatively influence GenAI execution success and constrain deployment quality and scaling ambition. Hughes et al. (2026) empirically confirm that staff skills positively influence GenAI use ($\beta = 0.390$, $p < 0.001$).

*Proposition 5 (P5), Governance Structures as Execution Enablers:* Robust AI governance and risk management structures, including clear accountability frameworks, AI usage policies, risk classification protocols, and ethical review mechanisms, positively influence GenAI execution capability by reducing policy-practice gaps and providing institutional scaffolding for confident deployment decisions (Neumann et al., 2026).

*Proposition 6 (P6), Change Capacity as a Moderating Factor:* Organisational change capacity moderates the relationship between technological complexity and GenAI execution outcomes, such that organisations with higher change capacity are better positioned to absorb and navigate technological execution barriers while maintaining implementation momentum. This proposition is grounded in Hughes et al.'s (2026) empirical finding that the interaction between complexity and change capacity significantly influences GenAI use ($\beta = 0.12$, $p < 0.05$).

*Proposition 7 (P7), Human-AI Collaboration Design as an Execution Requirement:* Explicit work redesign frameworks that define appropriate AI-assisted versus human-led task boundaries, and actively manage over-reliance risks, are a necessary condition for sustainable GenAI scaling in IT services organisations where diagnostic reasoning, service quality judgement, and escalation management are core delivery competencies (Basha & Rodríguez-Pérez, 2025; Kemell et al., 2025; Neumann et al., 2026).

*Proposition 8 (P8), Regulatory and Legal Uncertainty as a Scaling Constraint:* Unresolved regulatory uncertainty and legal liability ambiguity function as environmental barriers that make GenAI pilots commercially safer than production deployments, sustaining the execution gap in regulated IT services contexts (Neumann et al., 2026; Taeihagh, 2025).

*Proposition 9 (P9), Ethical Leadership and Legitimacy Safeguards as Value Enablers:* Organisations that explicitly embed ethical considerations, including bias mitigation, transparency mechanisms, and accountability structures, into their GenAI governance frameworks are better positioned to build stakeholder trust, satisfy client assurance requirements, and unlock the full performance potential of GenAI deployment (Kittipanya-Ngam et al., 2025; Kumar et al., 2025).

*Proposition 10 (P10), Resource Availability as a Cross-Dimensional Moderator:* Adequate financial, human, and technological resource availability moderates the impact of both technological and organisational execution barriers, with resource-rich organisations better positioned to invest in technical infrastructure, talent acquisition, training programmes, and governance development required for successful GenAI scaling. Resource constraints act as amplifiers of existing barriers, compounding their severity and extending the duration of the pilot equilibrium (Fernández & Uriarte, 2026; Kasireddy & Sreenivasu, 2025; Smith, 2025).

### 4.5.4 Conceptual Model Variables Table

Table B.1 (Appendix B) summarises the conceptual model variables, connecting each TOE dimension and its specific constructs to the associated GenAI execution barriers, proposed solutions, indicative survey and FDM measurement criteria, and key supporting literature. This table serves as the reference

architecture for the empirical phase of the study, providing the direct linkage between theoretical constructs, barrier measurement, expert validation criteria, interview probing themes, and thematic analysis coding categories.

### 4.5.5 Connecting the Conceptual Model to the Mixed-Methods Research Design

A distinctive contribution of this chapter is making explicit the methodological integration architecture, demonstrating precisely how the conceptual model informs and is tested by each component of the research design. The logic is sequential: the literature synthesis in Chapter 3 identifies the barrier-solution taxonomy in Table B.1 (Appendix B), which serves as the direct input to Phase 1 (the survey). Survey results provide construct-level quantitative data that feeds directly into Phase 2 (the FDM), which validates and prioritises the item set through expert consensus. FDM priority rankings then directly inform Phase 3 (semi-structured interviews), directing qualitative probing towards the areas of greatest practical significance. Phase 4 (thematic analysis) synthesises the interview findings back through the TOE framework, enabling triangulation across all four data sources and producing the final empirical conclusions. Each phase builds on the previous one, maintaining consistency across the research design, reduces the risk of fragmented findings, and substantially strengthens the validity and trustworthiness of the research conclusions (Ghauri et al., 2020; Leedy & Ormrod, 2021).

### 4.5.6 Visual Summary of the Conceptual Model

The logic of the conceptual model can be summarised in the following integrated causal and interactive pathway. The three TOE contextual dimensions each contain specific constructs that, when deficient or misaligned, generate corresponding categories of GenAI execution barriers. These barriers collectively produce the GenAI execution gap: the persistent inability of organisations to move GenAI initiatives from controlled pilots to enterprise-scale, value-producing, and governed operations. Organisations respond to these barriers through context-matched solutions, technical, managerial, and strategic, which, when effectively designed, implemented, and validated, reduce barrier intensity and improve execution capability. Cross-dimensional interaction effects, the compliance gap, the change capacity moderation effect, and resource-constraint amplification, add further analytical richness by capturing the systemic rather than additive nature of barrier dynamics.

- *Technology Context deficiencies* → Technological execution barriers (reliability, explainability, data quality, integration complexity, security vulnerabilities, infrastructure gaps, value ambiguity) → Technical solutions (RAG, data governance, modular architecture, security-by-design, cloud investment, ROI frameworks, maturity models)

- *Organisation Context deficiencies* → Organisational execution barriers (leadership gaps, change resistance, skills deficits, governance vacuums, resource constraints, human-AI collaboration risks) → Organisational solutions (strategic vision, upskilling, AI governance frameworks, change management, investment prioritisation, work redesign)

- *Environment Context pressures* → Environmental execution barriers (regulatory complexity, legal liability ambiguity, ethical exposure, competitive overreach, stakeholder demands, vendor immaturity) → Strategic solutions (compliance-by-design, contractual frameworks, ethical leadership, deliberate sequencing, industry partnerships, vendor management)

- *Cross-dimensional interactions* → Compliance gap (Environment × Organisation × Technology misalignment); Change capacity moderation (Organisation moderates Technology to Outcome); Resource amplification (Organisation constraints amplify Technology and Environment barriers) → Integrated, multi-level solutions addressing barrier co-production rather than isolated barrier categories

- *Methodological integration* → TOE constructs inform survey instrument (Phase 1) → Validated barrier-solution taxonomy feeds FDM expert consensus (Phase 2) → FDM priorities guide interview probing (Phase 3) → TOE framework structures thematic coding (Phase 4) → Triangulated findings produce IT services-specific, expert-validated execution gap model

This conceptual model constitutes theoretical scaffolding against which the empirical findings of this study are collected, interpreted, enriched, and refined. The ten research propositions derived from this model provide directional guidance for all four phases of the empirical investigation, ensuring that data collection and analysis remain anchored in theoretical expectations while remaining open to emergent insights from practitioners working at the frontier of GenAI execution in IT services.

# Chapter 5: Method

This chapter presents the methodological architecture of the study, explaining the philosophical foundations, research design, data collection strategies, analytical procedures, and validity measures employed to empirically investigate the barriers preventing IT service organisations from scaling Generative AI beyond pilot experimentation, and the solutions used to bridge the resulting execution gap. The chapter is structured around the four-phase sequential mixed-methods design that underpins this research, demonstrating how each phase is theoretically grounded in the TOE framework and how the outputs of each phase systematically inform the next.

## 5.1 Research Philosophy and Epistemological Position

Research methodology is necessarily grounded in epistemological commitments that determine what constitutes valid knowledge, how it is produced, and how it should be interpreted. This study adopts an interpretivist epistemological position, consistent with a tradition of organisational research that seeks to understand how individuals and groups make sense of technological phenomena within socially constructed, institutionally embedded contexts (Creswell & Creswell, 2017). The interpretivist stance is appropriate for the specific research problem under investigation: the GenAI execution gap is not a fixed, objectively measurable state but a socially produced phenomenon shaped by the perceptions, decisions, experiences, and contextual conditions of practitioners operating within complex organisations. As such, meaningful knowledge about this gap requires attending to subjective understandings, contextual mechanisms, and organisational dynamics that resist full quantification.

The study proceeds from an abductive logic of inquiry, which begins with empirical observation and prior theoretical framing, moves iteratively between deductive expectations derived from the TOE framework and inductive insights emerging from the data, and produces theoretical refinements rather than definitive confirmations or refutations (Patton, 2015). This abductive orientation is explicitly reflected in the analytical design: the TOE framework serves as the initial theoretical coding template for thematic analysis, while subsequent inductive coding stages allow emergent themes and mechanisms to develop beyond the a priori framework, extending or challenging the initial theoretical propositions where the data warrants. This approach, consistent with established practice in interpretivist and mixed-methods organisational research (Ghauri et al., 2020; Patton, 2015), ensures that the theoretical framework remains empirically responsive while maintaining analytical coherence across the four phases of the study.

## 5.2 Research Approach: Mixed-Methods Design

There are three primary research approaches available to researchers: qualitative, quantitative, and mixed methods (Creswell & Creswell, 2017). Quantitative methods involve surveys or experimental designs that provide numerical measurements for statistical analysis, enabling the identification of trends, patterns, and relationships through objective evaluation. Qualitative methods, by contrast, are exploratory approaches used to understand meanings, experiences, and concepts from participants' perspectives (Creswell & Poth, 2018). A mixed-methods approach integrates both qualitative and quantitative elements to leverage the strengths of each and provide a more comprehensive understanding of the research question (Creswell & Creswell, 2017).

This study employs a mixed-methods approach for reasons that are both epistemologically and empirically motivated. The research questions require simultaneous attention to two analytically distinct dimensions of the execution gap: the prevalence and relative severity of barriers and solutions across the IT services population, which demands quantitative measurement; and the contextual mechanisms, organisational dynamics, and practitioner experiences through which these barriers are produced and overcome, which demands qualitative exploration. Neither approach alone provides sufficient analytical leverage. A quantitative-only design would capture surface-level patterns without the interpretive depth

needed to understand why barriers persist and how solutions succeed or fail in practice. A qualitative-only design would risk producing rich but contextually isolated insights with limited generalisability to the broader IT services sector. The mixed-methods design resolves this tension by allowing quantitative and qualitative evidence to converge, complement, and cross-validate each other within a coherent analytical architecture (Creswell & Plano Clark, 2018).

More specifically, the study employs a sequential exploratory mixed-methods design with explanatory follow-up logic. In this design: (1) a survey questionnaire establishes an initial empirical baseline by measuring the perceived severity of barriers and the perceived effectiveness of solutions across the sample; (2) the Fuzzy Delphi Method applies structured expert consensus and prioritisation to the survey-derived item set, producing a validated and ranked barrier–solution hierarchy; (3) semi-structured interviews with IT services practitioners explore the highest-priority barriers and solutions in contextual depth; and (4) thematic analysis of interview transcripts identifies mechanisms, patterns, and cross-dimensional interaction effects that extend, refine, or challenge the prior empirical findings. The design is exploratory because the study seeks to identify, prioritise, and theorise GenAI scaling barriers and solutions in a relatively emerging research context. It also contains explanatory follow-up logic because the qualitative phase is used to interpret the mechanisms behind the FDM patterns.

## 5.3 Research Design Overview: Four-Phase Sequential Architecture

The four-phase research design is directly derived from the TOE framework's structured three-dimensional architecture. As articulated in Chapter 4, the TOE framework provides not merely a theoretical lens but a direct methodological scaffold: each of its seventeen operationalised constructs corresponds to measurement items in the survey instrument, evaluation criteria in the FDM, thematic probing questions in the interview guide, and first-level coding categories in the thematic analysis (Hughes et al., 2026). This structural alignment ensures coherence between theoretical constructs and empirical instruments across all four phases, enabling systematic comparison of barrier experiences and solution approaches across the Technology, Organisation, and Environment dimensions.

The overall research logic is as follows. The literature synthesis conducted in Chapter 3 identifies twenty barrier categories and a corresponding taxonomy of solutions, organised within the TOE framework and summarised in Table B.1 (Appendix B). This literature-derived taxonomy constitutes the direct input to Phase 1, the survey instrument. Survey results provide an initial empirical baseline, which then feeds into Phase 2, the FDM expert validation and prioritisation process. The FDM produces an expert-validated barrier–solution hierarchy, which directly informs Phase 3 by guiding the semi-structured interview protocol toward the areas of greatest practical and theoretical significance. Phase 4, thematic analysis, then synthesises the interview findings back through the TOE framework using theoretical template coding and inductive theme development. This four-phase design is sequential and exploratory in its overall purpose, while also incorporating explanatory follow-up logic through the qualitative interpretation of FDM patterns. It supports consistency between the different stages of the research, reduces the risk of fragmented findings, and strengthens the validity and trustworthiness of the research conclusions (Ghauri et al., 2020; Leedy & Ormrod, 2021).

To avoid ambiguity regarding participant numbers, the empirical design involved one broader respondent group and one smaller qualitative sub-sample. Phases 1 and 2 were based on 20 valid survey/FDM respondents, who provided structured ratings for the barrier and solution items. Phase 3 was based on 6 semi-structured expert interviews, selected from the 20 survey/FDM respondents, to provide deeper qualitative interpretation of the quantitative and FDM findings. Accordingly, the quantitative/FDM analysis is based on 20 respondents, while the qualitative thematic analysis is based on 6 interview participants and 6 interview transcripts.

## 5.4 Phase 1: Survey Questionnaire

### 5.4.1 Instrument Design and Structure

The survey instrument was designed to operationalise the seventeen TOE constructs identified in Section 4.4.2 as quantitatively measurable items, enabling systematic collection of empirical data on the perceived severity of GenAI execution barriers and the perceived effectiveness of corresponding solutions across the IT services population. The design followed the methodological approach of adapting and extending validated TOE-based survey instruments (Hughes et al., 2026) to the specific execution context of this study, with wording refined to reflect IT services deployment realities, client-facing service delivery environments, and the governance and regulatory considerations relevant to this sector.

The survey is structured into six sections (A to F). Section A collects demographic and professional profile information, including respondents' organisational role, sector, company size, years of experience with GenAI initiatives, and nature of GenAI involvement, enabling stratified analysis of responses across professional contexts. Sections B, C, and D constitute the substantive empirical core of the instrument, presenting barrier and solution items across the three TOE dimensions, respectively. Section B covers the Technology Context, comprising six construct blocks: model reliability and hallucination risk, data quality and privacy, integration complexity and technical debt, security vulnerabilities and IP risks, AI infrastructure readiness and scalability, and relative advantage clarity and value demonstration. Section C covers the Organisation Context, comprising six construct blocks: leadership commitment and strategic alignment, organisational readiness and change capacity, AI skills and capability gaps, governance structures and AI policy maturity, resource availability and investment capacity, and human-AI collaboration design and over-reliance risk management. Section D covers the Environment Context, comprising five construct blocks: regulatory frameworks and legal liability, ethical concerns and legitimacy expectations, competitive pressure and adoption urgency, customer and stakeholder pressure, and vendor ecosystems and AI platform maturity. Section E elicits respondents' holistic assessment of the overall GenAI execution gap and a weighted allocation of 100 points across the three TOE dimensions, capturing their overall judgement of relative barrier contribution. Section F provides open-ended items allowing respondents to identify barriers and solutions not captured in the structured instrument, supporting emergent discovery alongside deductive coverage.

Each construct within Sections B, C, and D is introduced by a brief contextual description that situates the items within the IT services execution context, thereby grounding abstract constructs in operationally recognisable terms for practitioner respondents. The instrument was administered online to support geographically dispersed participation and self-paced completion, consistent with online survey formats in survey research (Baburajan et al., 2022). The detailed data collection procedure and confidentiality and anonymity safeguards are described in Sections 5.4.4 and 5.10. The estimated completion time was 35 to 45 minutes, reflecting the substantive scope of the instrument and the level of reflective engagement required from respondents.

### 5.4.2 Measurement Scales and Construct Operationalisation

The survey instrument employs two parallel five-point Likert rating scales, applied consistently across all barrier and solution items throughout Sections B, C, and D. The first is the Barrier Severity Scale (BSS), which asks respondents to rate the extent to which each barrier item prevents or impedes the scaling of GenAI beyond pilots within their organisation or across the IT services sector more broadly: 1 = Not a barrier; 2 = Minor barrier; 3 = Moderate barrier; 4 = Significant barrier; 5 = Critical barrier. The second is the Solution Effectiveness Scale (SES), which asks respondents to rate the perceived effectiveness of each corresponding solution item in addressing the associated barrier: 1 = Not effective; 2 = Slightly effective; 3 = Moderately effective; 4 = Highly effective; 5 = Extremely effective. Respondents are instructed to

base their ratings on direct professional experience wherever possible, while also being permitted to draw on broader sector observations for barrier items where direct personal exposure is limited.

This dual-scale design captures two analytically distinct but complementary dimensions of the execution gap: the barrier dimension reflects the problem landscape as experienced by practitioners; the solution dimension reflects the perceived availability and practical value of remediation strategies. Together, these two dimensions enable the study to move beyond a purely descriptive account of what hinders GenAI scaling and towards an empirically grounded analysis of what can be done about it. This approach is methodologically consistent with established TOE-based survey instruments (Hughes et al., 2026) and ensures that survey responses can be systematically compared and aggregated across all seventeen TOE constructs.

Each TOE construct in the conceptual model corresponds to a distinct thematic block of barrier and solution items in the survey. The Technology Context construct of Model Reliability and Hallucination Risk, for example, generates survey items assessing the frequency of hallucinated outputs in client-facing work, the ability to explain or audit model outputs, inconsistency of outputs across repeated runs, and the overall maturity of models for live service delivery, alongside corresponding solution items addressing retrieval-augmented generation and grounding techniques, human-in-the-loop validation, standardised evaluation pipelines, and maturity-based deployment roadmaps. The Organisation Context construct of AI Skills and Capability Gaps similarly generates items on the availability of AI engineers and data scientists, prompt engineering competence and output evaluation skills, AI literacy among non-technical delivery staff, and the competitiveness of the talent market, with corresponding solutions addressing upskilling programmes, AI Centres of Excellence, strategic talent acquisition, and role-based AI training. This one-to-one correspondence between TOE constructs and survey thematic blocks ensures complete theoretical coverage and enables the FDM phase to directly process survey responses as construct-level inputs.

### 5.4.3 Sampling Strategy and Participant Selection

This study employed purposive sampling as the primary participant selection strategy for the survey phase. Purposive sampling is a non-probability technique in which participants are deliberately selected based on their expertise, experience, and characteristics relevant to the research objectives (Ghauri et al., 2020; Palinkas et al., 2015). This approach was chosen to ensure that survey respondents possessed substantive direct experience of GenAI initiatives within IT services contexts, and were therefore capable of providing informed, practice-grounded ratings of barrier severity and solution effectiveness. A convenience or random sampling approach would risk including respondents without meaningful GenAI implementation experience, severely reducing the construct validity and practical relevance of the responses (Palinkas et al., 2015).

Eligibility criteria for survey participation required respondents to: (1) be actively involved in GenAI experimentation, implementation, or governance within an IT services organisation; (2) hold a professional role with meaningful exposure to GenAI scaling decisions, including technical, strategic, managerial, or consulting roles; and (3) be employed within an IT services organisation, including system integrators, managed service providers, IT consulting firms, cloud and platform service providers, and SaaS organisations. The survey sample was additionally stratified by company size to ensure representation across small, medium, and large enterprises, enabling preliminary analysis of whether execution gap dynamics vary by organisational scale.

Participants were recruited through a multifaceted strategy combining outreach via professional networks, including LinkedIn, university connections, and industry-related organisations. Snowball sampling was additionally employed to supplement initial contacts and extend reach to senior executives and specialist practitioners with deeper GenAI scaling experience (Bell et al., 2019). The number of survey respondents

was determined with reference to the requirements of the Fuzzy Delphi Method, which constitutes Phase 2 of the analysis. Literature indicates that FDM applications can be conducted effectively with expert panels ranging from as few as nine to considerably larger groups, depending on research context and complexity (Ahmad & Wong, 2019; Petrudi et al., 2020). The survey sample was designed to meet the minimum recommended threshold for meaningful FDM analysis while remaining consistent with the purposive sampling logic of targeting information-rich, experience-bearing respondents rather than statistical representativeness.

The study acknowledges the inherent risk of selection bias in purposive sampling: participants drawn from organisations actively engaged with GenAI are more likely to hold informed but potentially optimistic views of solutions, while perspectives from organisations with no GenAI engagement may be underrepresented. To mitigate this, the recruitment strategy deliberately included participants from both organisations with successful GenAI scaling trajectories and those experiencing stalled or fragmented implementations, ensuring that barrier severity ratings captured genuine execution difficulty rather than a filtered success narrative.

The study did not impose a single-country sampling boundary because the research problem concerns GenAI scaling in IT services organisations whose client portfolios and delivery responsibilities frequently extend across national markets. Participants were therefore selected for their experience in globally operating or multi-market IT services contexts, rather than for representation of a specific country. This sampling logic is appropriate because the study examines organisational and service-delivery execution barriers, including data governance, integration complexity, client accountability, and AI governance, rather than producing a comparative analysis of national regulatory regimes.

### 5.4.4 Survey Data Collection Procedure

The survey was administered as a single-round, standardised self-completion questionnaire delivered via an online platform. A single-round format was adopted because the survey's primary function was to generate construct-level baseline data for the subsequent FDM analysis rather than to achieve iterative refinement of expert judgements. This design decision optimised respondent burden, reduced attrition across the data collection phase, and maintained clear methodological separation between the broad-sample survey and the focused expert validation stage. The survey was distributed online to eliminate geographical constraints on sampling and allow respondents to complete the instrument at their own pace, consistent with the use of online survey formats in survey research (Baburajan et al., 2022). The online administration also supported participant confidentiality and anonymity, with the specific ethical procedures for protecting participant identity described in Section 5.10.

Survey responses serve three analytical functions within the overall research design. First, they provide quantitative prevalence and severity data that allow the seventeen TOE construct-level barrier categories to be ranked by frequency and perceived impact across the sample, generating an empirical baseline profile of the execution gap. Second, the solution effectiveness ratings enable a preliminary cross-dimensional analysis of where respondents perceive the greatest solution gaps relative to barrier severity. Third, the TOE dimension weight allocation item in Section E provides an overall judgement of relative barrier contribution across the Technology, Organisation, and Environment contexts, enabling assessment of the construct validity of the theoretical framework against practitioner intuitions. These outputs feed directly into the FDM phase as the empirical item set to be subjected to expert validation and prioritisation.

## 5.5 Phase 2: Fuzzy Delphi Method (FDM)

### 5.5.1 Rationale for Choosing the FDM

The selection of the Fuzzy Delphi Method as the analytical instrument for Phase 2 reflects a deliberate choice among several candidate approaches, each of which was considered and rejected on methodological grounds. The alternatives evaluated included Latent Semantic Analysis (LSA), the classical Delphi method, the Analytic Hierarchy Process (AHP), and Structural Equation Modelling (SEM).

Latent Semantic Analysis is commonly used to identify hidden semantic patterns in large textual datasets (Leleu et al., 2011), but as the survey relies on structured, scale-based Likert items rather than qualitative text, LSA is not appropriate for processing the type of data collected in this study. The classical Delphi method, a well-established iterative consensus-building technique (Skulmoski et al., 2007; Dalkey & Helmer, 1963), assumes that participants' judgements are precise, stable, and capable of converging to crisp, definitive values, an assumption that is untenable when studying subjective practitioner perceptions of barrier severity and solution effectiveness, which inherently carry linguistic vagueness and inter-expert variability. Forcing these responses into precise numerical values would oversimplify the richness and uncertainty of the data (Hsu & Sandford, 2007). The Analytic Hierarchy Process has been widely applied in managerial decision-making (Vaidya & Kumar, 2006) but requires cognitively demanding pairwise comparisons among all items and assumes that respondents can provide internally consistent judgements, requirements that are unreasonable to impose on practitioners with varied familiarity across seventeen construct areas. Structural Equation Modelling is widely employed in management and organisational studies for testing complex relationships among variables (Hair et al., 2018), but demands large sample sizes, robust measurement models, and precisely specified quantitative indicators. Given the relatively small expert panel appropriate for the purposive FDM phase, and the exploratory rather than confirmatory intent of this phase, SEM (Structural Equation Modelling) is entirely unsuitable.

Unlike LSA, the FDM processes structured numerical responses while systematically accommodating their inherent vagueness. Unlike the classical Delphi method, it does not force respondents towards premature consensus or treat subjective judgements as precise but instead converts them into triangular fuzzy numbers that model the uncertainty intrinsic to human perception (Ishikawa et al., 1993; Zadeh, 1965). Unlike AHP, it avoids cognitively burdensome pairwise comparisons and remains analytically tractable with a small expert panel. Unlike SEM, it remains effective without requiring large sample sizes or highly formalised measurement models. These properties make the FDM particularly well suited to the expert validation and prioritisation task of Phase 2 (Bouzon et al., 2016; Nozari et al., 2021).

The FDM was developed by Ishikawa et al. (1993) as a modified version of Dalkey and Helmer's (1963) classical Delphi method, integrated with Zadeh's (1965) Fuzzy Set Theory. This integration equips FDM with the capacity to handle uncertainty, imprecision, and subjectivity in expert evaluations in a systematic and mathematically principled manner. FDM has been successfully applied in contexts including supply chain challenge identification (Bouzon et al., 2016), sustainability indicator development (Ahmad & Wong, 2019), humanitarian risk analysis (Petrudi et al., 2020), and technology adoption prioritisation (Hsu et al., 2010; Ocampo et al., 2018), consistently demonstrating its validity as a tool for consensus-based item prioritisation under conditions of expert judgement uncertainty.

### 5.5.2 Expert Panel Composition

The FDM expert panel was drawn from IT services industry professionals with strategic or operational experience in GenAI initiatives. Expert selection prioritised depth of GenAI execution experience over breadth of sector coverage, given that the FDM's purpose is to validate and prioritise items with respect to the specific IT services context. Eligibility criteria for FDM panel participation required: (1) active or

recent direct involvement in GenAI pilot or scaling initiatives within an IT services organisation; (2) professional seniority sufficient to have made or influenced decisions about GenAI investment, governance, or deployment; and (3) familiarity with the TOE-derived barrier and solution taxonomy sufficient to provide informed ratings across all seventeen construct areas.

The panel was recruited using a combination of purposive and snowball sampling methods, targeting individuals identified through the survey phase as possessing depth of experience, as well as additional specialists reached through professional networks. Literature on FDM applications indicates that a panel of between seven and sixty experts is adequate for reliable consensus formation (Ahmad & Wong, 2019; Petrudi et al., 2020), with smaller, more focused panels often producing higher-quality, more area-specific outputs in complex and specialised research contexts.

The final FDM panel composition is summarised in Table E.1 (Appendix E), which presents the profile of the 20 expert respondents who completed the survey and provided the structured ratings used in the FDM analysis. A full listing of all survey respondents by role and market is available in Table C.1 (Appendix C). The subsequent qualitative phase is documented separately in Table F.1 (Appendix F), which presents the 6 interview participants selected from the 20 survey/FDM respondents for semi-structured interviews. These 6 interviews were used to interpret, contextualise, and qualitatively validate the FDM findings; they did not replace or expand the 20-person FDM panel. This sequencing was intentionally designed to support methodological continuity between the quantitative/FDM and qualitative stages of the study.

### 5.5.3 FDM Implementation: Triangular Fuzzy Numbers (TFN)

The FDM is implemented in three sequential steps: the construction of Triangular Fuzzy Numbers from expert responses, the calculation of consensus and defuzzification, and the evaluation and acceptance of items against defined thresholds.

In the first step, expert responses on the five-point Barrier Severity Scale and the five-point Solution Effectiveness Scale are translated into Triangular Fuzzy Numbers. A TFN is defined by three parameters: *a* (the minimum possible value), *b* (the peak point, representing the value selected by the respondent), and *c* (the maximum possible value). The lower and upper bounds provide a fuzzy tolerance interval around the most likely value, modelling the inherent uncertainty in the respondent's judgement. For this study, the five-point linguistic scales are mapped to TFNs as shown in Table 5.1 below, following the approach of Bouzon et al. (2016) and Nozari et al. (2021) as applied to five-level scales.

Table 5.1: Triangular Fuzzy Number (TFN) Mapping for Five-Point Scales

| **BSS Level (Barrier)/ SES Level (Solution)** | **Linguistic Label** | **TFN (a, b, c)** |
|---|---|---|
| **1** | Not a barrier / Not effective | (0.00, 0.00, 0.25) |
| **2** | Minor barrier / Slightly effective | (0.00, 0.25, 0.50) |
| **3** | Moderate barrier / Moderately effective | (0.25, 0.50, 0.75) |
| **4** | Significant barrier / Highly effective | (0.50, 0.75, 1.00) |
| **5** | Critical barrier / Extremely effective | (0.75, 1.00, 1.00) |

The triangular membership function, $\mu(x)$, specifies the degree to which any value $x$ belongs to the fuzzy set defined by a given TFN, taking the value 0 outside the $[a, c]$ interval, rising linearly to 1 at the peak point $b$, and declining linearly back to 0 at the upper bound $c$. Since each survey item is rated by multiple expert respondents, a single aggregate TFN representing each item is calculated by taking the arithmetic mean of all individual TFNs, as expressed in Equation 1:

**Equation 1 (Average TFN):** $A^{-}_{j} = (a^{-}_{j}, b^{-}_{j}, c^{-}_{j}) = \left(\frac{\sum_{i=1}^{n} a_{ij}}{n}, \frac{\sum_{i=1}^{n} b_{ij}}{n}, \frac{\sum_{i=1}^{n} c_{ij}}{n}\right), \quad 1 \leq j \leq m$

where $n$ is the number of expert respondents and $m$ is the number of items (Bouzon et al., 2016; Nozari et al., 2021).

All quantitative data processing for the FDM phase, including survey data organisation, Triangular Fuzzy Number construction, average TFN calculation, distance measure computation, consensus threshold assessment, defuzzification, and priority ranking, was performed using Microsoft Excel, with structured calculation templates developed specifically for this study.

### 5.5.4 Consensus Assessment and Defuzzification

Following the calculation of aggregate TFNs, two analytically independent criteria are applied to assess the reliability and significance of each item: a consensus criterion and a prioritisation criterion. The consensus criterion determines whether an item is retained within the validated hierarchy, whereas the prioritisation criterion determines the relative ranking of retained items. These two functions are treated separately throughout the analysis to avoid conflating expert agreement with perceived importance or effectiveness.

For the consensus check, the Distance Threshold Criterion (Mohd Yusoff et al., 2024) is applied. For each item $j$, the distance $d$ between each individual expert's TFN and the group-averaged TFN is calculated using Equation 2:

*Equation 2 (Distance Measure):* $d_i = \sqrt{\frac{1}{3}\left[(a_j - a_i)^2 + (b_j - b_i)^2 + (c_j - c_i)^2\right]}, \quad 1 \leq i \leq n$

An average distance value $\bar{d}$ is then calculated across all expert respondents for each item. Consensus is considered achieved when the average $\bar{d}$ value for an item is less than or equal to 0.2 (Mohd Yusoff et al.,

2024). Therefore, items that do not achieve the consensus criterion ($\bar{d} > 0.20$) are excluded from the validated hierarchy on the grounds of insufficient expert convergence.

The consensus criterion therefore functions solely as a validation mechanism and does not indicate the relative importance of retained items.

For the importance and effectiveness assessment, each item's aggregate TFN is defuzzified into a single crisp value to enable direct comparison against a threshold. The weighted average defuzzification method is applied, as shown in Equation 3:

*Equation 3 (Defuzzification, Weighted Average):* $A_j^{crisp} = \frac{a_j + 2b_j + c_j}{4}, \quad 1 \leq j \leq m$

This method treats the peak value *b*, the respondents' most likely judgement, with double weighting relative to the boundary values, producing a crisp output that appropriately emphasises the central tendency of expert opinion while retaining the influence of the uncertainty bounds (Nozari et al., 2021).

### 5.5.5 Item Evaluation and Acceptance Threshold (Z)

The final step of the FDM applies a dual-criterion evaluation to determine which barrier and solution items are accepted, retained for further investigation, or excluded from the prioritised output. Two filtering criteria are applied in sequence.

Items that fail the consensus criterion (average $\bar{d} > 0.20$) are excluded from the prioritised output. For items that pass the consensus criterion, a defuzzified crisp threshold *Z* is defined as the average of all crisp values calculated via Equation 3 across all items. The resulting crisp value ranges from 0.00 to 1.00 and is used exclusively for the prioritisation and ranking of retained items. For barrier items, higher crisp values indicate higher perceived severity or criticality. For solution items, higher crisp values indicate higher perceived effectiveness and feasibility.

To support interpretation, crisp values are grouped into interpretation bands. For barrier items, values between 0.75 and 1.00 indicate critical barriers; 0.50 to 0.74 indicate significant barriers; 0.25 to 0.49 indicate moderate barriers; and 0.00 to 0.24 indicate low or minor barriers. For solution items, values between 0.75 and 1.00 indicate highly effective solutions; 0.50 to 0.74 indicate effective solutions; 0.25 to 0.49 indicate moderately effective solutions; and 0.00 to 0.24 indicate low or ineffective solutions.

Importantly, the crisp value plays no role in the consensus decision. An item may achieve strong consensus while receiving a comparatively lower crisp ranking or conversely receive a high crisp value but fail the consensus criterion due to disagreement among experts. The consensus criterion and prioritisation criterion are therefore analytically independent and must not be conflated when interpreting the findings.

The items achieving the consensus criterion constitute the final prioritised and expert-validated barrier–solution hierarchy. The retained items are subsequently ranked according to their defuzzified crisp values, enabling the identification of the most critical GenAI execution barriers and the most effective solutions within the IT services industry context. These prioritised findings directly inform the Phase 3 interview guide by identifying the highest-priority execution challenges warranting deeper qualitative exploration.

This FDM procedure therefore produces a validated and ranked hierarchy of GenAI execution barriers and solutions specific to the IT services industry context. The outcome directly addresses the third research gap identified in Section 3.6, namely the absence of IT service-specific expert prioritisation evidence and provides the empirical foundation for the subsequent qualitative investigation conducted in Phases 3 and 4.

## 5.6 Phase 3: Semi-Structured Interviews

### 5.6.1 Interview Guide Design

The semi-structured interview guide is designed to explore the mechanisms, contextual dynamics, and organisational responses associated with the highest-priority barriers and solutions identified through the FDM phase. The guide is structured around the three TOE dimensions, with questions designed to investigate how practitioners experience, make sense of, and navigate the barriers and solutions that the FDM has ranked as most critical in the IT services context.

Interview questions operationalise TOE constructs through concrete practice-proximate prompts, such as "what happened," "how did you respond," and "what governance mechanisms are in place", rather than abstract belief assessments. This design principle, consistent with the approach adopted by Hughes et al. (2026) and Neumann et al. (2026) in their TOE-grounded interview studies, shifts the focus of the interview from stated intentions or beliefs to actual organisational events, decisions, and practices. The advantage of this approach is twofold: it substantially enhances construct validity by grounding theoretical constructs in observable organisational behaviours; and it enables systematic cross-case comparison by generating comparable narratives of barrier experience and solution enactment across participants.

The FDM prioritisation results directly shape the interview guide in two ways. First, interview sections are ordered to weigh questioning towards the constructs and items that experts have rated as most critical, ensuring that limited interview time is directed towards the areas of greatest practical significance. Second, specific probing questions are tailored to the highest-priority barrier items, enabling the interviewer to move beyond surface-level descriptions and elicit the contextual mechanisms, organisational conditions, and enacted responses that explain why particular barriers are especially severe in IT services contexts and how specific solutions succeed or fail in practice.

### 5.6.2 Sampling Strategy and Participant Profile

This study employed purposive sampling as the primary participant selection strategy for the interview phase, supplemented by snowball sampling to extend access to senior practitioners with unique executive or strategic GenAI experience. Purposive sampling is widely recognised as the appropriate strategy when the research objective requires information-rich cases with direct experience of the phenomenon under study (Palinkas et al., 2015; Patton, 2015) and is particularly appropriate for exploratory mixed-methods research where the goal is depth of understanding rather than statistical representativeness.

Interview selection criteria required participants to: (1) be actively involved in GenAI initiatives within an IT services organisation, in either technical, strategic, operational, or governance roles; (2) have direct experience of the barriers to GenAI scaling, whether through stalled deployments, successful transitions from pilot to production, or governance and risk management challenges; (3) be employed within organisations of sufficient scale and GenAI ambition to have encountered the execution gap as a meaningful operational challenge; and (4) have participated in the Phase 1 survey, enabling continuity between the 20-person survey/FDM respondent group and the 6-person qualitative interview sample. The inclusion criterion of a minimum organisational size exceeding 100+ employees was applied to ensure contextual relevance, as this size IT services organisations operate the enterprise-scale governance structures, multi-client data environments, and complex legacy infrastructures most directly relevant to the execution gap phenomenon.

Snowball sampling enabled access to senior executives and strategic decision-makers, including chief AI sponsors, heads of automation and AI, and digital experience leaders, whose professional positions make them difficult to recruit through standard purposive outreach alone but whose insights into strategic and governance dimensions of the execution gap are particularly valuable (Bell et al., 2019).

The qualitative interview sample consisted of 6 participants selected from the broader 20-person survey/FDM respondent group and represented a range of IT services sub-sectors, geographic markets, and professional roles. Table F.1 (Appendix F) provides an overview of the interview participants' organisational affiliation, primary market, and role. The sample includes professionals from system integrators, managed service providers, IT consulting firms, cloud and platform service providers, and SaaS organisations, operating across Nordic, EU, US, and global markets. Professional roles range from strategy specialists, solution architects, and AI architecture heads to automation and AI leads, service desk GenAI specialists, and chief executive-level roles, providing broad coverage of both technical and strategic dimensions of the execution challenge.

Sample size was guided primarily by the principle of theoretical saturation, the point at which additional interviews cease to produce new categories, themes, or insights, rather than by a predetermined numerical target (Patton, 2015). A total of 6 interviews was considered sufficient for the qualitative phase because the later interviews reinforced and elaborated previously identified TOE-aligned patterns rather than generating entirely new theme categories. By the fifth and sixth interviews, the core themes of fragmented data governance, integration complexity, human oversight, organisational change resistance, client-side readiness, and regulatory contextuality had already emerged repeatedly across participants. The final interviews therefore contributed additional evidentiary depth and cross-case confirmation rather than new conceptual categories. Although purposive sampling restricts statistical generalisability, the 6-person qualitative interview sample, selected from the broader 20-person survey/FDM respondent group, was designed to capture meaningful variation in industry vertical, organisational scale, geographic market, and GenAI maturity stage (Leedy & Ormrod, 2021).

### 5.6.3 Interview Data Collection Procedure

Interviews were conducted via online video conferencing platforms, a mode that facilitated access to participants across multiple geographic markets while maintaining the conversational richness and interactive flexibility of face-to-face interaction. Each interview was conducted by the researchers individually, with a target duration of 45 to 60 minutes, consistent with the scope of the interview guide and the depth of contextual exploration required. All interviews were audio-recorded with participants' informed consent and subsequently transcribed verbatim in preparation for thematic analysis. The semi-structured format provided flexibility for detailed responses, allowing respondents to elaborate on their thoughts, diverge into contextually rich narratives, and introduce considerations not explicitly covered by the guide, without the rigid sequencing of a fully structured interview (Hsu & Sandford, 2007; Saunders et al., 2019).

Interview questions were ordered to move from broader contextual framing towards the specific FDM-prioritised constructs, following best practice for semi-structured interviews in technology adoption research (Hughes et al., 2026; Neumann et al., 2026). Probing follow-up questions were used throughout to deepen responses and explore the mechanisms behind initial answers, particularly where participants described specific barrier experiences or solution implementations. Notes were taken during each interview to record contextual cues and to flag significant points for thematic follow-up. Following each interview, a brief reflective memo was prepared documenting initial analytical impressions, emergent themes, and areas warranting further exploration in subsequent interviews, a practice consistent with the iterative and reflexive character of qualitative data collection in interpretivist research (Patton, 2015).

## 5.7 Phase 4: Thematic Analysis

### 5.7.1 Analytical Framework

Interview transcripts are analysed using Braun and Clarke's (2006) six-phase thematic analysis framework, the most widely applied systematic approach for identifying, analysing, and reporting patterns

or themes within qualitative data. The six sequential phases are: (1) familiarisation with the data through reading, re-reading, and initial note-taking; (2) generating initial codes by systematically labelling recurring ideas, experiences, and concepts across the dataset; (3) searching for themes by grouping codes into broader candidate themes; (4) reviewing themes against both the coded extracts and the full transcripts to assess their coherence and distinctiveness; (5) defining and naming themes to produce clear, analytically precise descriptions; and (6) producing the final report through written synthesis of the thematic structure, supported by selected participant quotations. This structured process ensures systematic and transparent engagement with the qualitative data, reducing the risk of selective interpretation and enhancing the credibility of the analytical conclusions (Braun & Clarke, 2006; Miles et al., 2014; Saldaña, 2021).

### 5.7.2 Coding Strategy: Theoretical Template and Inductive Dimensions

The coding strategy in this study employs an explicitly abductive approach that combines theoretical template coding with inductive axial and selective coding, reflecting the study's epistemological position and the dual purpose of the thematic analysis phase. The TOE framework was used as an initial theoretical template to organise first-level coding, while the subsequent coding stages remained open to emergent themes that refined, extended, or challenged the initial framework.

At the first level, the TOE framework served as the theoretical coding template, following established practice in theory-informed qualitative analysis and TOE-grounded technology adoption research (Hughes et al., 2026; Neumann et al., 2026). Initial codes were mapped to the three TOE dimension categories: Technology, Organisation, and Environment. They were also mapped to the construct-level sub-categories operationalised in Section 4.4.2 and the conceptual model variables table in Appendix B. This theoretical template ensured that interview data was systematically organised around the study's conceptual framework, enabling direct comparison of barrier experiences and solution approaches across participants and systematic triangulation against the survey and FDM findings.

At the second and third levels, coding proceeded inductively. Axial coding identified relationships between first-level codes, particularly cross-dimensional interaction effects such as the compliance gap, Environment–Organisation–Technology misalignment, and the change capacity moderation effect, Organisation moderating Technology–Outcome relationships. Some of these relationships were anticipated by the conceptual model, while others emerged more strongly through participant accounts. Selective coding then identified the core narrative or central phenomenon organising the thematic structure, enabling the development of a refined, empirically grounded account of the GenAI execution gap. This abductive movement between a theory-informed template and inductive theme development ensured that the TOE framework remained analytically useful without forcing all findings into predetermined categories.

The labels of open, axial, and selective coding are used here as descriptive analytical terms to characterise successive levels of coding abstraction within Braun and Clarke's (2006) thematic analysis framework, rather than indicating the adoption of a formal Grounded Theory methodology (Strauss & Corbin, 1990).

NVivo, a leading computer-assisted qualitative data analysis platform, is utilised to manage the coding process, organise coded data across the coding hierarchy, enable systematic retrieval of coded extracts by construct and theme, and facilitate visualisation of thematic relationships (Zamawe, 2015). NVivo substantially enhances the rigour, transparency, and auditability of the analytical process by providing a structured digital record of coding decisions that can be reviewed, revised, and reported systematically (Zamawe, 2015).

## 5.8 Triangulation and Methodological Integration

A distinctive methodological contribution of this study is its explicit integration of four empirically distinct data sources, survey responses, FDM expert consensus outputs, semi-structured interviews, and thematic analysis findings, into a coherent triangulated analytical architecture. Triangulation, as advocated by Creswell and Creswell (2017), Denzin (1977), and Patton (1999), refers to the integration of multiple data sources, methods, and perspectives to enhance credibility, validity, and comprehensiveness of research findings. In this study, triangulation operates at three levels.

At the level of data triangulation, the survey (broad sample, structured ratings), the FDM (structured consensus), and the interviews (purposive sample, qualitative depth) each provide different but complementary perspectives on the same underlying phenomenon. Convergence across these three data sources strengthens confidence in findings; divergence points to contextually important variations that enrich rather than undermine the analysis.

At the level of methodological triangulation, the combination of quantitative and qualitative methods ensures that no single methodological limitation, such as the subjectivity of qualitative coding or the aggregation loss inherent in quantitative scaling, dominates the final conclusions. The sequential design, in which each phase informs the next, ensures that the strengths of earlier phases compensate for the limitations of later phases.

At the level of theoretical triangulation, the TOE framework serves simultaneously as the theoretical template coding structure for thematic analysis, the construct architecture for the survey instrument, and the item organisation for the FDM evaluation criteria, creating a single theoretical reference point against which all four phases of empirical evidence are compared. This structural alignment, often described as methodological coherence in the mixed-methods literature (Creswell & Plano Clark, 2018), substantially strengthens the internal validity of the overall research design (Ghauri et al., 2020; Leedy & Ormrod, 2021).

## 5.9 Reliability, Validity, and Trustworthiness

Ensuring the reliability, validity, and trustworthiness of research findings is a fundamental obligation of rigorous scholarship and is particularly important in mixed-methods research, where quantitative and qualitative quality criteria coexist and must be jointly satisfied (Creswell & Creswell, 2017).

For the quantitative components of the study, the survey instrument and the FDM, reliability is assessed in terms of the internal consistency and stability of measurement. The use of validated TOE-based survey constructs (Hughes et al., 2026), adapted with careful attention to construct-level wording, reduces the risk of construct invalidity. The FDM's use of Triangular Fuzzy Numbers (TFNs) to formally model the inherent vagueness of individual expert judgement, combined with group-level aggregation and distance-based consensus calculation, directly enhances the reliability of the consensus assessment. By representing the uncertainty bounds surrounding each expert's rating as a fuzzy range rather than a precise point estimate, the FDM reduces the distorting influence of individual biases and idiosyncratic outlier ratings without requiring iterative revision rounds (Bouzon et al., 2016; Nozari et al., 2021). The application of the distance threshold criterion further ensures that retained items reflect genuine convergence of expert opinion, while the defuzzified crisp values support the prioritisation and ranking of the validated barriers and solutions according to their perceived severity, effectiveness, and feasibility.

For the qualitative components, the interviews and thematic analysis, trustworthiness is evaluated through the criteria of credibility, transferability, dependability, and confirmability, as established in Lincoln and Guba's (1985) framework for qualitative quality. Credibility is supported through purposive sampling of information-rich cases (Patton, 2015), use of concrete practice-proximate interview prompts that reduce social desirability bias, and multi-source data triangulation that cross-validates qualitative findings against

quantitative evidence. Transferability is supported by the provision of sufficiently detailed descriptions of the research context, participant profiles, and analytical procedures to allow readers to judge the applicability of findings to related IT services contexts. Dependability is enhanced through the use of a structured, documented analytical process in NVivo, enabling independent audit of coding decisions. Confirmability is supported by the explicit documentation of the abductive analytical logic and coding framework, reducing the risk of unacknowledged researcher bias shaping analytical outcomes. The study adheres throughout to the ethical guidelines for research ethics in the social sciences and humanities (Swedish Research Council, 2017), ensuring that all data collection and analysis procedures maintain the highest standards of methodological integrity.

Construct validity was strengthened through the systematic operationalisation of TOE constructs into survey items, FDM evaluation criteria, interview guide questions, and thematic coding categories, maintaining the same theoretical constructs across all four research phases and reducing the risk of construct drift. Practice-proximate interview prompts grounded abstract constructs in observable organisational behaviours, further supporting construct validity in the qualitative phase.

External validity is constrained by the purposive expert sample drawn from IT services organisations operating at pilot or production stage, and by the study's deliberate sectoral focus. The findings should therefore be interpreted as analytically generalisable to globally operating IT services contexts of similar organisational scale rather than statistically generalisable to all industries, organisation types, or geographic markets. Sufficient contextual detail is provided throughout to enable readers to assess the applicability of findings to related contexts.

Conclusion validity is supported by multi-source triangulation across four empirical phases, enabling convergence of quantitative FDM findings with qualitative thematic evidence before conclusions are drawn. Where conclusions are stated strongly, such as the identification of commercial model incentive misalignment as a structural barrier, they are grounded in cross-participant qualitative evidence corroborated by the FDM holistic weight allocation data, rather than in single-source claims. Readers should nonetheless note that the evidential base is expert-representative rather than statistically representative, and conclusions should be interpreted accordingly.

## 5.10 Ethical Considerations

This study follows the ethical principles for good research practice established by the Swedish Research Council (Swedish Research Council, 2017), which define four core requirements governing research involving human participants: the information requirement, the consent requirement, the confidentiality requirement, and the data use requirement. These principles guided the design and execution of both the survey and interview phases of the study.

*Information requirement*

In accordance with the information requirement, all participants were provided with clear and transparent information about the purpose of the study, the nature of their participation, the type of data collected, and the voluntary character of participation. Survey participants received this information in the introductory section of the online questionnaire, while interview participants received written information in advance and a verbal explanation at the start of the interview.

*Consent requirement*

The consent requirement was fulfilled by ensuring that participation was entirely voluntary and based on informed consent. Survey consent was obtained electronically prior to questionnaire completion, with participants required to explicitly confirm their agreement before proceeding. Interview participants

provided informed consent prior to the interview and explicitly agreed to audio recording, with the right to withdraw at any point without negative consequences.

*Confidentiality requirement*

In line with the confidentiality requirement, all collected data was handled in a manner that prevented identification of individual participants or their organisations. Survey responses were collected anonymously, with no personally identifiable information recorded. Interview data were pseudonymised, with participant identifiers replaced by codes and any potentially identifying organisational details removed during transcription and analysis.

*Data use (purpose and limitation) requirement*

The data use requirement was addressed by ensuring that all collected data were used exclusively for the purposes of this thesis and not shared with third parties or reused in other research projects. Data was stored securely and accessed only by the research team. Interview recordings and transcripts were retained solely for analytical purposes and will be deleted following the completion and examination of the thesis, in accordance with good research practice.

## 5.11 Researcher Reflexivity

Given the interpretivist orientation of this study, researcher reflexivity was important for strengthening the credibility of the qualitative analysis. The researchers acknowledge that their academic interest in GenAI adoption and professional exposure to digital transformation and IT services contexts may have shaped their sensitivity to themes such as governance, organisational change, cultural resistance, and execution barriers. In addition, the use of purposive and snowball sampling through professional and academic networks may have created a risk of selection bias toward participants already engaged with GenAI initiatives.

To mitigate these risks, the interview guide was anchored in the TOE framework and the FDM-prioritised findings. Participants were asked to provide concrete examples rather than general opinions, and thematic coding combined a theoretical template with inductive coding to allow unexpected themes to emerge. The qualitative findings were also triangulated against the FDM results to assess convergence and divergence across the data. These steps helped ensure that the analysis remained grounded in participant accounts while acknowledging the researchers' interpretive role.

## 5.12 Limitations

Several limitations of the methodology should be explicitly acknowledged. First, the use of purposive and snowball sampling for both the survey and interview phases necessarily restricts statistical generalisability. The sample is theoretically representative rather than statistically representative of the IT services population, and findings should be interpreted as offering contextually grounded insights rather than population-level statistics. Organisations or practitioners with no meaningful GenAI engagement are unlikely to be captured in the sample, which may produce a degree of selection bias towards organisations that have at least reached the pilot stage of GenAI experimentation.

Second, the FDM relies on expert judgements that are subjective and potentially influenced by organisational context, professional role, and individual experience with specific GenAI initiatives. While the TFN and consensus procedures formally model and partially control for this subjectivity, they cannot eliminate it entirely. Although TFNs and consensus thresholds reduce judgement variability, the FDM remains dependent on subjective expert ratings.

Third, the qualitative thematic analysis, while rigorously structured around Braun and Clarke's (2006) six-phase framework using a hybrid deductive-inductive approach with the TOE framework as the initial

theoretical coding template, introduces an element of interpretive subjectivity that is inherent to qualitative methods. The use of NVivo and documented coding procedures substantially reduces this risk, and triangulation against quantitative findings further constrains the range of plausible interpretations, but the analytical conclusions remain interpretive rather than deterministic.

Fourth, the study focuses specifically on the IT services industry, which limits the direct transferability of findings to other enterprise sectors, including financial services, healthcare, manufacturing, or the public sector, where the regulatory environment, organisational culture, and specific GenAI use cases differ materially. This sectoral focus is simultaneously a strength, enabling depth and specificity of contextualisation, and a constraint on generalisation. Future research could extend the empirical investigation to comparative multi-sector designs to assess the degree to which the barrier-solution hierarchy identified in this study is IT services-specific or more broadly generalisable across enterprise technology adoption contexts.

Fifth, a further limitation concerns geographic boundary conditions. Although the study draws on participants with global or multi-market IT services experience, it does not claim statistical representativeness across all regions or national IT services markets. Regulatory frameworks, client expectations, labour-market conditions, and data governance requirements differ across jurisdictions. Therefore, the findings should be interpreted as analytically generalisable to globally operating IT services contexts rather than universally representative of every geographic market. This limitation is particularly relevant to the Environment dimension, where regulatory and legal barriers were found to vary by jurisdiction and client context.

Notwithstanding these limitations, the mixed-method four-phase design, with its explicit methodological integration, multi-source triangulation, and TOE framework alignment, produces a level of empirical rigour and contextual validity that substantially exceeds what any single-method approach could achieve. The resulting findings offer a robust, expert-validated, and qualitatively enriched account of the GenAI execution gap in IT services, one that is specifically designed to be both theoretically grounded and practically actionable for the organisations and practitioners who encounter this challenge in their day-to-day work.

## 5.13 Use of AI Tools for Language and Writing Support

The use of artificial intelligence (AI) tools in academic research is increasingly recognised as a supportive mechanism for enhancing clarity, structure, and language quality. In line with principles of methodological transparency, this study explicitly acknowledges the limited use of generative AI tools during the writing process.

AI tools (e.g., ChatGPT (Chat Generative Pre-trained Transformer) and Copilot) were utilised in a supportive capacity only, primarily for language refinement, sentence restructuring, and improving overall readability. They were not used to generate original research content, empirical findings, or analytical insights. All core elements of the study, including research design, data collection, analysis, and interpretation, remain the intellectual contribution and responsibility of the authors.

This approach is aligned with Blekinge Institute of Technology's requirements for academic integrity and independent authorship, under which the use of supporting tools must not replace the author's own analysis, interpretation, or responsibility for the submitted work (Blekinge Institute of Technology, n.d.). Furthermore, methodological literature emphasises that research processes must transparently disclose the tools used in data production and analysis (Leedy & Ormrod, 2021), and that transparency is fundamental to ensuring trustworthiness and rigor in research (Lincoln & Guba, 1985).

Accordingly, all AI-assisted outputs were critically reviewed and validated by the researchers to ensure accuracy, coherence, and alignment with academic standards.

# Chapter 6: Data Analysis

## 6.1 Introduction

This chapter presents the empirical findings from the Fuzzy Delphi Method (FDM) phase of the study, which constitutes Phase 2 of the four-phase sequential mixed-methods research design established in Chapter 5. The purpose of the chapter is to validate, prioritise, and analytically interpret the barriers that prevent IT service organisations from scaling Generative AI (GenAI) beyond pilot stages, and the solutions through which these barriers may be overcome, using the Technology-Organisation-Environment (TOE) framework as the organising theoretical lens.

The chapter directly addresses both research questions guiding this study. RQ1 asks: what barriers prevent IT service organisations from scaling GenAI beyond pilot stages? RQ2 asks: what solutions are used to overcome these barriers and bridge the GenAI execution gap? Together, the FDM results provide a validated, expert-prioritised hierarchy of barriers and solutions specific to the IT services context, structured across the three TOE dimensions of Technology, Organisation, and Environment.

The analytical approach employed in this chapter follows the FDM methodology established in Chapter 5. Expert responses, collected via the Phase 1 survey instrument, were converted into Triangular Fuzzy Numbers (TFNs). Two derived metrics govern all analytical decisions in this chapter. First, the average distance value ($\bar{d}$) is used exclusively as the consensus criterion: items with $\bar{d} \leq 0.20$ are retained in the validated hierarchy (KEEP); items with $\bar{d} > 0.20$ are removed (REMOVE). Second, the average crisp value is used exclusively for the prioritisation and ranking of retained items: higher crisp values indicate higher perceived barrier severity or higher perceived solution effectiveness. These two functions must not be conflated; the consensus criterion and the prioritisation criterion are analytically independent and conflating them would produce misleading conclusions about the relative importance of retained items.

To strengthen the accessibility of the empirical findings, this chapter presents compact summary tables of the FDM results most directly connected to the research questions. The full item-level and aggregation tables remain in Appendix E to preserve transparency, auditability, and methodological detail. This structure allows the reader to engage directly with the validated, expert-prioritised barrier and solution hierarchy while still having access to the complete supporting dataset.

The chapter is structured as follows. Section 6.2 presents the participant profile of the expert panel. Section 6.3 explains the analytical procedure and core formulas. Section 6.4 addresses dataset reliability and data processing quality. Section 6.5 presents item-level FDM results for all 122 barriers and solution items. Section 6.6 aggregates item-level results to the construct level. Section 6.7 aggregates construct-level results to the TOE dimension level. Section 6.8 analyses expert TOE dimension weight allocations as a supplementary holistic judgement dataset. Sections 6.9 and 6.10 present the prioritised top-barrier and top-solution hierarchies respectively. Section 6.11 reports supplementary insights from open-ended survey contributions. Section 6.12 synthesises the findings against RQ1, RQ2, and the research propositions developed in Chapter 4. The FDM findings also informed the interview focus used in Phase 3, as described in Section 5.6.1.

## 6.2 Participant Profile

The FDM expert panel comprised *20 valid survey respondents*, all drawn from the IT services industry and selected based on direct professional involvement in GenAI strategy, architecture, programme management, technology leadership, or executive decision-making. These 20 respondents provided the

structured ratings used for the FDM analysis in Chapter 6. From this broader respondent group, 6 participants were selected for the semi-structured interviews analysed separately in Chapter 7. The panel was recruited through purposive and snowball sampling methods, targeting individuals whose professional positions provide genuine exposure to the challenges of moving GenAI from pilot experimentation into governed, enterprise-scale operations. The profile of the FDM panel is summarised in Table E.1 in Appendix E.

The composition of the expert panel provides a strong foundation for FDM analysis. Eighty per cent of respondents held a directly responsible or closely involved role in their organisation's GenAI strategy, ensuring that ratings reflect substantive operational and strategic experience rather than peripheral observation. The predominance of respondents with more than ten years of professional experience (80%) adds depth to the judgements, while the distribution across system integrators, managed service providers, IT consulting firms, cloud service providers, and SaaS organisations reflects the heterogeneous nature of the IT services sector itself. The geographic diversity of the panel, spanning EU, North American, Asia-Pacific, and UK (United Kingdom) markets, is analytically significant and will be revisited when interpreting the Environment dimension results, where cross-jurisdictional variation in regulatory and competitive environments is expected to produce higher d-values and lower consensus. Critically, most respondents operated in both internal and client-facing GenAI contexts, meaning their barrier and solution ratings capture the full accountability complexity of IT services: the organisation must govern GenAI not only for internal use, but also for outputs that directly affect client deliverables, contracts, and reputational trust.

## 6.3 Analytical Procedure and Decision Logic

This section provides a concise account of the FDM computation pipeline applied to the survey data, enabling the results that follow to be correctly interpreted. The full methodological justification for each step is provided in Chapter 5; the purpose here is to document the decision logic applied in producing the results tables.

*Step 1: TFN Construction.* Each respondent's five-point linguistic rating (1 = Not a barrier / Not effective; 5 = Critical barrier / Extremely effective) was converted into a Triangular Fuzzy Number (TFN) using the mapping established in Chapter 5 (Table 5.1), where rating level 1 maps to (0.00, 0.00, 0.25), level 2 to (0.00, 0.25, 0.50), level 3 to (0.25, 0.50, 0.75), level 4 to (0.50, 0.75, 1.00), and level 5 to (0.75, 1.00, 1.00). This mapping preserves the linguistic vagueness of expert judgement within a mathematically tractable triangular membership function.

*Step 2: Average TFN.* For each item *j*, the group-level Average TFN was calculated as the arithmetic mean of all *n* individual TFNs:

$$A^{-}_{j} = \left(a^{-}_{j},\, b^{-}_{j},\, c^{-}_{j}\right) = \left(\frac{\sum_{i=1}^{n} a_{ij}}{n},\, \frac{\sum_{i=1}^{n} b_{ij}}{n},\, \frac{\sum_{i=1}^{n} c_{ij}}{n}\right)$$

where *n* is the number of valid respondents for item *j* and *m* is the total number of items.

*Step 3: Distance Value and Average d.* For each respondent *i*, the distance *di* between that respondent's individual TFN and the group Average TFN was calculated using:

$$d_i = \sqrt{\frac{1}{3}\left[\left(a^{-}_{j} - a_i\right)^2 + \left(b^{-}_{j} - b_i\right)^2 + \left(c^{-}_{j} - c_i\right)^2\right]}$$

The average distance value $\bar{d}$ for item $j$ is then the arithmetic mean of all individual distance values across the panel.

*Step 4: Consensus Decision (KEEP / REMOVE).* If $\bar{d}^{-} \leq 0.20$, the item achieves consensus and is retained (KEEP). If $\bar{d}^{-} > 0.20$, the item is removed from the validated hierarchy (REMOVE). This threshold follows Mohd Yusoff et al. (2024) as operationalised in Chapter 5.

*Step 5: Defuzzification and Crisp Value.* For each retained item, the Average TFN was defuzzified using the weighted average method:

$$A_{crisp_j} = \frac{\bar{a}_j + 2\bar{b}_j + \bar{c}_j}{4}$$

The crisp value ranges from 0.00 to 1.00. For barrier items, higher crisp values indicate higher perceived severity or criticality. For solution items, higher crisp values indicate higher perceived effectiveness. The crisp value is used exclusively for ranking and prioritisation of retained items; it plays no role in the consensus decision.

*Crisp Interpretation Bands.* For barrier items: 0.75–1.00 = Critical; 0.50–0.74 = Significant; 0.25–0.49 = Moderate; 0.00–0.24 = Low or Minor. For solution items: 0.75–1.00 = Highly effective; 0.50–0.74 = Effective; 0.25–0.49 = Moderately effective; 0.00–0.24 = Low or Not effective.

The complete FDM computation pipeline described above was executed in Microsoft Excel, with the full dataset spanning 20 respondents and 122 items maintained in a structured analytical workbook, the confirmed range for which is documented in Section 6.4.

## 6.4 Reliability of the FDM Dataset and Data Processing

The confirmed analytical range covered 20 valid respondents and 122 FDM items, comprising 67 barrier items and 55 solution items structured across the TOE framework. The total possible response cells in this range were 2,440 (20 respondents × 122 items). Of these, 2,435 cells contained valid responses that were successfully converted into TFNs. Only five source cells were blank, representing a non-response rate of 0.20 per cent. These five missing cells were excluded only from the valid respondent count for the affected item; no imputation or fabrication of values was performed. All 122 items yielded valid Average TFN, average $\bar{d}$, and average crisp values, providing a complete analytical foundation for the FDM prioritisation. The near-complete response matrix substantially strengthens confidence in the robustness of the findings and ensures that the computed consensus and crisp statistics reflect genuine expert judgement rather than data handling artefacts.

## 6.5 Item-Level FDM Results: Detailed Barriers and Solutions

Table E.2 presents the item-level FDM results for all 122 barrier and solution items, organised by TOE dimension and construct. Average $\bar{d}$ governs the consensus decision; average crisp governs prioritisation within the retained set. The global rank column indicates each item's relative crisp position across all 122 items combined; separate priority rankings for barriers and solutions are provided in Tables E.6 and E.7 respectively. Items are grouped by TOE dimension and construct for readability; within each construct, barriers precede solutions.

The FDM results constitute a central empirical contribution of this thesis, a summary of the item-level results is presented in Table 6.1. The full item-level FDM results for all 122 barriers and solution items remain available in Appendix E for transparency and auditability.

Table 6.1: Summary of FDM Item-Level Results

| Item group | Total items assessed | Retained items | Main interpretation |
|---|---|---|---|
| **Barriers** | 67 | 11 | Retained barriers were concentrated mainly in Technology and Organisation |
| **Solutions** | 55 | 44 | Experts showed stronger consensus on solutions than on barriers |
| **Total** | 122 | 55 | The FDM retained 55 consensus-supported items |

Table 6.1 shows that expert consensus was stronger for solutions than for barriers. This indicates that while experts differed in how they experienced specific GenAI scaling barriers, they converged more strongly on the types of interventions considered effective across IT services contexts.

## 6.6 Construct-Level Aggregation: B1-B6, C1-C6, D1-D5

Table E.3 presents the construct-level aggregation of all retained items. Construct-level mean crisp and mean $\bar{d}$ are computed exclusively from retained items. Constructs from which no items were retained are excluded from the barriers' aggregation, as no retained data exists for comparison. Mean $\bar{d}$ is reported as a descriptive indicator of average consensus quality among retained items within each construct; it is not used as a new consensus threshold at this level.

The construct-level results reveal several analytically important patterns. Among barrier constructs, Data Quality, Availability and Privacy (B2) has the highest mean retained crisp (0.681), driven by B2.2 (fragmented data governance across business units or client environments), despite achieving consensus on only one of its four items. The fact that a single item attained consensus, while three closely related items did not, indicates that fragmented data governance is identified as the most universally severe data challenge, whereas poor data quality per se, privacy obligations, and resource constraints for data curation are understood to vary substantially in severity across organisations and deployment contexts.

Integration Complexity and Technical Debt (B3) is the only barrier construct to achieve 100 per cent consensus coverage, with all four items retained and a mean crisp of 0.603. This result is analytically significant: it suggests that integration and technical debt challenges are both severe and consistently experienced across the panel, regardless of organisational type, size, or geographic market. In IT services contexts specifically, where GenAI must integrate with multi-client architectures, legacy ITSM workflows, and complex API ecosystems, this result confirms that the architectural and technical debt bottleneck is not a context-specific anomaly but a sector-wide structural challenge.

Among solution constructs, the highest-ranking group is Customer and Stakeholder Pressure solutions (D4S, mean crisp 0.732), which covers client co-design of governance frameworks and AI-enabled service assurance. This finding is directly relevant to the IT services context: client-facing service providers cannot scale GenAI unilaterally; they must negotiate governance, accountability, and assurance frameworks with clients. The second-highest solution construct, Vendor Ecosystem solutions (D5S, mean crisp 0.730), confirms that vendor management strategy is not a peripheral concern but a central element of responsible GenAI scaling. Together, D4S and D5S establish that environmental-facing solutions are the most universally valued class of interventions, even though environmental barriers produced no

consensus retention, an asymmetry that carries important implications for how IT service organisations should prioritise their GenAI scaling efforts.

It is also notable that Governance Structures and AI Policy Maturity (C4S) produced zero-retained solution items. This does not mean that governance solutions are unimportant; indeed, all four C4S solutions had crisp scores in the effective range (0.625 to 0.678). Rather, it means that experts hold meaningfully divergent views about which specific governance mechanism is most effective. This could reflect genuine variation in the governance frameworks respondents have encountered (NIST AI RMF (National Institute of Standards and Technology Artificial Intelligence Risk Management Framework), ISO/IEC (International Organisation for Standardization/ International Electrotechnical Commission) 42001, proprietary models), or it could indicate that governance maturity is too contingent on organisational culture and sector-specific regulatory context to produce universal consensus on a single approach.

## 6.7 TOE-Level Aggregation: Technology, Organisation, and Environment

Table 6.2 presents a compact TOE-level summary of retained barriers and solutions. The full TOE-level aggregation tables remain available in Appendix E as Tables E.4 and E.5.

Table 6.2: TOE-Level Summary of Retained Barriers and Solutions

| TOE dimension | Retained barriers | Mean barrier crisp | Retained solutions | Mean solution crisp | Key interpretation |
|---|---|---|---|---|---|
| **Technology** | 7 | 0.575 | 19 | 0.696 | Strongest and most specific retained barrier signal |
| **Organisation** | 4 | 0.499 | 14 | 0.694 | Capability absorption layer; organisational barriers are more systemic |
| **Environment** | 0 | — | 11 | 0.710 | No retained barriers due to contextual variability; strongest solution mean |

Table 6.2 shows that Technology produced the strongest retained barrier signal, while Environment produced the highest retained solution mean. This asymmetry is analytically important: experts did not agree on universal environmental barriers, but they did agree that environmental-facing solutions, including client co-design, service assurance, and vendor ecosystem management, are highly effective across multi-market IT services contexts.

The TOE-level results generate three findings that merit detailed analytical attention.

*Technology produces the most visible and specific barrier signal.* Technology barriers have the highest retention rate (29.2% of proposed items retained), the highest mean retained barrier crisp (0.575, in the significant range), and the largest absolute count of retained barrier items (7 items). This confirms that technological execution challenges are the most consistently experienced across the IT services sector, regardless of organisational size, geographic market, or GenAI deployment stage. The Technology

dimension problems are not merely technical nuisances; they are operational thresholds: organisations cannot responsibly deploy GenAI in client-facing IT services contexts if they cannot assure data governance integrity, manage integration and legacy architecture complexity, guarantee output consistency, and maintain auditability. Technology, in this analysis, is the floor, the set of minimum conditions that must be satisfied before GenAI can be scaled responsibly.

*Organisation functions as the capability absorption layer.* With a 16.0% barrier retention rate and a mean retained barrier crisp of 0.499 (on the boundary between moderate and significant), the Organisation dimension presents a more contextually variable barrier picture than Technology. The retained organisational barriers are concentrated in change capacity, AI culture, and prompt evaluation skills rather than in governance, leadership, or resource availability. This is analytically interesting: it suggests that it is not the absence of strategic vision or budget that most consistently blocks scaling, but rather the organisation's inability to translate technical enablement into operational readiness. Staff who lack change capacity and output evaluation capability amplify technical barriers rather than absorbing them. This cross-dimensional amplification effect aligns with Proposition 6 (Chapter 4), which posits that organisational change capacity moderates the impact of technological complexity barriers.

*Environment functions as the legitimacy and risk-allocation layer, with paradoxical solution consensus.* The most analytically striking TOE-level finding is the combination of zero barrier retention and the highest mean retained solution crisp (0.710) in the Environment dimension. This paradox, experts diverge completely on environmental barrier severity yet strongly agree on environmental solution effectiveness, can be explained as follows. Environmental barriers (regulatory uncertainty, legal liability, competitive pressure, client demands, vendor immaturity) are experienced differently depending on jurisdiction, client industry, and market context. An organisation operating in the Nordic public sector faces regulatory and ethical pressures that differ materially from those of a US (United States) cloud services provider or an APAC (Asia Pacific) managed services firm. Expert opinion diverges because the barriers genuinely manifest differently across contexts. Environmental solutions (multi-vendor strategy, client co-design, service assurance frameworks, compliance-by-design), however, are effective across all those contexts because they address the underlying logic of external risk and accountability management rather than any particular regulatory regime. This finding has important practical implications: organisations should invest in environmental solution mechanisms as cross-contextual risk reduction tools, even when they cannot precisely quantify which environmental barriers they face most severely. This finding supports the study's boundary logic: the research does not seek to rank geographic markets against each other, but to identify execution patterns that recur across globally operating IT services contexts while recognising that environmental pressures remain jurisdictionally contingent.

## 6.8 TOE Dimension Weight Allocation: Expert Holistic Judgements

Survey Section E4 invited respondents to distribute 100 points across the three TOE dimensions to reflect their holistic judgement of which dimension contributes most to the GenAI execution gap. Unlike the FDM barrier ratings, which assess specific items, the weight allocation captures overall dimensional judgement. Fifteen of the 20 respondents provided weight allocations that summed to 100 points or very close to 100. The remaining five respondents provided incomplete or ambiguous entries that could not be reliably processed and were excluded from the dimensional average calculation. The results of the valid responses are summarised below, with approximate averages calculated from the 15 valid responses.

The approximate mean weight allocations across valid respondents are *Organisation: 38.0 points; Technology: 35.6 points; Environment: 26.3 points.* These figures indicate that, at the holistic level of expert judgement, Organisation is seen as the most consequential single contributor to the GenAI execution gap, narrowly leading Technology, with Environment seen as less dominant but still substantial.

This result creates an analytically important tension with the FDM barrier retention data. The item-level FDM results show Technology producing more retained barriers with a higher mean retained barrier crisp; yet the holistic weight allocation places Organisation as the lead contributor. Several interpretations warrant consideration. First, it is possible that Technology barriers are more operationally visible and specific, while organisational barriers are more systemic and slow-moving, experts may believe that organisational factors *create the conditions* in which technology problems become unmanageable, even when specific organisational barriers do not achieve individual item-level consensus. Second, the weight allocation may reflect a forward-looking rather than descriptive judgement: respondents may be indicating where they believe investment is most needed rather than where they observe the most acute current barrier. Third, the discrepancy could reflect the FDM consensus filter's effect: because organisational barrier items produced high $\bar{d}$ values (experts disagree on their severity), they are removed from the validated hierarchy despite being individually perceived as significant by many respondents. The holistic weight allocation is not subject to this filter and may therefore reveal latent organisational concerns that the item-level consensus process is too strict to surface.

This divergence between item-level FDM barrier consensus and holistic TOE weight allocation is itself a contribution: it reveals the limits of item-level expert consensus as a sole analytical lens and underscores the value of the Phase 3 interviews in exploring how organisational dynamics shape technology barrier severity in practice.

Respondents who allocated highest weight to Technology (one respondent: Technology 50%, Organisation 20%, Environment 30%; another: Technology 55%, Organisation 30%, Environment 15%) appeared to be senior technical specialists (architects, AI engineers) whose day-to-day work involves direct confrontation with technical barriers. Respondents allocating highest weight to Organisation (multiple allocations of Organisation 45%–50%) tended to hold programme management, strategy, or executive roles where the human, cultural, and governance dimensions of GenAI adoption are most salient. Environment allocations clustered in the 20–30% range, reinforcing the finding from the FDM that environmental pressures are perceived as real but secondary to technological and organisational factors in terms of moment-to-moment scaling difficulty.

## 6.9 Prioritisation of the Most Critical Barriers

Table 6.3 presents the 11 retained barriers ranked by average crisp value, separately from solutions. These are the barrier items that achieved both consensus ($\bar{d} \leq 0.20$) and the highest perceived severity among retained items. The full supporting FDM calculations remain available in Appendix E.

Table 6.3: Expert-Validated Priority Barriers to GenAI Scaling

| Rank | Item | TOE dimension | Barrier summary | Avg. crisp | Interpretation |
|---|---|---|---|---|---|
| **1** | B2.2 | Technology | Fragmented data governance across business units or client environments | 0.681 | Significant barrier |
| **2** | B3.2 | Technology | Technical debt increases time, cost, and risk of scaling | 0.653 | Significant barrier |
| **3** | B3.1 | Technology | Legacy systems make GenAI integration complex and costly | 0.644 | Significant barrier |

| | | | | | |
|---|---|---|---|---|---|
| **4** | B3.4 | Technology | Weak API management limits repeatable service delivery integration | 0.562 | Significant barrier |
| **5** | B3.3 | Technology | Multi-client tenant isolation and segregated environments increase complexity | 0.553 | Significant barrier |
| **6** | B1.3 | Technology | Output inconsistency makes quality assurance difficult at scale | 0.525 | Significant barrier |
| **7** | C2.4 | Organisation | Low AI readiness and weak AI culture reduce adoption willingness | 0.506 | Significant barrier |
| **8** | C2.3 | Organisation | GenAI-related change exceeds change management capacity | 0.503 | Significant barrier |
| **9** | C3.2 | Organisation | Weak prompt engineering and output evaluation skills create quality risks | 0.497 | Moderate barrier |
| **10** | C2.1 | Organisation | Cultural resistance and fear of job displacement limit embedding | 0.491 | Moderate barrier |
| **11** | B6.3 | Technology | Confusion about GenAI relative to other AI or automation tools affects investment decisions | 0.406 | Moderate barrier |

Table 6.3 makes the validated barrier hierarchy visible in the main findings chapter. The ranking shows that the execution gap is anchored first in Technology barriers, especially fragmented data governance and integration complexity. However, the presence of Organisation barriers in ranks 7 to 10 confirms that technical barriers are amplified when staff readiness, change capacity, AI culture, and output evaluation skills are underdeveloped.

The retained barrier hierarchy reveals an architectural bottleneck at the heart of the GenAI execution gap in IT services. The top barrier, B2.2 (fragmented data governance, crisp = 0.681), is not simply a data quality problem; in IT services organisations that simultaneously serve multiple clients with distinct data classification policies, access controls, and sovereignty requirements, fragmented governance is an architectural reality rather than an oversight. Each client engagement creates a separate data universe, and GenAI systems that need to learn from, or operate across, multiple client contexts face compounded governance complexity that pilots rarely surface but production deployment consistently encounters.

The cluster of four B3 items in positions 2 through 5 (technical debt, legacy architecture, API management, and multi-client tenant isolation) tells a coherent story about the IT services architectural environment. Unlike standalone enterprise adopters, IT services organisations typically maintain extensive legacy system portfolios on behalf of clients, inherit technical debt from long-term managed service relationships, and must manage data isolation requirements across clients within shared infrastructure. This makes GenAI integration not merely a matter of connecting APIs but a comprehensive architectural challenge that requires re-engineering service delivery models that were

designed without GenAI in mind. The 100% consensus coverage of B3, four of four items retained, is the strongest indicator in the entire dataset of a structurally unavoidable barrier class.

The entry of three C2 items (positions 7, 8, and 10) and one C3 item (position 9) into the retained barrier hierarchy illustrates the cross-dimensional interaction effect described in Section 6.7. Technology barriers create execution risk; organisational barriers determine whether that risk can be competently managed. Staff who lack AI readiness (C2.4) and change capacity (C2.3), and who exhibit cultural resistance (C2.1), will be less able to operationalise technical controls, interpret model outputs critically, or escalate issues appropriately when AI systems produce unexpected results. The inclusion of prompt engineering and output evaluation skills (C3.2) at position 9 is particularly significant for IT services: it is not enough to deploy a GenAI model in a service workflow; delivery staff must be capable of recognising when outputs are unreliable, inappropriate for client use, or require human override. Without this epistemic capability, the organisation amplifies technical risks rather than mitigating them.

B6.3, conceptual confusion about what GenAI is relative to other AI and automation tools, ranks last among retained barriers with the only moderate-band crisp score (0.406), but it is analytically the most strategically neglected finding. Investment confusion, stakeholder misalignment, and inappropriate benchmarking of GenAI against narrower automation tools are soft barriers that do not appear in technical architecture reviews but profoundly affect whether organisations make coherent, sustained resource commitments to scaling.

## 6.10 Prioritisation of the Most Effective Solutions

Table 6.4 presents the top 10 retained solutions ranked by average crisp value. All 44 retained solutions are available in the full item-level Table E.7 in Appendix E; the top 10 are presented here, and they represent the most accessible summary of the expert-validated solution hierarchy.

Table 6.4: Expert-Validated Priority Solutions for Bridging the GenAI Execution Gap

| Rank | Item | TOE dimension | Solution summary | Avg. crisp | Interpretation |
|---|---|---|---|---|---|
| **1** | D5.S2 | Environment | Multi-vendor ecosystem strategy to reduce platform dependency | 0.772 | Highly effective |
| **2** | B1.S2 | Technology | Human-in-the-loop validation before client use or service embedding | 0.759 | Highly effective |
| **3** | B4.S1 | Technology | Security-by-design controls at prompt, data, and model levels | 0.753 | Highly effective |
| **4** | C2.S3 | Organisation | Iterative pilot-to-scale frameworks to build confidence and change capacity | 0.747 | Effective |
| **5** | D4.S1 | Environment | Client co-design of governance, acceptable use, and auditability frameworks | 0.734 | Effective |

| | | | | | |
|---|---|---|---|---|---|
| **6** | D4.S2 | Environment | AI-enabled service assurance to evidence quality, compliance, and risk management | 0.730 | Effective |
| **7** | B1.S3 | Technology | Standardised evaluation pipelines for GenAI output monitoring | 0.725 | Effective |
| **8** | D5.S1 | Environment | Strategic vendor assessment and platform maturity evaluation | 0.722 | Effective |
| **9** | C1.S2 | Organisation | Integrating GenAI into strategic planning, OKRs (Objective and Key Result), and transformation roadmaps | 0.719 | Effective |
| **10** | C3.S2 | Organisation | AI Centre of Excellence to coordinate standards, expertise, and scaling practices | 0.719 | Effective |

Table 6.4 shows that the highest-ranked solutions are cross-dimensional rather than purely technical. The strongest solution is multi-vendor ecosystem strategy, followed by human-in-the-loop validation and security-by-design. This pattern indicates that bridging the GenAI execution gap requires simultaneous attention to technology controls, organisational change capacity, client accountability, and vendor ecosystem governance.

The top 10 solution hierarchy deserves analytical attention beyond mere ranking. Four of the top 10 solutions are from the Environment TOE dimension (D5.S2, D4.S1, D4.S2, D5.S1), reinforcing the finding from Section 6.6 that environmental solution constructs attract the highest mean retained crisp. This is the most analytically unexpected result of the FDM phase: the leading solutions for bridging the GenAI execution gap are not primarily technical controls, but rather ecosystem management and client accountability frameworks. Three of the top 10 solutions are from the Technology dimension (B1.S2, B4.S1, B1.S3), and three are from the Organisation dimension (C2.S3, C1.S2, C3.S2).

The top-ranked solution, D5.S2 (multi-vendor ecosystem strategy, crisp = 0.772), achieves the highest crisp score of all 122 items in the study. This result reflects a mature strategic understanding of GenAI platform risk: at the time of data collection, GenAI capabilities were evolving rapidly across competing platforms, foundation model providers were revising terms of service, and data sovereignty concerns were complicating single-vendor relationships. Experts broadly concur that diversifying platform dependencies is a foundational risk management strategy that reduces concentration risk, improves regulatory flexibility, and prevents vendor lock-in from constraining deployment choices. For IT services firms managing GenAI on behalf of multiple clients, vendor diversification also provides the flexibility to match the most appropriate platform to each client's compliance, performance, and cost requirements.

The second-ranked solution, B1.S2 (human-in-the-loop validation, crisp = 0.759), and third-ranked B4.S1 (security-by-design, crisp = 0.753) together establish a clear operational philosophy: in IT services contexts, GenAI cannot be deployed on the assumption that model outputs are inherently reliable or secure. The most universally endorsed operational mechanisms are those that treat GenAI as a probabilistic tool requiring human oversight and security governance rather than as an autonomous decision-making system. This finding has direct implications for service design, staffing models, and

contractual frameworks: any responsible GenAI deployment in IT services must include human validation checkpoints and security controls from inception, not as optional add-ons to scale readiness.

C2.S3 (iterative pilot-to-scale frameworks, crisp = 0.747) and C1.S2 (integrating GenAI into strategic planning, crisp = 0.719) address the organisational governance side of scaling. Incrementalism, building capability and confidence through staged deployment rather than attempting wholesale transformation, reduces the psychological and operational shock of GenAI adoption and allows organisations to develop the change capacity that the retained barriers identify as most lacking. Strategic alignment ensures that GenAI scaling programmes have the executive mandate and sustained investment that is required to move from pilot fragmentation to coordinated production deployment.

The AI Centre of Excellence (C3.S2, crisp = 0.719) as the tenth solution reflects a well-established organisational design principle: centralised expertise centres reduce duplication, ensure standards are applied consistently across projects, and accelerate the diffusion of validated GenAI practices across business units. In IT services organisations with multiple service lines and client verticals, this structural approach provides the institutional scaffolding for consistent, governed, cross-programme GenAI execution that individual project teams cannot achieve on their own.

## 6.11 Supplementary Insights: Open-Ended Survey Contributions

Survey Sections F3a (missed barriers) and F3b (missed solutions) invited respondents to identify barriers and solutions that were not captured in the structured FDM instrument. Responses were provided by 10 of the 20 respondents, representing a 50% participation rate for the open-ended sections. The contributions provide qualitative enrichment to the FDM findings and identify potential areas for instrument refinement in future research.

Three barrier themes recurred across multiple respondents and were not fully covered by the structured instrument. First, *ROI ambiguity and value attribution* was identified by three respondents as a critical or significant barrier. Unlike B6 (Relative Advantage Clarity), which addresses the difficulty of building a business case for scaling, these responses emphasised the ongoing challenge of attributing measurable value, in revenue, cost savings, or customer experience improvement, to live GenAI deployments after the pilot stage has ended. This suggests the study's B6 construct may need to be split into a prospective (pre-deployment business case) and a retrospective (post-deployment value attribution) dimension in future iterations.

Second, *LLMOps and operational AI infrastructure maturity*, encompassing model monitoring, prompt lifecycle management, evaluation tooling, and production observability, was identified by at least two respondents as a critical gap. This barrier is partially captured by B1.3 (output inconsistency) and B1.4 (model maturity), but the open-ended responses suggest that the operational infrastructure required to run GenAI at production scale (LLMOps tooling, continuous evaluation pipelines, automated drift detection) constitutes a distinct capability domain that merits its own construct in the TOE taxonomy.

Third*, cost unpredictability and financial governance*, relating to token pricing volatility, infrastructure scaling costs, and the difficulty of budgeting for AI-enabled service delivery, was flagged by multiple respondents as a significant barrier to sustained investment beyond pilots. This was not included in the C5 (Resource Availability) construct, which focused on budget availability and investment appetite rather than cost variability and financial forecasting challenges specific to AI workloads.

On the solutions side, respondents proposed *standardised pilot-to-production operating models with clear ownership and value KPIs, enterprise data product governance frameworks, GenAIOps/LLMOps platform standardisation, and value engineering and FinOps (Financial Operations) for AI* as important solution categories not fully represented in the structured instrument. One respondent proposed a venture-capital-style portfolio approach to GenAI pilots, killing weak initiatives early and accelerating those with

proven traction, as an under-utilised governance mechanism for managing the execution gap. These contributions indicate that the solution taxonomy may require augmentation in subsequent research iterations, particularly around financial governance and operational AI management disciplines.

## 6.12 Summary of Findings and Connection to Research Questions

This chapter has presented the complete FDM results for 122 barriers and solution items across the Technology, Organisation, and Environment TOE dimensions. The findings directly address both research questions guiding this study.

*In response to RQ1,* what barriers prevent IT service organisations from scaling GenAI beyond pilot stages? The FDM results produce a validated hierarchy of 11 consensus-supported barriers. The most critical barriers are concentrated in two Technology constructs: fragmented data governance (B2.2) and the cluster of integration complexity and technical debt challenges (B3.1, B3.2, B3.3, B3.4). These are followed by output inconsistency (B1.3) and a group of organisational readiness constraints (C2.1, C2.3, C2.4) and an AI skill deficit (C3.2). The data does not support a single-cause account of the execution gap. Rather, the retained barrier hierarchy reveals a structured set of compounding constraints: data governance fragmentation prevents reliable data pipelines; technical debt and legacy architecture increase integration risk and cost; inconsistent outputs undermine client trust; and organisational change capacity limitations prevent the human capital development needed to manage these technical challenges responsibly. Together, these barriers explain why pilot success does not consistently translate into production deployment: the conditions required for scalable, governed, client-safe GenAI delivery are architecturally and organisationally more demanding than those required for successful experimentation. The complete absence of retained Environmental barriers does not indicate that the external context is unimportant; rather, it reflects the jurisdictional and market variability of those pressures, a finding that has direct methodological implications for how environmental barriers should be assessed in future research using geographically and sectorally diverse expert panels.

*In response to RQ2,* what solutions or enablers are most likely to bridge the execution gap? The FDM results produce a validated hierarchy of 44 consensus-supported solutions across all three TOE dimensions, with the top 10 detailed in Section 6.10. The most analytically significant finding is that the highest-ranked solutions are concentrated in the Environment dimension, not the Technology dimension. Multi-vendor ecosystem strategy (D5.S2, crisp = 0.772), client co-design of governance frameworks (D4.S1, crisp = 0.734), and AI-enabled service assurance (D4.S2, crisp = 0.730) collectively define an outward-facing, ecosystem-level solution logic: bridging the execution gap in IT services requires managing client relationships, vendor dependencies, and external accountability structures, not merely resolving internal technical problems. Human-in-the-loop validation (B1.S2, crisp = 0.759) and security-by-design (B4.S1, crisp = 0.753) establish the operational philosophy that must underpin GenAI deployment at the item level: probabilistic AI outputs require human governance, not autonomous operation. Organisational solutions, iterative pilot-to-scale frameworks (C2.S3), strategic alignment (C1.S2), and an AI Centre of Excellence (C3.S2), address the capability and governance infrastructure that converts technical enablement into sustained organisational performance. The overall solution hierarchy is therefore cross-dimensional: closing the execution gap requires simultaneous action on technological controls, organisational capability, and ecosystem management rather than sequential or siloed intervention.

*Retention asymmetry as a meta-finding.* A structural finding that cuts across both research questions is the pronounced asymmetry between barrier and solution retention rates: 16.4% of proposed barriers were retained compared with 80% of proposed solutions. This asymmetry is not a methodological artefact; it reflects genuine expert characteristics. Experts share a more differentiated and contextually variable view of which specific barriers they face most severely, reflecting the heterogeneity of their organisations, markets, and deployment stages, but converge strongly on which solutions are broadly effective across

those heterogeneous contexts. This has an important practical implication: organisations need not wait for perfect consensus on which barriers they face before acting. The solution frameworks validated in this study are sufficiently universal to justify implementation even in contexts where individual barrier severity is contested. Proportionate investment in the top-ranked solutions represents a defensible strategy regardless of whether an organisation's dominant execution constraint is primarily technological, organisational, or environmental.

*Divergence between item-level consensus and holistic weight allocation. The* tension between item-level FDM barrier retention (Technology dominant, seven retained barriers; Organisation second, four retained) and expert holistic weight allocation (Organisation leading at 38.0 points, Technology at 35.6 points) constitutes an additional analytical contribution. The FDM consensus filter surfaces what experts agree is universally significant; the weight allocation reveals what experts believe is most consequential as a systemic category. The discrepancy suggests that organisational barriers may be more systemic and contextually embedded than item-level consensus can fully capture, and that their full causal significance, particularly as amplifiers of technological risk, requires interpretive investigation that the quantitative FDM instrument cannot provide alone.

*Connection to Phase 3.* The FDM findings generate a targeted and theoretically grounded interview agenda for Phase 3 of this study. The 11 retained barriers and the top 10 solutions provide structured anchor points for semi-structured interviewing: practitioners can be asked to contextualise these consensus findings within their own organisational experience, to account for how barrier severity varies by organisational profile (size, geography, sector), and to elaborate on the mechanisms by which the highest-ranked solutions are applied in practice. The Environmental paradox, barrier non-consensus and solution consensus, warrants specific exploration: interviewees will be asked to describe how they perceive and manage regulatory and ecosystem uncertainty in the absence of a settled shared understanding of environmental risk. The divergence between holistic dimension weight and item-level retention provides a further interview theme: understanding whether experts locate organisational barriers at the systemic or cultural level that item-level survey instruments are poorly positioned to capture. The complete Phase 3 interview guide, incorporating FDM-derived propositions and constructs, is presented in Chapter 5 (Section 5.6.1).

# Chapter 7: Thematic Analysis

This chapter presents the findings from Phase 3 of the research: the qualitative thematic analysis of 6 semi-structured interviews conducted with IT services practitioners. These 6 interviews form the qualitative dataset for the thematic analysis and were conducted with participants selected from the 20 survey/FDM respondents used in Chapter 6. Using Braun and Clarke's (2006) six-phase thematic analysis framework, with the Technology–Organisation–Environment (TOE) framework as the initial theoretical coding template, the analysis synthesises practitioner experience into a coherent set of themes that contextualise, validate, challenge, and extend the Fuzzy Delphi Method (FDM) findings reported in Chapter 6. The chapter directly addresses both research questions: RQ1 (what barriers prevent IT service organisations from scaling GenAI beyond pilot stages?) and RQ2 (what solutions or enablers are most likely to bridge the execution gap?). It concludes with an integrated mixed-methods triangulation that identifies points of convergence, refinement, and productive tension between the quantitative and qualitative evidence streams.

## 7.1 Analytical Approach and Methodological Recap

The thematic analysis was conducted in accordance with Braun and Clarke's (2006) six-phase framework: familiarisation with data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the final report. As established in Chapter 5, the coding strategy employed a hybrid theoretical template–inductive design. The three TOE dimensions provided the initial theoretical coding template, ensuring systematic alignment with the research propositions and FDM-derived constructs. An inductive second level was applied within and across these template categories to allow practitioner framings, emergent constructs, and unexpected tensions to surface from the data without being forced into predetermined categories.

All six interview transcripts were processed sequentially, with an initial round of open coding applied to the full text of each transcript. A second round of focused coding grouped open codes into candidate themes aligned with the TOE dimensions. Participant validation logic was operationalised through Section 5 of each interview, in which participants reacted directly to selected FDM findings. This dual structure, inductive coding followed by deductive alignment and participant validation logic, supports both the credibility and the analytical rigour of the findings (Lincoln & Guba, 1985; Braun & Clarke, 2006).

The analysis proceeded from data familiarisation through thematic saturation. Saturation was assessed across the six participants at the construct level: when no new TOE-coded theme categories were emerging and when subsequent interviews were generating additional evidentiary detail rather than new conceptual categories, the analysis was considered substantively complete (Guest et al., 2006).

In presenting the themes, the analysis prioritises cross-participant evidence rather than isolated individual accounts. Where one participant provided a particularly detailed example, the interpretation was checked against the remaining interviews to determine whether the theme was confirmed, refined, contrasted, or contextualised by other participants. This approach supports cross-case saturation and ensures that individual quotations are used to illustrate themes rather than define them.

## 7.2 Participant Overview

Table F.1 in Appendix F summarises the 6 interview participants, their organisational profiles, and their primary GenAI involvement. These participants were selected from the broader 20-person survey/FDM respondent group and constitute the qualitative sample for the thematic analysis. Pseudonyms are used throughout this chapter to protect participants' confidentiality, in accordance with the ethical protocols outlined in Chapter 5. The panel spans roles across enterprise agentic AI architecture, managed IT service product ownership, technology strategy and transformation consulting, pre-sales cloud infrastructure, IT

consulting delivery, and programme management. Collectively, the participants represent the IT services value chain from vendor-side delivery and pre-sales through to client-side strategy and programme governance, providing the cross-contextual coverage necessary for the triangulation of FDM findings with lived practitioner experience.

## 7.3 Initial Coding and Theme Generation

Initial open coding across the six interview transcripts generated 187 distinct codes. Focused coding reduced these to 42 candidate theme clusters, which were subsequently reviewed for thematic coherence, cross-participant saturation, and analytical alignment with the TOE framework. After the theme review phase, 11 final themes were identified and named: four within the Technology dimension, three within the Organisation dimension, two within the Environment dimension, and two cross-dimensional interaction themes.

To make the cross-case basis of the thematic analysis visible in the main text, Table 7.1 summarises the participants' support for each final theme and indicates its relationship to the FDM barrier or solution hierarchy. The table does not imply that each participant discussed every theme with equal depth. Rather, it shows how each theme was supported, refined, or contextualised across multiple interviews. The detailed coding summary is provided in Appendix F, Table F.2, while the full mixed-methods triangulation matrix is provided in Appendix F, Table F.3.

Table 7.1: Cross-Participant Support and FDM Alignment of Final Themes

| Theme code | Theme label | TOE dimension | Contributing participants | FDM relationship |
|---|---|---|---|---|
| **T1** | The Data Foundation Paradox | Technology | A, R1, D, H, S | Validates B2.2 and refines severity framing |
| **T2** | The Integration Bottleneck | Technology | R1, D, A, R2 | Validates B3.1, B3.2, B3.3, and B3.4 |
| **T3** | Output Reliability and the Human Oversight Imperative | Technology | A, D, H, R1, R2 | Validates B1.3 and supports B1.S2 |
| **T4** | The Governance Architecture Gap | Technology / Organisation | A, R2, H | Extends B6.3 through risk-tiered governance |
| **O1** | The Frozen Middle | Organisation | S, A, H, D, R1 | Validates C2.1 and C2.4; explains organisational weight |
| **O2** | Strategic Ambiguity and the ROI Credibility Gap | Organisation | A, D, R2, H, S | Validates B6.3 and refines C1.S2 |

| | | | | |
|---|---|---|---|---|
| **O3** | The AI Council, CoE, and Upskilling Triad | Organisation | A, S, H, D, R2 | Supports C3.S2, C1.S3, and C3.2 |
| **E1** | Client-Side Readiness as the External Scaling Ceiling | Environment | D, H, R1, S | Explains environmental barrier non-consensus; supports D4.S1 and D4.S2 |
| **E2** | Regulatory Contextuality and Universal Solution Logic | Environment | D, H, A, R2 | Explains the environmental paradox; supports D1.S1–D1.S3 |
| **X1** | Security Gatekeeping as a Cross-Dimensional Amplifier | Cross-dimensional | D, A, R2, H | Operationalises the compliance gap |
| **X2** | Iterative Scaling vs. Pilot Purgatory | Cross-dimensional | D, A, H, S, R2 | Validates C2.S3 and refines pilot-to-scale logic |

The following sections develop these themes in detail, using participant accounts to explain how the FDM-prioritised barriers and solutions operate in practice.

## 7.4 Technology Dimension Themes

The Technology dimension produced four themes. Together, these themes validate the FDM's finding that Technology barriers represent the most operationally visible and consistently experienced constraint on GenAI scaling in IT services. Qualitative evidence deepens this finding by revealing the mechanisms behind the quantitative rankings and by identifying a significant emergent theme, risk-tiered governance architecture, that the structured FDM instrument only partially captured.

### 7.4.1 Theme T1: The Data Foundation Paradox

Fragmented data governance across client environments (FDM item B2.2, crisp = 0.681, ranked first among all retained barriers) was the single most consistently discussed theme across the Technology dimension. However, the qualitative evidence introduces an important analytical distinction that the FDM crisp score alone cannot surface: participants did not universally characterise data governance fragmentation as a fatal barrier. Rather, a productive tension emerged between those who experienced it as a structural, architecturally unavoidable constraint and those who characterised it as a resolvable challenge provided the organisation commits to building the right data foundation early.

A, whose role involves designing enterprise agentic AI architectures for clients across manufacturing, Financial Services, and energy sectors, offered the most technically precise account of this distinction. She noted that the absence of a unified data layer was the most common reason GenAI pilots fail to transition to production: "The common thing is that everyone is onboard at this AI transformation... but the backbone is the data layer. AI agent will perform exceptionally well if your data layer is sorted." She described the typical enterprise pattern as one in which organisations focus exclusively on the visible AI layer, models, interfaces, and agents, while neglecting the data infrastructure

beneath it. Critically, however, she observed a recent and encouraging shift: "First time I saw two consecutive customers, they came and they said that in their RFP (Request for Proposal), it was a unified data layer they want to set up." This signals emergent data maturity within the client base, with a growing recognition that data consolidation is a prerequisite rather than an optional refinement of GenAI scaling.

D provided a complementary perspective grounded in multi-client managed service delivery. From his experience managing a portfolio of over 300 customers across diverse regulatory and operational environments, he described data governance as fundamentally variable rather than universally absent: "Password reset for customer A doesn't necessarily copy the password reset for customer B... the differences in data governance in general are very different customer by customer." This multi-client fragmentation creates a compounding complexity that single-client enterprise pilots do not encounter: each client environment constitutes a separate data universe with distinct access controls, sovereignty requirements, and classification policies. D confirmed this as "heavily impacting" the transition from pilot to scale, particularly in terms of the operational overhead of maintaining change management coherence across multiple client environments simultaneously.

H, approaching the question from a healthcare consulting context, offered a nuanced refinement that distinguishes between two aspects of data governance that the FDM's B2.2 item conflates: structural quality issues (clean, structured, unified data) and re-identification or privacy risk issues that become more acute at lower levels of aggregation. He observed that at enterprise level, aggregating patient data across a large network reduces re-identification risk substantially, whereas processing data at a division or practice level creates higher individual-level exposure with less governance infrastructure: "When you're doing at a patient level aggregation on a division level, then there is a risk... a higher risk of you know exposing or re-identification of the patients. But there is no governance that is separately added at the division level." This finding suggests that the FDM's ranking of fragmented data governance as the top barrier may undersell the risk-differentiation problem: the challenge is not merely that governance is absent but that governance frameworks are not calibrated to the specific risk profile of each deployment context.

R1 added an IT services-specific perspective by emphasising that data is often divided between centralised organisational repositories and decentralised, customer-specific environments, with access shaped by contract-specific permissions. S provided a useful contrasting case, noting that data fragmentation was less constraining where the required observability data was already available, secured, and directly connected to the selected use case.

Taken together, these accounts validate B2.2 as the highest-ranked FDM barrier while extending its interpretation beyond a general data-quality problem. The qualitative evidence shows that fragmented data governance becomes most severe where client contracts, access rights, security approvals, and environment-specific governance prevent the creation of reusable data foundations. The resolution pathway is therefore not a single technical fix, but an architectural and governance commitment to data platform maturity before GenAI deployment is scaled. This finding aligns with A’s practical emphasis on enterprise data product frameworks and metadata layers as foundational prerequisites, which also appeared in the FDM open-ended contributions in Section 6.11 but were not fully captured in the structured item hierarchy.

### 7.4.2 Theme T2: The Integration Bottleneck

The four B3 items (B3.1 to B3.4) achieved 100 per cent consensus coverage in the FDM, the only barrier construct to do so, signalling that integration complexity and technical debt constitute a structurally unavoidable barrier class across the IT services sector. The interview evidence corroborates this signal and provides mechanistic detail about why the bottleneck is so persistent.

R1's account was the most architecturally precise. Drawing on experience in legacy ITSM environments, he described what he termed the "integration paradox": legacy on-premises infrastructure may be functionally adequate for existing service workflows, but it fundamentally lacks the real-time data processing and API orchestration capabilities that GenAI workloads require. Where a pilot can be constructed in an isolated sandbox environment with clean data feeds and controlled API access, production deployment requires integration with live customer systems, ITSM platforms, identity management systems, ERP (Enterprise Resource Planning) layers, and communication channels, that were never designed to accommodate AI-driven data flows. The cost and complexity of retrofitting these architectures escalates rapidly as the number of client environments grows.

D confirmed this pattern from his direct operational experience, describing integration timelines as one of the most persistent sources of project delay: "Integrations are taking long. For example, in my projects in general, I would wish them to be much faster and much sooner implemented." He clarified, however, that the delay was more frequently driven by security approval processes than by technical infeasibility: "The integrations are taking long... it is technically doable. But my biggest worry is the security, security concerns of customers who are in the end supposed to provide this API." D's account was reinforced by other participants, who framed integration as both a technical and organisational problem. R1 distinguished between legacy infrastructure and what he described as a legacy mindset, where stakeholders may support AI in principle but remain reluctant to open data access or create APIs. R2 added a production-oriented view, explaining that integration becomes more demanding once a solution must handle real-time load, infrastructure choices, vector database design, LLM (Large Language Model) service selection, and cost-performance trade-offs. A similarly linked integration complexity to the maturity of the underlying data and metadata layer. Together, these accounts show that the integration bottleneck is not purely a technical-debt problem, but an interaction between legacy systems, security approvals, infrastructure scalability, and organisational willingness to expose data and interfaces. This reframing of the technical debt bottleneck as partially a governance and approval challenge rather than a purely architectural one has important implications: it relocates part of the solution logic from the Technology dimension (API architecture improvement) to the cross-dimensional environment-technology interaction (Theme X1, discussed in Section 7.7.1).

A added a further dimension to the integration theme by noting that enterprise clients are frequently reluctant to allow direct bidirectional API integration with their core ERP (Enterprise Resource Planning) systems, SAP (Systems, Applications, and Products (enterprise software)), Salesforce, and similar platforms, citing security and governance concerns: "The governance comes into picture because their digital security or CISO (Chief Information Security Officer) won't approve your architecture if you have a direct bidirectional communication with the enterprise platforms." The result is a technically suboptimal but governance-acceptable architecture that relies on batch-processed data copied to an intermediate environment: "Through JDBC (Java Database Connectivity) connector, they have just extended their SAP data to one of the EC2 (Elastic Compute Cloud) instances... they said that okay, it's okay if we are not having the real time thing. We are dependent on batch processing." This architectural workaround illustrates the "integration paradox" in concrete terms: the desire for technically optimal GenAI integration collides with security governance constraints, and the resolution typically involves compromising on real-time capabilities, reducing the potential business value of the AI implementation in the process.

This theme validates the B3 integration-complexity cluster by showing that technical debt becomes most restrictive when legacy systems, data access restrictions, and organisational risk concerns interact.

### 7.4.3 Theme T3: Output Reliability and the Human Oversight Imperative

FDM item B1.3 (output inconsistency, crisp = 0.525) ranked sixth among retained barriers, while B1.S2 (human-in-the-loop validation, crisp = 0.759) ranked second among all retained solutions. The interview

data strongly validates the solution ranking while adding important contextual nuance to how human oversight should be designed.

A's account was the most technically detailed. She distinguished between contexts where output inconsistency has been substantially resolved through advances in retrieval-augmented generation (RAG) techniques, temperature control parameters, and confidence scoring mechanisms, and contexts where human validation remains indispensable regardless of model maturity. She observed that "now we have different types of RAG (Retrieval-Augmented Generation) which gives you the proper data grounding almost everything," but cautioned against using this technical maturity as a justification for removing human oversight universally. Her practical principle was risk-contextual: for low stakes, standardised outputs (a positive customer feedback response), automation without human review is acceptable; for high-stakes, emotionally sensitive, or brand-critical outputs (for example in healthcare context), human validation before delivery is non-negotiable: "make sure that you have a human in loop so that whatever it is responding, the human should read it before sending because it's a critical." This contextual calibration of human oversight is more sophisticated than the FDM item B1.S2 implies: rather than a binary presence or absence of human-in-the-loop, effective governance requires a risk-tiered design that allocates human review proportionately to output consequence.

D's experience provided a quantitative dimension to the reliability theme. Working with Microsoft Copilot Studio deployments, he described a planned threshold of 10 per cent failure rate across automated interactions: "If I am speaking about expectations of how much automated we will bring, we said 60% until 2030. If we speak about the recent rollout... we expect 10% of the threshold. So from 20,000 tickets... we expect that robotics and AI will fail in 2000 cases." This explicit modelling of expected failure rates, and the corresponding design of human fallback mechanisms, represents a mature approach to human-in-the-loop design that goes beyond treating human validation as a temporary patch on unreliable models. It positions human oversight as a permanent governance feature of responsible AI deployment, consistent with the FDM's strong consensus around B1.S2.

H's contribution to this theme was more indirect but analytically significant. He observed that the perceived versus actual benefits of GenAI are frequently misaligned, with the consequence that expectations around output reliability are set unrealistically high during sales and consulting engagements: "The perceived or promised benefits are sometimes overestimated... when it actually comes to seeing how much of a benefit it will have, it doesn't really follow through in a lot of cases." This expectation gap, which maps onto FDM item B6.3 (conceptual confusion about what GenAI is relative to other AI tools), means that clients may expect autonomous, consistent AI outputs but receive probabilistic, threshold-dependent results that require sustained human monitoring. Managing this expectation gap is therefore as much a commercial and communication challenge as it is a technical one.

Across the interviews, human oversight appeared less as a temporary limitation of GenAI and more as a risk-control design principle. R2 described healthcare and pharmacy use cases in which generated outputs require review because errors may create patient harm. H made a similar point in the context of agentic payments, where transaction sensitivity and national banking regulation require explicit validation before execution. D added a service desk perspective, explaining that automation can reduce human workload but still requires fallback thresholds where language recognition, security checks, or system uncertainty cause the AI process to fail.

This theme validates B1.3 and supports B1.S2 by showing that HITL (Human-in-the-Loop) should be calibrated by risk level, sector, and consequence severity rather than applied as a generic control.

### 7.4.4 Theme T4: The Governance Architecture Gap

This theme emerged inductively from the interviews and represents a finding that extends beyond the structured FDM taxonomy. Multiple participants, most prominently A and R2, identified a critical structural flaw in how organisations approach AI governance: they apply a single, uniform governance framework across fundamentally different categories of AI deployment, creating both excessive friction for low-risk use cases and insufficient scrutiny for high-risk ones.

A articulated this concern most directly, drawing on her experience at an energy industry where she was involved in the responsible AI and governance team: "A lot of good AI use cases used to die" because digital security assessment processes applied the same level of scrutiny to a low-risk internal productivity tool as to a high-risk client-facing agentic system. Her proposed solution was a risk-tiered governance architecture that distinguishes between at least three categories of AI deployment: traditional machine learning models running in self-contained organisational environments (minimal governance scrutiny), SaaS-based AI tools embedded in third-party platforms (moderate governance scrutiny focused on data exposure risks), and bespoke cloud-based agentic AI systems processing sensitive client data (full governance review). She observed: "You cannot assess all the three sorts of AI in one thing... assessing AI, different sorts of AI in one single lens are the problem, I guess."

This governance gap was also visible in R2's account of responsible AI review, where use cases were assessed for risk, data attributes, bias, fairness, accuracy, and deployment approval before scaling. H identified a related issue at the organisational boundary: governance may exist at enterprise level, but may not be sufficiently translated into division-level processes where data is processed at a more granular and sensitive level. These accounts support A's argument that governance must be calibrated by use case type, risk level, data sensitivity, and deployment context.

This finding represents a substantive refinement of FDM item B6.3 (conceptual confusion about what GenAI is relative to other AI tools, crisp = 0.406, ranked last among retained barriers). While the FDM framed this confusion primarily as an investment and stakeholder communication problem, the interview evidence reveals a deeper governance design consequence: organisations that cannot differentiate between AI types cannot calibrate their governance frameworks proportionately, resulting in both over-governance (slow approvals that kill viable use cases) and under-governance (insufficient scrutiny for genuinely risky deployments). The practical implication, an AI placement framework that categorises deployments before governance review, is a solution construct that does not appear explicitly in the FDM's retained solution hierarchy but was identified as critical by multiple participants, particularly those with client-side delivery experience.

This theme extends B6.3 by showing that governance ambiguity is not only a matter of understanding GenAI as a technology, but also of designing risk-tiered governance structures that are practical at the point of deployment.

## 7.5 Organisation Dimension Themes

The Organisation dimension produced three themes. Collectively, these themes address the most analytically complex finding in the FDM dataset: the discrepancy between expert holistic weight allocation (Organisation rated as the highest contributor to the execution gap at 38%) and item-level consensus (only four organisational barriers retained, the lowest retention rate of any dimension). The interview evidence provides a substantive explanation for this discrepancy and reveals that organisational barriers operate at a systemic, cultural, and structural level that item-level survey instruments are not well positioned to capture.

### 7.5.1 Theme O1: The Frozen Middle

While FDM items C2.1 (cultural resistance) and C2.4 (low AI readiness) captured the presence of organisational change barriers at the staff level, the interviews revealed a more precise and analytically significant location for this resistance: the operational middle management layer. This finding, introduced most clearly by S and corroborated by A and H, identifies a structural dynamic that item-level survey questions about "cultural resistance" are unlikely to surface.

S, drawing on his experience as an IT programme delivery manager, described a pattern that he termed "the frozen middle": senior leadership is enthusiastic about GenAI and willing to allocate budget for pilots and proofs of concept (POC), while frontline delivery staff are broadly adaptive to technology change. The friction resides in the intermediate management layer, the operational managers who own the billable hour model, who are measured on utilisation metrics, and who perceive GenAI-enabled automation as a threat to the headcount and revenue structures their performance is evaluated against. As he observed, the incentive system within IT service delivery does not naturally reward the disruption of the model that generates margin: managers who reduce headcount through automation may find themselves managing smaller teams and lower revenue bases, regardless of whether the value delivered to clients has increased.

A described a closely related pattern from the vendor side, observing that enterprise clients frequently exhibit what she characterised as “FOMO (Fear of Missing Out)-driven” GenAI adoption. In this pattern, senior leadership announces GenAI strategies, allocates budgets, and creates AI Centre of Excellence, but operational embedding stalls because the business divisions responsible for execution have not changed their incentive structures or measurement frameworks:

*I have seen like almost every time this is a very common trend... they will set up a small AI CoE (Centre of Excellence) and everything. They will hire people and all of a sudden they will realise that it is a cost to the organisation and they can't bear it and they will go in hibernation mode.*

This hibernation cycle, from enthusiastic launch to operational retreat, maps directly onto the FDM’s finding that the breadth of GenAI-related change exceeds current change management capacity (C2.3). It also provides a more precise structural account: the issue is not merely change capacity, but incentive misalignment between the strategic and operational layers of the organisation.

H offered a complementary perspective from his consulting work, noting that workshop facilitation and client engagement required careful navigation of the FTE (Full time equivalent)-reduction framing of GenAI: "We do not position it as something that is going to lead to effort reduction, rather productivity gain is how we try to show it." This reframing strategy, presenting GenAI as augmentation rather than replacement, reflects a pragmatic response to the frozen middle dynamic but does not resolve the underlying incentive misalignment. Executive-level conversations reintroduce the FTE-reduction framing as a key performance indicator, creating a communication contradiction that delivery managers are expected to navigate. He also cited a specific example from a Heavy Manufacturing Sector where employees with 20-25 year careers in SAP-based operations resisted the implementation of a centralised shared services model: "It does lead to a large transformation and process redesign, and then the operating model might also change in some cases... whether they easily adopt those new operating models" remains the defining organisational question.

D and R1 further refined this theme by showing that resistance also appears at the client and stakeholder boundary. D described customer-side financial resistance, particularly when clients interpret automation primarily through immediate savings rather than service quality or speed improvement. R1 located a related barrier in stakeholder trust, where decision-makers may support AI in principle but hesitate to

release data or accept automation because of legal, contractual, hallucination, or penalty concerns. Read together with S's more positive account of leadership support and user-driven use-case development, these findings suggest that resistance is not inevitable. It becomes more pronounced where incentive structures, trust conditions, and commercial expectations are misaligned.
This theme provides the most compelling qualitative explanation for the FDM's discrepancy between holistic weight allocation and item-level consensus. The frozen middle dynamic is systemic and culturally embedded rather than item-specific: it shapes the severity of multiple identified barriers without being reducible to any single survey item. Respondents who allocated high weight to the Organisation dimension in the FDM holistic judgement were likely sensing this systemic dynamic without being able to express it through the binary severity ratings of individual items.

This theme validates C2.1 and C2.4 while extending the FDM results by locating resistance not only in employee attitudes, but also in commercial incentives, client expectations, and trust conditions.

### 7.5.2 Theme O2: Strategic Ambiguity and the ROI Credibility Gap

This theme addresses one of the most frequently cited concerns across the interviews: the absence of credible, specific, measurable ROI expectations as a prerequisite for sustained GenAI investment beyond the pilot phase. The theme connects to FDM item B6.3 (conceptual confusion about GenAI relative value, crisp = 0.406) and to the open-ended FDM contributions identifying ROI ambiguity as a critical omitted barrier (Section 6.11), but it emerged as a far more central concern in the qualitative evidence than its FDM ranking suggested.

D's account was the most candid and detailed. He described a period in his organisation's history, approximately five to six years ago, in which automation investments were made without governance requirements for ROI articulation or outcome measurement: "We have burned ourselves by investing into robotizations, automations... the investment was there. The actual automation and robotization proceeded, but in the end, it had no impact, no ROI, anything... no monitoring, no anything." The consequence was financially and strategically costly: recurring expenditure on automation tooling produced no measurable FTE savings, cost reductions, or client experience improvements, eroding executive confidence in AI investment at precisely the moment when GenAI capabilities were beginning to mature. The corrective response, requiring a thorough return on investment strategy, analytics and reporting infrastructure, and FTE-impact modelling as prerequisites for investment approval, represents a governance maturation that the FDM's solution hierarchy identifies under C1.S2 (integrating GenAI into OKRs and strategic planning, crisp = 0.719) but that the interviews reveal as grounded in a specific history of investment failure.

A identified a structurally identical pattern from the client side, using the concept of "success criteria" as the critical differentiating factor between GenAI projects that successfully move to production and those that stall at MVP (Minimum Viable Product) stage: "Most of the time what happens, customer, they know that they will build an agent. However, we don't know once the agent is built what we are going to measure." Her successful production deployment with a global airline was distinguished by the investment of significant effort in the assessment phase to define micro-level KPIs before development began: "We tagged every single KPI at a very micro level so that we are covering all the aspects of that agent." The implication is that ROI credibility is not merely a retrospective reporting challenge but a prospective design requirement: if measurable success criteria are not established before development, no post-deployment data can demonstrate sufficient value to justify scaling investment.

R2 and S added a further dimension to this theme: the credibility gap between what GenAI can demonstrably deliver and what client organisations expect it to deliver, particularly in relation to the velocity of transformation. Both participants noted that clients frequently enter GenAI engagements with expectations shaped by analyst reports, vendor marketing, and media coverage, expectations that do not

account for the integration complexity, data governance requirements, and organisational change investment that production-scale deployment entails. Managing this expectation gap is a commercial and strategic challenge that IT service providers must address explicitly, both in the scoping of client engagements and in the design of value communication frameworks that distinguish between pilot-phase and production-scale ROI profiles.

The ROI credibility gap was also evident in participants' discussion of infrastructure and client demand. S explained that his organisation avoided scaling GPU (Graphics Processing Unit) capacity until there was clear client demand and budget justification, indicating that technical scalability decisions were directly tied to commercial confidence. A described successful production movement as dependent on business-aligned use-case selection, measurable KPIs, and governance work completed during the assessment phase. D added that customers may evaluate automation through immediate savings expectations, narrowing the perceived value of GenAI to cost reduction. Together, these accounts show that ROI credibility functions as a coordination mechanism between technical readiness, governance readiness, and client-side investment commitment.

This theme validates B6.3 and refines C1.S2 by showing that value demonstration must be linked to measurable outcomes, governance readiness, and client-side investment commitment.

### 7.5.3 Theme O3: The AI Council, CoE, and Upskilling Triad

The FDM identified an AI Centre of Excellence (C3.S2, crisp = 0.719) and strategic integration of GenAI into OKRs (C1.S2, crisp = 0.719) as joint tenth-ranked solutions. The interviews significantly enrich this finding by revealing a more differentiated and sequenced governance infrastructure than the FDM's solution items individually suggest. A, in particular, proposed a three-layer governance architecture that represents a practical elaboration of the CoE concept into an actionable institutional design.

A argued that an AI Centre of Excellence, as conventionally understood, focuses primarily on delivery capability: it builds agents, develops standards, and scales proven practices across business units. This function is necessary but insufficient. Before a CoE can operate effectively, an AI Council must be established at the enterprise level with a different and more foundational mandate: "Before AICOE (Artificial Intelligence Centre of Excellence), I should set up an AI Council, which should have the responsible AI team, AI ethics team, then the governance team, AI architecture approval team." The AI Council serves as the governance foundation, determining which use cases are approved, under what conditions, and with what accountability frameworks, while the CoE serves as the delivery engine that executes within those boundaries. Without the AI Council's governance function, CoE-delivered agents enter a governance vacuum in which security, ethics, and accountability requirements are either addressed ad hoc (creating delays and inconsistencies) or ignored (creating risk).

The interviews suggest that the AI Council and AI CoE should not be treated as interchangeable structures. A distinguished between the AI Council as the body responsible for responsible AI, ethics, legal, compliance, and enterprise assurance, and the AI CoE as the delivery-oriented unit that develops or enables AI solutions. R2's account of responsible AI review supports this distinction, showing how risk assessment and approval mechanisms must precede deployment. S added the capability-building perspective, describing continuous training and user contribution to new use cases as important conditions for adoption. The theme therefore points to a sequencing logic: governance authority and risk ownership should be established before the CoE is expected to scale delivery.

H echoed the layering logic from a consulting perspective, describing the "AI pod" concept as a practical organisational design for GenAI delivery: an interdisciplinary team combining a change and adoption lead, data engineers, AI engineers, application developers, and business analysts, with the operating model explicitly designed to address both technical delivery and organisational change

simultaneously. He noted that the capacity requirements of this pod model could themselves constitute a barrier if organisations attempted to assemble it too quickly or without phased investment: "If the capacity requirements are very high, it may not be in their best interest to prioritise those initiatives first."

On upskilling, both A and D noted that the skill gap challenge (FDM item C3.2, crisp = 0.497) manifested differently across organisational contexts in ways that the item-level FDM score obscures. In D's service desk environment, the primary workforce challenge was not training existing staff in GenAI but managing the displacement anxiety created by automation's reduction of transactional labour: "Our teams and our people are sensing the development of the time... a lot of people are actually even attempting to get scared that the AI and automations and technologies are going to replace them." Effective upskilling in this context is as much about change communication and psychological safety as it is about technical AI literacy. A, by contrast, described a context in which prompt engineering capability and agent design skills were in active short supply, requiring structured enterprise framework adoption (such as LangChain or AWS Amazon Web Services Agent Core) to reduce the dependency on rare, deep AI engineering expertise for standard use case deployment.

This theme supports C3.S2 and C1.S3 while validating C3.2 by showing that capability building requires both governance authority and practical upskilling infrastructure

## 7.6 Environment Dimension Themes

The Environment dimension produced two themes that directly address the most analytically striking finding in the FDM dataset: the Environmental paradox of zero barrier consensus combined with the highest mean retained solution effectiveness (0.710). The interview evidence provides a substantive explanation for this paradox that quantitative analysis alone cannot supply.

### 7.6.1 Theme E1: Client-Side Readiness as the External Scaling Ceiling

A theme that emerged consistently, and that was not explicitly represented in the FDM's structured item taxonomy, was the role of client-side GenAI readiness as an external constraint on scaling that IT service providers cannot unilaterally resolve. This theme extends the FDM's Environmental dimension beyond regulatory and vendor ecosystem pressures to encompass the client organisation itself as a source of external bottleneck.

D was the most direct in articulating this dynamic. He described a pattern in which some clients actively welcome GenAI-enabled automation ("just do best on our environment; just make us more modern") while others, particularly public-sector clients operating under legacy contractual and political constraints, remain " scared of cloud-based solutions" and resistant to the data access and API permissions that GenAI integration requires. The key insight is that the IT service provider's scaling ambitions are bound by the client's organisational readiness: "Sometimes, customers are simply saying hey, where is the AI? Where is the automation? How does it impact our volumes?" while others demand "a thorough description of how things are working." Both postures create challenges, one through excessive urgency that overrides governance, the other through excessive caution that delays technical enablement.

D’s account was reinforced by other participants who located the scaling ceiling on the client side. R1 emphasised that data access depends on contract-specific permissions and on how different client and delivery stakeholders interpret those contractual clauses. H added that enterprise-level AI capability does not necessarily translate into division-level governance readiness, especially where sensitive data is processed locally. S further linked client readiness to demand and budget, noting that infrastructure expansion is difficult to justify without clear customer commitment. These accounts show that client-side readiness is not only a technical maturity issue; it includes contract interpretation, budget availability, governance capability, and operational access.

H described an analogous pattern from healthcare consulting, where a nationwide healthcare network's GenAI ambitions at enterprise level were constrained by the absence of division-level governance infrastructure to support safe deployment at the point of care:

*At an enterprise level, they have AI initiatives... but when you're trying to socialise the idea of a use case and get alignment on which use cases... there's not alignment on onboarding these platforms or the skills because... at a division level, they may not have the right means to safely deploy a use case or they cannot get the governance for it.*

The multi-level alignment problem indicates that enterprise-level AI governance frameworks do not automatically cascade to operational divisions with different risk profiles. This is therefore a client-side readiness challenge that IT service providers must diagnose and address before they can responsibly deploy GenAI at scale.

R2 extended this theme to the presales context, describing how client readiness assessments, evaluating data maturity, platform readiness, and capability infrastructure, are now a foundational step in his engagement methodology. The implication is that client-side readiness has become an explicit environmental variable in GenAI scaling decisions: IT service providers that ignore it risk committing to production deployment in environments that are architecturally and organisationally unprepared, creating precisely the pilot stagnation that this research investigates.

This theme connects directly to the FDM's highest-ranked solutions in the Environmental dimension, client co-design of governance frameworks (D4.S1, crisp = 0.734) and AI-enabled service assurance (D4.S2, crisp = 0.730), by providing the experiential mechanism behind their effectiveness: when IT service providers actively involve clients in designing the governance frameworks that will govern GenAI deployment, they simultaneously assess and develop client-side readiness rather than treating it as a fixed external constraint. The co-design process is therefore both a governance mechanism and a readiness-building intervention.

This theme explains the absence of retained Environment barriers in the FDM results while supporting D4.S1 and D4.S2, because client co-design and service assurance address context-specific readiness constraints.

### 7.6.2 Theme E2: Regulatory Contextuality and the Universal Solution Logic

The FDM's Environmental paradox, zero barrier consensus, highest solution effectiveness, is directly explained by this theme. Participants consistently described regulatory and compliance pressures as highly contextual, varying by jurisdiction, sector, and organisational type, in a manner that prevents any single regulatory barrier from achieving universal consensus across a geographically and sectorally diverse expert panel. At the same time, they endorsed environmental solution mechanisms as broadly applicable regardless of which specific regulatory regime they operated under.

D's experience in the Nordic public sector illustrates the specific nature of jurisdiction of regulatory barriers most clearly. He described the DORA (Digital Operational Resilience Act) compliance requirement as highly relevant to larger public-sector clients in Sweden but effectively invisible to smaller commercial clients: "From my side, I know that these kinds of specifics are coming in place, but so far this is covered directly by Microsoft who does this for us. And either it's customers who don't demand on this... or it's actually a showstopper for those bigger customers and public customers." The same regulatory development produces fundamentally different barrier experiences across different client segments within the same geographic market, precisely the kind of within-market variability that drives high average distance values in the FDM's consensus calculation, even when the regulatory environment itself is objectively uniform.

H, drawing on experiences spanning healthcare in the United States, financial services in Asia, and technology consulting across multiple markets, observed that regulatory barriers are not merely contextual but multiply so: they vary by industry (healthcare, banking, manufacturing), by geography (Central Bank guidelines in the Asia, EU,...), and by the specific type of GenAI use case being deployed (agentic payment processing carries very different regulatory requirements than internal document summarisation). He noted that the central banks in Asia mandates around agentic payment processing, requiring specific validation protocols before AI-initiated transactions can be executed, are highly relevant in the Asian fintech context but entirely absent in markets with different payment governance frameworks: "It depends on the policies of a country, like regulations of the central bank, those things like, for example, in Asia, they have certain regulations where it will be very difficult to implement." This multi-dimensional contextuality is precisely why Environmental barrier items produced no FDM consensus: the barriers genuinely manifest differently across the panel's jurisdictional and sectoral diversity.

A contributed a forward-looking dimension to this theme, observing that the governance process within individual enterprise organisations is currently asymmetric: either excessively stringent governance that kills viable use cases, or negligible governance that exposes the organisation to unmanaged risk. The solution she endorsed, a risk-tiered, AI-type-sensitive governance architecture, is effective across all regulatory contexts because it addresses the underlying logic of risk proportionality rather than any specific regulatory requirement. This universality of solution logic at the same time as regulatory barriers remain contextually variable provides the most precise qualitative explanation for the FDM Environmental paradox identified in Chapter 6.

R2's healthcare and pharmacy examples further support this interpretation. In these contexts, the regulatory issue was not a generic environmental barrier, but the consequence severity of specific GenAI outputs, particularly where incorrect recommendations could affect patient safety. This reinforces the cross-case pattern identified by D, H, and A: environmental barriers are experienced through specific combinations of jurisdiction, sector, client type, data sensitivity, and use-case risk. As a result, they are difficult to validate as universal barrier items, even though environmental-facing solutions such as compliance-by-design, legal review, client co-design, and risk-tiered governance remain broadly applicable.
This finding also reinforces the clarified scope of the thesis. The study examines GenAI scaling in globally operating, multi-market IT services contexts, while treating geography and regulation as contextual variables rather than fixed comparative categories. The theme therefore explains the Environmental paradox in the FDM results: environmental barrier experience is context-specific, whereas environmental-facing solutions remain broadly valuable across regulated and client-facing IT services contexts.

## 7.7 Cross-Dimensional Interaction Themes

The conceptual model developed in Chapter 4 posited three cross-dimensional interaction effects: the compliance gap (Environment x Organisation x Technology misalignment), the change capacity moderation effect (Organisation moderating Technology-to-Outcome), and resource-barrier amplification (Organisation constraints amplifying Technology and Environment barriers). The interview evidence generated two cross-dimensional themes that operationalise the first two of these effects in practitioner terms.

### 7.7.1 Theme X1: Security Gatekeeping as a Cross-Dimensional Amplifier

Across multiple interviews, a specific cross-dimensional dynamic emerged that the conceptual model's compliance gap construct anticipates but that the FDM's item-level structure cannot fully capture: security and compliance approval processes, which belong to the Environment dimension, systematically amplify

the severity of Technology dimension integration barriers by extending timelines, constraining architecture choices, and creating approval uncertainty that discourages investment in technically viable use cases.

Security gatekeeping appeared as a cross-dimensional amplifier because it converted technical integration work into organisational and environmental approval work. D indicated that legacy technical gaps were often manageable, whereas security approval created the more significant obstacle. H's healthcare example shows why this occurs: the same data may be lower risk when aggregated at enterprise level but higher risk when processed at patient or division level. A's governance account adds that undifferentiated security assessment can delay or terminate viable AI use cases if low-, medium-, and high-risk deployments are reviewed through the same process.

D provided the most operationally specific account. He described integration delays not primarily as a consequence of technical complexity (which he characterised as manageable) but as a consequence of security approval processes at the client side: "The owners, admins of the APIs are refusing without a proper explanation to provide the APIs and set up the integrations rather than there would be a technical, technical issue. So, the integrations are taking long." What appears in the FDM data as a Technology barrier (B3.4: weak API management, crisp = 0.562) is revealed in the qualitative evidence to be at least partially driven by an environmental compliance mechanism, customer security teams' refusal to grant API access, that operates externally to both the technology and the organisation. The technology barrier and the environmental compliance constraint are not additive but interactive: the compliance constraint transforms a manageable technical integration into an extended, governance-dependent process that can delay deployments by months.

A's account of the Architecture Review Board (ARB) and digital security team approval processes at an energy company illustrates the same dynamic from the client side: "You will be in the loop for a good amount of time because you will have a good amount of Q&A sessions with the ARB... separately with the digital security team where you will be explaining this is what we need. But then they will say no, and then you have to come up with a workaround or alternative architecture." The consequence is a technically suboptimal but governance-acceptable solution, the batch-processing workaround described in Theme T2, that constrains the potential value of the GenAI deployment. In this interaction, the Environment dimension (compliance and security governance) directly shapes the Technology dimension outcome (architecture choice and integration capability).

This cross-dimensional amplification effect has important implications for how practitioners and researchers should conceptualise the Integration Complexity and Technical Debt barrier cluster (B3). The 100 per cent consensus coverage of B3 in the FDM reflects genuine agreement that integration and technical debt create barriers. But qualitative evidence reveals that a significant portion of the integration bottleneck is not technical in origin, it is a governance and security approval problem wearing a technical mask. Solutions that address only the technical dimension (modular API-first architecture, phased integration roadmaps) will be necessary but insufficient unless they are complemented by environmental solutions that address the approval and access-control dimensions of integration delay.

This theme operationalises the compliance gap in the conceptual model by showing how environmental security requirements amplify technology integration barriers through organisational approval processes.

### 7.7.2 Theme X2: Iterative Scaling vs. Pilot Purgatory

FDM's fourth-ranked solution, iterative pilot-to-scale frameworks (C2.S3, crisp = 0.747), reflects strong expert consensus that incremental GenAI integration is more effective than big-bang deployment attempts. The interviews validate this consensus but introduce a critical qualification: iteration without defined governance decision gates does not lead to scaling. It leads to "pilot purgatory", a state in

which organisations accumulate pilots indefinitely without transitioning any of them to governed production deployment.

D described the pilot purgatory dynamic from direct experience. The period in which his organisation invested in automation without ROI governance requirements resulted in a proliferation of automation “modules” that were individually functional but collectively unaccountable:

*A particular service desk team said, oh okay, great, now we have saved 200 tickets. So, some of our people have yet enough time for lunches or stuff like this. And this wasn't a good thing because the governance process behind that was completely missing.*

The projects therefore existed in a perpetual pilot-equivalent state: technically operational, but lacking the business outcome measurement, stakeholder accountability, and investment justification frameworks required for leadership to authorise full-scale production deployment.
A characterised the same dynamic from the client side as the FOMO-driven deployment cycle: enterprises announce GenAI strategies and create CoEs, generate pilots, then retreat into "hibernation mode" when ROI does not materialise spontaneously. The transition from pilot to production is not automatic; it requires an explicit governance decision gate, a structured moment at which pilot results are evaluated against pre-defined success criteria and a governance-sanctioned decision is made to scale, iterate, or terminate. Without this gate, the iterative framework that the FDM endorses as a top solution collapse into unstructured experimentation.

D’s account of the current investment governance model at his organisation illustrates what the governance decision gate looks like in practice:

*We needed to bring the return of investment strategy... there were analyses done, it was realised what volumes we are expecting to have affected, we had selected a time frame, we had requested a thorough analytics and reporting system to be established in a place.*

The ROI governance requirement, established as a prerequisite for investment approval, functions as a forcing mechanism that compels pilot teams to articulate testable business value hypotheses before scaling resources are committed. This practical instantiation of C1.S2, integrating GenAI into strategic OKRs, and C2.S3, iterative pilot-to-scale frameworks, illustrates that the two solutions are more effective in combination than either is individually.
This practical instantiation of C1.S2 (integrating GenAI into strategic OKRs) and C2.S3 (iterative pilot-to-scale frameworks) illustrates that the two solutions are more effective in combination than either is individually.

D's separate warning about over-customisation also bears directly on Theme X2. He described a pattern in which the desire for highly customised AI solutions, each tailored to the specific requirements of individual client environments, prevents the development of standardised, replicable deployment patterns that can be scaled efficiently across a portfolio. The most scalable path, in his view, is the development of a small number of standardised "MVPs" (minimum viable products) that can be adapted incrementally to client contexts rather than rebuilt from scratch for each client. This finding qualifies the FDM's support for "internal tooling" and customisation: while bespoke solutions may maximise fit for individual deployments, they create the "pilot purgatory" dynamic at portfolio scale by preventing the accumulation of reusable deployment knowledge.

The interviews suggest that iterative scaling is effective only when each stage has a decision logic. A described successful movement to production as beginning with structured assessment, business-aligned

use-case selection, KPI definition, and governance preparation before MVP sign-off. S described a similar staged logic in which use cases were first tested internally, stabilised in departmental production, and then considered for customer-facing deployment once demand and infrastructure requirements became clearer. R2's account of several solutions already operating in real-time pharmacy systems, while others remained under development, further supports the view that scaling is not a single transition but a staged maturity process.

This theme validates C2.S3 while refining it by showing that iterative pilot-to-scale frameworks require explicit governance decision gates, measurable value criteria, and client demand validation.

## 7.8 Mixed-Methods Triangulation

The integration of qualitative thematic analysis findings with the FDM-derived barrier and solution hierarchy generates a set of triangulation outcomes that together constitute the empirical synthesis of this thesis. Table 7.2 presents a compact triangulation summary showing how the qualitative themes validate, refine, or extend the FDM findings. The integration of qualitative thematic analysis findings with the FDM-derived barrier and solution hierarchy generates the empirical synthesis of this thesis. The qualitative findings validate the strongest FDM results, particularly B2.2, B3.1–B3.4, B1.3, and B1.S2, while also extending the FDM hierarchy through mechanisms that were less visible in the structured survey instrument, including the frozen middle, client-side readiness, risk-tiered governance, and security gatekeeping. A full theme-by-theme triangulation matrix is provided in Appendix F, Table F.3.

Table 7.2: Mixed-Methods Triangulation Summary

| Qualitative theme | Main FDM connection | Triangulation outcome |
|---|---|---|
| **T1: Data Foundation Paradox** | B2.2 fragmented data governance | Validates and refines top-ranked barrier |
| **T2: Integration Bottleneck** | B3.1–B3.4 integration complexity and technical debt | Strongly validates 100% consensus barrier construct |
| **T3: Output Reliability and Human Oversight** | B1.3 and B1.S2 | Validates human-in-the-loop as high-priority solution |
| **T4: Governance Architecture Gap** | B6.3 and governance solution items | Extends FDM by identifying risk-tiered governance |
| **O1: Frozen Middle** | C2.1, C2.3, C2.4 | Extends organisational barriers into incentive misalignment |
| **O2: ROI Credibility Gap** | B6.3 and open-ended FDM responses | Refines value-demonstration barrier |
| **O3: AI Council, CoE, and Upskilling Triad** | C1.S2 and C3.S2 | Refines organisational solution architecture |
| **E1: Client-Side Readiness** | D4.S1 and D4.S2 | Extends environmental dimension |

| | | |
|---|---|---|
| **E2: Regulatory Contextuality** | Environment barrier non-consensus | Explains environmental paradox |
| **X1: Security Gatekeeping** | B3 and environmental compliance | Shows cross-dimensional barrier amplification |
| **X2: Iterative Scaling vs. Pilot Purgatory** | C2.S3 | Refines iterative scaling solution |

Table 7.2 demonstrates that the qualitative phase did not merely repeat the FDM results. Instead, it validated the strongest quantitative findings while also extending them through mechanisms that the structured survey instrument could not fully capture, including risk-tiered governance, client-side readiness, the frozen middle, and the commercial incentive logic of the billable-hour model.

## 7.9 Key Emergent Findings Not Captured in the FDM Taxonomy

Beyond validating and refining the FDM hierarchy, the thematic analysis generated three analytically significant findings that did not appear in the structured FDM item taxonomy and represent original qualitative contributions to the study's empirical conclusions.

First, the *risk-tiered governance architecture,* the principle that AI deployments must be categorised by type and risk profile before governance frameworks are applied, rather than subjected to uniform review, emerged as a critical solution construct not represented in the FDM's retained hierarchy. A's three-category framework (self-contained ML (Machine Learning)) models, SaaS-embedded AI, and bespoke agentic AI systems) provides an actionable design principle for organisations seeking to reduce governance friction for low-risk deployments while maintaining rigour for high-risk ones. This finding extends the FDM's construct B6.3 from a stakeholder communication problem into a governance design problem with direct implications for the AI Council and AI CoE structures identified in Theme O3.

Second, the *client-side readiness bottleneck* represents a distinct environmental barrier category that the FDM's structured instrument did not explicitly address. The FDM's Customer and Stakeholder Pressure construct (D4) was framed primarily around client demands and expectations, but the qualitative evidence reveals that client readiness, specifically, the client's data maturity, platform infrastructure, and division-level governance capability, functions as an external scaling ceiling that is independent of the IT service provider's own capabilities. This finding suggests that future research iterations should include an explicit Client Maturity construct within the Environmental dimension.

Third, the *incentive misalignment* dynamic underlying Theme O1 (the frozen middle) represents the most significant structural explanation for the discrepancy between the FDM's holistic Organisation weight allocation and its item-level consensus results. The billable-hour model of IT service delivery creates a structural disincentive for operational managers to embrace GenAI-enabled automation, because the model rewards headcount utilisation over value efficiency. This finding suggests that bridging the GenAI execution gap in IT services requires not only the technical and governance interventions identified in the FDM solution hierarchy but also a commercial model evolution, from headcount-based to outcome-based or value-based pricing, that realigns operational incentives with the efficiency gains that GenAI enables. This commercial model dimension is entirely absent from the FDM taxonomy and represents a genuinely novel contribution of the qualitative phase.

## 7.10 Summary of Thematic Findings and Connection to Research Questions

This chapter has presented eleven themes derived from the thematic analysis of six practitioner interviews, organised within the TOE framework and triangulated against the FDM findings from Chapter 6. The evidence addresses both research questions directly.

In response to *RQ1*, what barriers prevent IT service organisations from scaling GenAI beyond pilot stages? The thematic analysis confirms and elaborates upon the FDM's validated barrier hierarchy while introducing three significant qualitative additions. Fragmented data governance (Theme T1) operates as a structural architectural constraint in multi-client environments that goes beyond data quality management to encompass the entire deployment readiness prerequisite. Integration complexity (Theme T2) is shown to be partially driven by environmental compliance approval processes (Theme X1) rather than purely by technical debt. Cultural resistance (Theme O1) is located most precisely in the operational middle management layer, where incentive misalignment with the billable-hour model prevents the organisational change necessary to embed GenAI in sustained operations. ROI ambiguity (Theme O2) functions as both a pilot-stalling mechanism and an investment credibility problem that prevents executive commitment to scaling. Client-side readiness (Theme E1) functions as an external scaling ceiling that IT service providers cannot unilaterally overcome.

In response to *RQ2*, what solutions are most likely to bridge the execution gap? the thematic analysis validates the FDM's top-ranked solutions while adding important design details and sequencing logic. Human-in-the-loop validation (Theme T3) is most effective when designed proportionately to output risk rather than applied universally. An AI Council providing governance foundations must precede the AI CoE's delivery functions (Theme O3). Iterative pilot-to-scale frameworks require explicit governance decision gates to prevent collapse into pilot purgatory (Theme X2). Client co-design (Theme E1) is effective both as a governance mechanism and as a client readiness-building intervention. Three solution mechanisms not represented in the FDM hierarchy emerge from the qualitative data: risk-tiered AI placement frameworks (Theme T4), commercial model evolution from headcount-based to value-based pricing (Theme O1), and multi-level alignment structures bridging enterprise AI governance to operational division delivery (Theme E1).

Together, the quantitative FDM findings and the qualitative thematic analysis produce a coherent and mutually enriching picture of the GenAI execution gap in IT services: a multi-dimensional, socio-technical, and ecosystem-level challenge whose resolution requires simultaneous action across technological controls, organisational capability and incentive structures, and external relationship and governance management, rather than any single-dimensional technical or organisational fix. These findings are synthesized with broader literature and translated into theoretical and practical conclusions in Chapters 8.

Overall, the thematic analysis strengthens the FDM findings by demonstrating that the GenAI execution gap is reproduced across multiple organisational and client contexts, but not always through identical mechanisms. The cross-case evidence shows that Technology barriers are the most operationally visible, Organisation barriers are the most structurally embedded, and Environment barriers are the most context-dependent. This explains why the FDM produced strong item-level consensus for Technology, higher holistic weight for Organisation, and limited universal consensus for Environment barriers. The qualitative phase therefore does not merely confirm the FDM hierarchy; it explains the mechanisms through which the hierarchy operates in practice.

# Chapter 8: Discussion

## 8.1 Introduction and Chapter Purpose

This chapter synthesises the empirical, theoretical, and practical contributions of the thesis, provides direct answers to the two research questions, and draws the study to an academically and analytically complete conclusion. The chapter is structured as follows. Section 8.2 answers the research questions directly, drawing on the integrated findings from the FDM and thematic analysis phases. Section 8.3 discusses the proposition outcomes, evaluating the extent to which each research proposition was supported, partially supported, refined, or challenged by the empirical evidence. Section 8.4 presents the theoretical contributions of the study. Section 8.5 translates the integrated findings into practical implications for IT service organisation and GenAI practitioners, including a staged implementation pathway for sequencing interventions across foundation and readiness, governance and assurance, scaling and value realisation, and commercial incentive alignment. Section 8.6 reflects on the methodological contributions and limitations of the study. Section 8.7 identifies priority directions for future research. Section 8.8 provides a brief concluding statement.

## 8.2 Answering the Research Questions

This thesis was guided by two interconnected research questions. Their answers are presented below, synthesising evidence from all four phases of the study.

### 8.2.1 RQ1: What barriers prevent IT service organisations from scaling GenAI beyond pilot stages?

The study produces a validated, expert-prioritised hierarchy of eleven barriers that collectively constitute the structural anatomy of the GenAI execution gap in IT services. These barriers are not randomly distributed; they cluster into three analytically coherent groups that reveal a layered picture of execution constraint.

The first and most universally experienced group comprises architectural and integration constraints at the Technology dimension. Fragmented data governance across client environments (B2.2) ranks as the single most critical barrier across the dataset, reflecting a structural condition of multi-client IT service delivery in which each client engagement constitutes a separate data universe with distinct access controls, sovereignty requirements, and classification obligations. The integration complexity and technical debt cluster (B3.1 to B3.4) was the only construct to achieve 100 per cent consensus coverage, making it the strongest consensus signal in the dataset, confirming it as the strongest consensus signal, confirming that legacy architecture incompatibility, technical debt, API management limitations, and multi-client tenant isolation are not context-specific challenges but sector-wide structural constraints. Output inconsistency (B1.3) adds a reliability dimension: GenAI outputs in client-facing service contexts cannot be treated as deterministic, and the validation burden that results from probabilistic outputs increases operational cost and constrains responsible deployment scope.

The second group comprises organisational capability and readiness constraints: cultural resistance (C2.1), inadequate change capacity (C2.3), low AI staff readiness (C2.4), and deficiencies in prompt engineering and output evaluation skills (C3.2). Although these items achieved lower individual crisp scores than the Technology cluster, expert holistic weight allocation placed the Organisation dimension as the leading contributor to the execution gap at 38.0 per cent of total weight. Thematic analysis resolved this apparent tension: organisational barriers operate at a systemic and culturally embedded level that item-level survey instruments capture only partially. The frozen middle dynamic, in which operational middle management layers resist GenAI-enabled automation because their performance incentives are aligned to headcount utilisation rather than value efficiency, represents the most structurally significant organisational barrier identified by the thematic analysis. It is not captured by any single FDM item

precisely because it is not an item-level phenomenon; it is a systemic condition embedded in the commercial architecture of IT service delivery. The ROI credibility gap, whereby organisations lack credible, pre-specified success metrics for scaling investment decisions, operates as a closely related stalling mechanism that prevents sustained executive commitment to production-scale deployment even where pilot results appear promising.

The third barrier group is smaller in scale but analytically significant. B6.3 (conceptual confusion about what GenAI is relative to other AI and automation tools, crisp = 0.406) is the only retained item in the moderate crisp band. Its inclusion confirms that definitional ambiguity imposes a real, if subtle, execution cost, not as a primary operational barrier, but as a governance design problem with downstream consequences for investment sequencing, stakeholder alignment, and the calibration of oversight frameworks. Organisations that cannot differentiate between AI deployment types cannot calibrate governance proportionately, producing both over-governance of low-risk tools and under-governance of genuinely high-risk deployments.

The complete absence of retained Environmental barriers warrants careful interpretation. This finding does not indicate that external forces are irrelevant to the execution gap; rather, it reflects the jurisdictional, sectoral, and deployment-context variability of regulatory, legal, and competitive pressures across the geographically and sectorally diverse expert panel. Environmental barriers are real and consequential, but they manifest differently depending on whether an organisation operates in Nordic public-sector markets, US commercial cloud contexts, or Asian fintech environments. High average distance values in the Environmental dimension should therefore be understood as a finding about the context-dependence of external constraints, not as evidence of their absence.

### 8.2.2 RQ2: What Solutions Are Used to Overcome These Barriers and Bridge the GenAI Execution Gap?

The FDM phase validated forty-four consensus-supported solutions across all three TOE dimensions, with the solution hierarchy producing several analytically significant findings that inform both practical strategy and theoretical interpretation.

The most analytically unexpected result is that the highest-ranked solutions are concentrated in the Environment dimension rather than the Technology dimension. The top-ranked solution across all 122 items is multi-vendor ecosystem strategy (D5.S2, crisp = 0.772), followed by client co-design of governance frameworks (D4.S1, crisp = 0.734) and AI-enabled service assurance (D4.S2, crisp = 0.730). This cluster of ecosystem and relationship management solutions establishes an outward-facing, stakeholder-centric solution logic: bridging the execution gap in IT services requires managing client relationships, vendor dependencies, and external accountability structures as much as, and in aggregate more immediately than, resolving internal technical problems. This finding is consistent with the structural nature of IT services delivery, in which GenAI cannot be deployed unilaterally but must be negotiated, governed, and assured within client-provider relationships that were not originally designed to accommodate AI-generated outputs.

At the operational level, human-in-the-loop validation (B1.S2, crisp = 0.759) and security-by-design (B4.S1, crisp = 0.753) establish the foundational operational philosophy for responsible GenAI deployment in IT services: probabilistic AI outputs require proportionate human governance, and security must be embedded from inception rather than retrofitted. Thematic analysis extends this finding by contributing to the risk-tiered design principle: the intensity of human oversight should be calibrated to the risk profile of the deployment context rather than applied uniformly, with high-stakes client-facing or diagnostically sensitive outputs requiring mandatory human validation and low-stakes internal productivity outputs permitting automated release without individual review.

The organisational solution cluster, iterative pilot-to-scale frameworks (C2.S3, crisp = 0.747), integration of GenAI into strategic planning and OKRs (C1.S2, crisp = 0.719), and an AI Centre of Excellence (C3.S2, crisp = 0.719), addresses the governance and capability infrastructure required to convert technical enablement into sustained organisational performance. Thematic analysis qualifies the iterative scaling solution with a critical design requirement: governance decision gates must be embedded within iterative frameworks to prevent the accumulation of pilots in a state of perpetual experimentation without production deployment commitment. The AI Centre of Excellence concept is further elaborated into a two-layer institutional design: an AI Council providing governance foundations and risk-classification protocols must precede and supervise the CoE's delivery function to prevent CoE-generated deployments from operating in a governance vacuum.

The overall solution hierarchy is cross-dimensional in character: closing the execution gap requires simultaneous and coordinated action across technological controls, organisational capability and incentive structures, and ecosystem management, not sequential or siloed intervention along a single TOE dimension. This cross-dimensional solution imperative is the defining strategic implication of the thesis for management practice.

## 8.3 Research Proposition Outcomes

This section evaluates the ten research propositions developed in Chapter 4 against the integrated empirical evidence from the FDM and thematic analysis phases. Each proposition is assessed as supported, partially supported, refined, or challenged.

*P1 (Technological Complexity as a Barrier): SUPPORTED.* The FDM retained seven Technology dimension barriers, the largest single-dimension retention count, and the Technology dimension produced the highest mean retained barrier crisp value (0.575). The thematic analysis further elaborated the architectural mechanisms behind the two dominant barrier clusters (fragmented data governance and integration complexity). The evidence fully supports the proposition that technological complexity constitutes a primary barrier to GenAI execution in IT services, with the qualification that the most critical barriers are specifically data governance architecture deficiencies and legacy integration constraints rather than model capability limitations per se.

*P2 (Relative Advantage Clarity as an Adoption Enabler): PARTIALLY SUPPORTED.* FDM item B6.3 was retained but ranked last among consensus-supported barriers in the moderate crisp band (0.406), suggesting that relative advantage confusion is a real but comparatively lower-priority barrier relative to architectural and integration constraints. However, thematic analysis identified the ROI credibility gap as a substantially more central concern: multiple participants described the absence of pre-specified success criteria as a decisive factor preventing pilot-to-production transition. The proposition is directionally correct but requires refinement. The barrier is not merely communicative (the inability to articulate advantage) but structural (the absence of measurable, pre-committed success thresholds that compel investment governance decisions at the scaling juncture).

*P3 (Leadership Commitment as an Organisational Enabler): PARTIALLY SUPPORTED. The* FDM produced no retained barrier items in the Leadership Commitment construct (C1), and no C1 barrier items achieved individual consensus. This does not mean leadership commitment is unimportant; rather, it appears to be a threshold condition whose presence is insufficient on its own to guarantee production-scale deployment. Thematic analysis specifically identifies the frozen middle dynamic as the mechanism through which leadership enthusiasm is absorbed without producing operational embedding: senior leadership commits, operational management resists, and the execution gap persists. The proposition therefore requires a supplementary condition; leadership commitment must be accompanied by commercial incentive realignment at the operational management layer.

*P4 (Skills Gaps as an Organisational Barrier): SUPPORTED.* FDM item C3.2 (prompt engineering and output evaluation skills deficiency) was retained at a moderate-significant crisp level (0.497), and thematic analysis confirmed that skills challenges manifest across a spectrum, from technical AI capability deficits in organisations building bespoke agents to displacement anxiety and psychological safety concerns in service desk environments facing transactional automation. The Hughes et al. (2026) empirical coefficient ($\beta = 0.390$, $p < 0.001$) provides robust independent corroboration.

*P5 (Governance Structures as Execution Enablers): REFINED.* The FDM result for the Governance Structures solution construct (C4S) is analytically distinctive: all four solution items achieved crisp scores in the effective range (0.625–0.678), yet none achieved consensus (all $\bar{d}$ values exceeded the 0.20 threshold). This indicates not that governance solutions are ineffective, but that expert opinion is genuinely divided on which specific governance mechanism is most effective in practice. Thematic analysis explains this divergence: governance frameworks are highly context-dependent, varying by organisational culture, regulatory regime, client industry, and the specific risk profile of AI deployment types. The proposition is refined from a general endorsement of governance structures to a more precise finding: governance structures are necessary but insufficient unless designed as risk-tiered, deployment-type-sensitive frameworks rather than uniform policy instruments applied indiscriminately across all AI categories.

*P6 (Change Capacity as a Moderating Factor): SUPPORTED.* The tension between FDM item-level Technology dimension dominance and holistic Organisation weight allocation leadership (38.0 per cent) is itself the primary evidence for this moderation effect: experts agree that Technology produces the most specific and visible barriers, yet attribute the greatest systemic contribution to the Organisation dimension, consistent with the interpretation that organisational change capacity shapes the severity of technological barriers rather than acting independently. The frozen middle theme provides the most precise qualitative instantiation of this mechanism: operational managers with misaligned incentives amplify technical barriers rather than absorbing them, extending the duration of pilot stagnation beyond what technical complexity alone would predict.

*P7 (Human-AI Collaboration Design as an Execution Requirement): SUPPORTED, with refinement.* B1.S2 (human-in-the-loop validation) ranked second highest-rated solution across all 122 items (crisp = 0.759), providing strong quantitative support. Thematic analysis extends the proposition by identifying the risk-tiered design principle as the operationally appropriate application: rather than uniform human validation of all outputs, responsible deployment requires oversight intensity calibrated to the consequences of output error, with high-stakes contexts receiving mandatory review and low-stakes contexts permitting automated release. This refinement is practically important for organisations concerned about the operational scalability of human-in-the-loop requirements as deployment scope expands.

*P8 (Regulatory and Legal Uncertainty as a Scaling Constraint): REFINED.* The Environmental paradox, zero retained barriers despite the highest mean retained solution effectiveness, substantially qualifies this proposition. Regulatory uncertainty functions as a scaling constraint, but it does so in a contextually variable manner that prevents universal consensus across jurisdictionally and sectorally diverse expert panels. The proposition's implicit assumption that regulatory uncertainty is a universally felt barrier requires qualification: it is a differentially felt barrier whose severity depends on geographic market, client industry, and use case risk classification. Environmental solutions, however, are universally applicable and should be implemented regardless of whether specific regulatory barriers are currently salient.

*P9 (Ethical Leadership and Legitimacy Safeguards as Value Enablers): PARTIALLY SUPPORTED.* Ethical AI governance is acknowledged across the evidence base as a legitimacy-building mechanism, but it did not emerge as a high-priority barrier or solution item in the FDM rankings for this IT services

context. This likely reflects the pragmatic, operationally focused orientation of the expert panel rather than a dismissal of ethical governance as unimportant in principle. The proposition is retained as theoretically valid but is assessed as less immediately actionable than P5 and P7 for organisations in the early stages of bridging the execution gap.

*P10 (Resource Availability as a Cross-Dimensional Moderator): PARTIALLY SUPPORTED.* No C5 (Resource Availability) barrier items were retained in the FDM consensus hierarchy, suggesting that resource constraints do not function as universally experienced discrete barriers for organisations that have cleared the investment threshold required to reach the pilot stage. However, open-ended survey contributions identified cost unpredictability and LLMOps infrastructure investment as significant omitted barriers, and the frozen middle theme reveals that resource allocation decisions are structurally distorted by commercial model incentives rather than aggregate resource insufficiency. The proposition is retained as directionally valid, with the refinement that resource availability in mature IT service organisations operates less as an absolute constraint and more as a sequencing and governance challenge shaped by ROI uncertainty and commercial model misalignment.

## 8.4 Theoretical Contributions

This thesis makes four distinct theoretical contributions to the management and technology adoption literature.

*First, an IT services-specific empirical account of the GenAI execution gap. The* study produces a validated, expert-prioritised barrier-solution hierarchy specific to IT services organisations, a sector characterised by multi-client accountability structures, service continuity obligations, and legacy architecture complexity that generate execution challenges not captured by broader enterprise AI adoption studies. The finding that fragmented data governance across client environments is the single highest-priority barrier, and that integration complexity achieves the strongest consensus signal in the entire dataset, establishes an IT services-specific fingerprint of the execution gap that differentiates this study from prior enterprise and software-organisation research (Hughes et al., 2026; Kemell et al., 2025; Neumann et al., 2026).

*Second, identification of the Environmental paradox and its explanatory implications for TOE research.* The systematic divergence between Environmental dimension barrier non-consensus and Environmental dimension solution consensus constitutes a novel finding for TOE-grounded research. It implies that environmental barriers should be treated as contextually variable inputs rather than universally stable constructs in multi-jurisdictional, multi-sectoral research designs, while environmental solutions can be confidently recommended as cross-contextual risk reduction mechanisms. This finding has methodological implications for future TOE-based studies conducted across diverse geographic and regulatory environments.

*Third, identification of incentive misalignment as a structural barrier not captured by the TOE framework.* The frozen middle dynamic, in which the commercial architecture of IT service delivery systematically misaligns operational management incentives with the automation value that GenAI enables, is a novel theoretical contribution that extends beyond the standard TOE construct set. The TOE framework's Organisation Context captures culture, capability, and governance maturity, but does not explicitly model the commercial pricing model of service delivery as a structural enabler or constraint. This study's identification of commercial model architecture as a determinant of the execution gap represents an opportunity for future theoretical extension of the TOE framework in professional services and managed service contexts.

*Fourth, operationalisation of risk-tiered governance architecture as an emergent construct.* The concept of governance proportionality, calibrating oversight intensity to the risk profile of the AI deployment type rather than applying uniform governance frameworks, emerged across multiple participants and

represents a practically grounded theoretical proposition for future research on AI governance design. This construct extends the TOE framework's Governance Structures construct (C4) from a binary presence or absence of governance policy to a nuanced, risk-differentiated institutional design principle with direct implications for how organisations can simultaneously reduce friction for low-risk deployments and maintain appropriate rigour for high-risk ones.

Taken together, these theoretical contributions suggest that the GenAI execution gap should not be understood only as a set of isolated technological, organisational, or environmental barriers. Rather, the findings indicate a dependency structure between the TOE dimensions. Technology establishes the operational foundation for scaling; Organisation determines whether that foundation can be absorbed, governed, and embedded; and Environment shapes the external accountability conditions under which scaling becomes legitimate and commercially viable. The incentive misalignment finding extends this logic by showing that the commercial model of IT service delivery operates as an organisational condition that can either support or obstruct GenAI-enabled value creation.

## 8.5 Practical Implications

The findings generate practical implications for IT service organisations seeking to move GenAI initiatives from pilot experimentation to governed, enterprise-scale deployment. Rather than presenting these implications as separate recommendations, this section organises them into a staged implementation pathway. The pathway should not be understood as an additional methodological phase of the study, nor as a rigid maturity model formally tested by the FDM. It is a managerial synthesis derived from the integrated findings. The FDM results identify which barriers and solutions were prioritised by experts, while the thematic analysis explains the mechanisms through which these barriers and solutions operate in practice.

The staged pathway follows the dependency structure identified in the findings. Foundational readiness should be addressed first, because data governance and integration complexity form the most visible operational barriers. Governance and assurance mechanisms should then be established to determine whether pilots can move into risk-bearing production environments. Scaling and value realisation should guide expansion beyond isolated pilots. Commercial incentive alignment should be developed as a parallel and progressively maturing workstream, because the frozen middle dynamic may otherwise prevent technically feasible GenAI initiatives from becoming embedded in service delivery practice.

Table 8.1: Staged Implementation Pathway for Bridging the GenAI Execution Gap

| Stage | Main implementation focus | Core actions | Relationship to empirical findings |
|---|---|---|---|
| **Stage 1: Foundation and Readiness** | Technical and client-side readiness | Strengthen data governance, metadata management, API readiness, access-control clarity, data sovereignty arrangements, and client readiness assessment | Derived from B2.2, B3.1–B3.4, Theme T1, Theme T2, and Theme E1 |
| **Stage 2: Governance, Assurance, and Ecosystem Risk Management** | Risk control, production decision rights, and vendor dependency management | Apply risk-tiered governance, HITL validation, security-by-design controls, responsible AI review, governance decision gates, multi-vendor strategy, and contractual data sovereignty provisions | Derived from B1.S2, B4.S1, D5.S2, C2.S3, Theme T3, Theme T4, Theme X1, and Theme X2 |
| **Stage 3: Scaling and Value Realisation** | Movement from pilots to repeatable production deployment | Define KPIs, ROI logic, OKRs, reusable service components, client co-design practices, service assurance mechanisms, and repeatable multi-platform deployment patterns | Derived from C1.S2, C2.S3, D4.S1, D4.S2, Theme O2, Theme E1, and Theme X2 |
| **Stage 4: Commercial Model and Incentive Alignment** | Frozen middle and billable-hour model adjustment | Review utilisation metrics, headcount-based incentives, pricing models, value attribution, and movement toward value-based or hybrid commercial models | Derived from Theme O1, the frozen middle finding, and the theoretical contribution on billable-hour incentive misalignment |

Note. The stages represent a practical sequencing logic derived from the integrated FDM and thematic findings. They should not be interpreted as additional research phases or as a rigid linear maturity model. Some activities, particularly governance design, vendor ecosystem assessment, and commercial incentive review, may begin in parallel but mature at different points in the scaling process.

*Stage 1: Foundation and Readiness*

The first stage concerns the technical and client-side foundations required before GenAI initiatives can be scaled responsibly. The dominant FDM finding, that fragmented data governance is the highest-priority barrier and integration complexity produces the strongest consensus signal, indicates that GenAI scaling readiness begins with the data and integration layer rather than the model layer. IT service organisations should therefore prioritise unified data governance, metadata management, API readiness, access-control clarity, and data sovereignty arrangements with clients before investing heavily in advanced foundation model selection.

This stage should also include client readiness assessment as a standard element of GenAI engagement design. The qualitative findings show that client-side maturity functions as an external scaling ceiling that IT service providers cannot unilaterally overcome. Client data maturity, platform readiness, governance infrastructure, contractual permissions, and division-level decision rights should therefore be assessed during pre-sales, scoping, and assessment phases. This reduces the risk of committing to production deployment in environments that are technically promising but not organisationally or contractually ready for scale.

*Stage 2: Governance, Assurance, and Ecosystem Risk Management*

The second stage concerns the governance mechanisms required to move from technically feasible pilots to controlled production deployment. The findings show that iterative scaling frameworks are effective only when supported by explicit governance decision gates. These gates should define when a pilot is scaled, redesigned, or terminated, based on pre-agreed success criteria, risk assessment, ownership, and measurable business outcomes. Without such gates, iterative experimentation risks becoming pilot purgatory, where technically functional pilots accumulate without becoming governed production capabilities.

This stage should also include risk-tiered AI governance. A uniform governance model applied to all AI deployments creates excessive friction for low-risk use cases and insufficient scrutiny for high-risk agentic systems. IT service organisations should therefore classify GenAI deployments by risk level, data sensitivity, client exposure, and operational consequence. Human-in-the-loop validation, security-by-design controls, auditability requirements, and responsible AI review should then be calibrated proportionately to the risk profile of each deployment type.

Finally, ecosystem risk management should be treated as part of governance and assurance rather than as a separate procurement concern. The FDM identified multi-vendor ecosystem strategy as the highest-ranked solution across all 122 items. This indicates that IT service organisations should avoid excessive dependency on a single foundation model provider or cloud AI platform, particularly where GenAI capabilities affect client deliverables, contractual SLAs, regulatory flexibility, and data sovereignty obligations. Vendor diversification strategies, contractual data protection provisions, and platform exit options should therefore be embedded in AI procurement and governance decisions from inception.

*Stage 3: Scaling and Value Realisation*

The third stage concerns the transition from governed pilots to repeatable, measurable, and portfolio-level scaling. The findings indicate that GenAI initiatives require value metrics before scaling decisions are made, not only after deployment. Organisations should define success criteria during the assessment and MVP stages, linking GenAI outcomes to service quality, SLA improvement, risk reduction, customer experience, productivity, or cost efficiency. These metrics should be connected to strategic planning and OKRs so that GenAI scaling is treated as an operational transformation programme rather than a collection of isolated technical experiments.

At this stage, IT service organisations should also build reusable scaling patterns. The qualitative findings suggest that excessive customisation can reproduce pilot fragmentation across client environments. Scalable deployment therefore requires standardised GenAI service components, reusable governance templates, common evaluation pipelines, multi-vendor deployment options, and repeatable client co-design practices that can be adapted across different contracts without being rebuilt from the beginning for each client.

*Stage 4: Commercial Model and Incentive Alignment*

The final stage addresses the organisational and commercial conditions that determine whether GenAI scaling can be sustained. This stage should not be interpreted as the last issue to be considered; rather, it should begin early as a diagnostic workstream and mature as evidence of value, client acceptance, and governance requirements becomes clearer. The frozen middle finding shows that leadership enthusiasm and technical feasibility are insufficient when operational management incentives remain tied to utilisation, headcount, or billable-hour logic. In such contexts, GenAI-enabled automation may appear to threaten the performance measures through which middle managers are evaluated, even when it increases client value or service quality.

IT service organisations should therefore review how pricing models, performance metrics, delivery incentives, and value attribution mechanisms align with GenAI-enabled productivity gains. This may require gradual movement from headcount-based or effort-based commercial models toward value-based, outcome-based, or hybrid pricing structures. The practical implication is that bridging the GenAI execution gap is not only a matter of better technology or stronger governance. It also requires commercial model adaptation so that the value created by automation is not structurally misaligned with the incentives of the delivery organisation.

The staged pathway should be interpreted as a flexible implementation logic rather than a universal sequence. In practice, the stages may overlap. Governance design, vendor ecosystem assessment, and commercial incentive review may begin during foundation-building, while full commercial model adaptation may mature only after value evidence and client expectations become clearer. However, the sequence remains analytically useful because it reflects the dependency structure identified in the findings: GenAI scaling requires data and integration readiness before production deployment, governance and assurance before risk-bearing use, measurable value logic before portfolio expansion, and incentive alignment before sustained organisational embedding.

## 8.6 Methodological Reflections and Limitations

The sequential four-phase mixed-methods design employed in this study represents a methodologically coherent and epistemologically appropriate approach to investigating the GenAI execution gap. The combination of a quantitative survey baseline, FDM expert consensus analysis, six individual semi-structured expert interviews, and thematic analysis enabled a level of empirical triangulation that no single-method approach could have achieved. The structured integration of the TOE framework across all four phases, as both a theoretical organising lens and a methodological coding scaffold, ensures consistency of construct coverage and systematic comparison of evidence across phases.

The FDM phase's single-round design, in which survey respondents' Likert-scale ratings were directly converted into Triangular Fuzzy Numbers and subjected to consensus and defuzzification analysis without iterative expert revision rounds, is an important methodological feature of this study that should be understood when interpreting the FDM results. The single-round design was chosen to maintain clear phase separation between the broad-sample survey and the focused individual expert interviews, and to optimise respondent burden. This design produces a cross-sectional consensus picture of barrier and solution severity at the point of data collection. The theoretical and mathematical basis for single-round FDM analysis is well established in the literature (Bouzon et al., 2016; Nozari et al., 2021), and the use of Triangular Fuzzy Numbers to model the inherent vagueness of expert judgement remains fully justified within this design.

Several limitations should be explicitly acknowledged. First, purposive and snowball sampling restricts statistical generalisability. Findings should be interpreted as contextually grounded insights rather than population-level statistics. Second, the qualitative phase involved six individual expert interviews,

providing depth rather than breadth of coverage. While thematic saturation was assessed and achieved at the construct level, interpretive conclusions are necessarily bound by the professional experiences and organisational contexts of the six participants. Third, the sectoral focus on IT services limits direct transferability to healthcare, financial services, manufacturing, or the public sector, where structural conditions and regulatory environments differ materially. Fourth, the cross-sectional design captures barrier and solution perceptions at a single point during a period of rapid GenAI platform evolution; findings may shift as technology matures, regulatory frameworks are clarified, and organisations accumulate more production-scale deployment experience.

A further limitation concerns the staged implementation pathway presented in Section 8.5. This pathway is derived from the integrated FDM and thematic findings, but it was not independently tested as a formal maturity model or longitudinal implementation sequence. It should therefore be interpreted as a practical synthesis of the study's findings rather than as a statistically validated or universally generalisable implementation model. Future research could examine whether the proposed sequence holds across different organisational settings and over time.

## 8.7 Directions for Future Research

The findings of this study generate a well-defined agenda for future empirical investigation across six priority directions.

*Longitudinal study designs.* The cross-sectional nature of this study captures barrier and solution perceptions at a single moment in time. Future research should employ longitudinal designs to track how barrier severity and solution effectiveness evolve as organisations progress from pilot to production and through successive cycles of GenAI programme maturation. Longitudinal evidence would provide substantially stronger support for the causal relationships implied by the conceptual model and would allow the change capacity moderation effect and the frozen middle dynamic to be observed as dynamic organisational processes rather than static conditions.

*Cross-industry comparative studies.* To what degree are fragmented data governance, integration complexity, and incentive misalignment barriers specific to IT services, and to what degree are the features of enterprise GenAI scaling more broadly? Comparative studies spanning healthcare, financial services, manufacturing, and the public sector would test the generalisability of the barrier-solution hierarchy developed in this study and identify sector-specific barrier profiles that warrant tailored governance and intervention strategies.

*AI governance maturity research.* The findings on governance structures, particularly the zero-consensus outcome for C4S solution items and the risk-tiered governance architecture construct emerging from the qualitative phase, point to an urgent need for research on AI governance maturity models. How do organisations progress through stages of AI governance development? What governance architectures are associated with sustained GenAI scaling success? Research that maps governance maturity stages to GenAI execution outcomes would provide practitioners and policymakers with actionable guidance on governance investment sequencing that this study's cross-sectional design cannot supply.

*Post-deployment operational studies.* This study focuses on barriers that prevent scaling beyond pilots; it does not investigate what happens after production deployment is achieved. Future research should examine post-deployment challenges, including LLMOps infrastructure requirements, prompt lifecycle management, continuous evaluation pipelines, drift detection, and the value attribution challenge identified in the open-ended survey contributions. The financial governance of AI workloads, including token pricing volatility, infrastructure cost management, and FinOps disciplines for AI, represents a particularly under researched but practically significant area for multi-client IT service delivery.

*Commercial model transformation research.* The incentive misalignment finding that the billable-hour commercial architecture creates structural disincentives for operational GenAI embedding, is the most novel theoretical contribution of the qualitative phase and carries the highest practical implications for the industry. Dedicated empirical research on the transition from headcount-based to value-based or outcome-based pricing models in IT services, including the barriers to commercial model transformation and the conditions under which clients accept value-based contractual structures, would directly address a structural gap that current technology adoption frameworks do not model.

*Human-AI governance and organisational transformation research.* The risk-tiered human oversight principle and the client-side readiness bottleneck both point towards a broader research agenda on human-AI governance design in complex, multi-stakeholder service environments. How should organisations design human oversight frameworks that scale proportionately without becoming prohibitively expensive? How does client-side AI governance maturity influence IT service providers' deployment decisions? What organisational structures beyond the AI CoE model are most effective at bridging enterprise AI governance with division-level operational implementation? These questions sit at the intersection of technology governance, organisational design, and professional services management and represent a rich agenda for interdisciplinary research.

Future research could also test the staged implementation pathway proposed in Section 8.5 through longitudinal or comparative case study designs. Such research could examine whether organisations that address foundational readiness, governance and assurance, value realisation, and commercial incentive alignment in a structured sequence are more successful in moving GenAI initiatives from pilot to production. This would help determine whether the pathway functions as a transferable implementation logic across IT services organisations or whether it requires adaptation by market, client segment, or service delivery model.

## 8.8 Concluding Statement

This thesis sets out to investigate why IT service organisations with demonstrable GenAI pilot success continue to encounter barriers when attempting to move toward enterprise-scale, governed deployment. The findings show that the GenAI execution gap is not fundamentally a technology capability problem. It is a multi-dimensional execution problem produced by the interaction of fragmented data governance, integration complexity, organisational readiness, client-side maturity, governance design, vendor ecosystem dependency, and commercial incentive structures.

The study contributes an IT services-specific account of this execution gap by producing an expert-validated barrier–solution hierarchy and enriching it through qualitative thematic analysis. The findings show that fragmented data governance and integration complexity form the most visible operational barriers, while organisational incentive misalignment explains why technically feasible initiatives often fail to become embedded in delivery practice. The frozen middle dynamic is central to this explanation: where middle management remains accountable for utilisation, headcount, and billable delivery capacity, GenAI-enabled efficiency can become organisationally difficult to absorb even when it is strategically desirable.

The implications for the IT services industry are therefore significant. GenAI scaling will not be determined solely by access to stronger models or more advanced tools. It will depend on whether IT service organisations can build the data foundations, governance architectures, vendor strategies, client assurance mechanisms, and commercial models required to convert AI capability into accountable service value. The industry may therefore face a gradual shift from labour-volume-based delivery models toward more outcome-oriented and value-based forms of service contracting, particularly as clients increasingly expect AI-enabled productivity gains to be reflected in service quality, speed, and cost structures.

In this sense, the GenAI execution gap represents more than a temporary implementation challenge. It signals a broader transition in the logic of IT service delivery. The evidence suggests that organisations treating GenAI as an isolated automation tool are more likely to encounter persistent pilot fragmentation, governance delays, vendor dependency, and incentive misalignment. By contrast, organisations approaching GenAI as a catalyst for redesigning data architecture, governance maturity, client collaboration, vendor ecosystem strategy, and commercial value creation may be better positioned to move from experimentation to sustained, responsible, and scalable AI-enabled service delivery.

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# Appendix A: Comprehensive Overview of GenAI Execution Barriers and Solutions

Table A.1: Comprehensive Overview of GenAI Execution Barriers and Solutions

| Barrier | Barrier Description | Solution Approach | Solution Description | Key Literature |
|---|---|---|---|---|
| **Data quality and privacy** | Enterprise data is fragmented, incomplete, biased, or sensitive, limiting the safe and reliable use of GenAI in operational and client-facing service contexts. | Data governance and quality controls | Establish enterprise data standards, access controls, privacy-by-design mechanisms, data quality pipelines, and governance controls before scaling GenAI beyond pilots. | Voorneveld (2024); Smith (2025); Pushpakumara & Ahsan (2025); Taeihagh (2025) |
| **Integration complexity** | GenAI is difficult to embed into legacy IT systems, ITSM workflows, API ecosystems, and multi-client service architectures, creating a gap between pilot feasibility and production readiness. | Modular architecture and integration readiness | Use API-first architectures, phased deployment pathways, middleware orchestration, and architecture readiness assessments to integrate GenAI into existing service processes without disrupting operational stability. | Neumann et al. (2026); Kasireddy & Sreenivasu (2025); Voorneveld (2024) |
| **Security vulnerabilities** | GenAI introduces new security risks, including prompt injection, data leakage, misuse of sensitive client information, and exposure through cloud-hosted or third-party AI services. | Security-by-design controls | Apply prompt-level safeguards, data anonymisation, vendor data sovereignty agreements, secure deployment patterns, and GenAI-specific security controls from the start of implementation. | Taeihagh (2025); Neumann et al. (2026); Kemell et al. (2025) |
| **Model hallucinations** | Large language models can generate inaccurate, fabricated, or misleading outputs, creating a validation burden that increases as GenAI moves into operational or client-facing use cases. | RAG and output validation | Ground GenAI outputs in verified enterprise knowledge sources through Retrieval-Augmented Generation, standardised evaluation pipelines, and human-in-the-loop validation routines. | Pushpakumara & Ahsan (2025); Basha & Rodríguez-Pérez (2025); Simaremare & Edison (2024); Neumann et al. (2026) |
| **Explainability limitations** | GenAI outputs may be difficult to explain, audit, or reproduce, making it hard for practitioners and clients to know when outputs should be trusted. | Explainability and auditability mechanisms | Embed explainability mechanisms into deployment design, use context-specific tolerance frameworks, and sequence use cases from low-risk to high-risk based on auditability and client exposure. | Hughes et al. (2026); Voorneveld (2024); Taeihagh (2025) |

| | | | | |
|---|---|---|---|---|
| **Technology immaturity** | GenAI tooling, standards, and operational practices remain unstable in some enterprise contexts, making large-scale deployment risky when governance and infrastructure maturity are insufficient. | Staged or maturity-based adoption | Adopt GenAI incrementally through maturity-based deployment pathways, beginning with low-risk use cases and expanding only when infrastructure, governance, and assurance conditions are stable. | Smith (2025); Kasireddy & Sreenivasu (2025); Voorneveld (2024) |
| **Monitoring and reproducibility challenges** | GenAI outputs may vary across runs and over time, making it difficult to ensure consistent performance, trace decisions, and reproduce outputs in audited service environments. | Standardised evaluation and monitoring pipelines | Document prompts, data sources, model settings, and evaluation criteria; continuously monitor outputs, performance drift, and reproducibility across operational contexts. | Kasireddy & Sreenivasu (2025); Pushpakumara & Ahsan (2025); Neumann et al. (2026) |
| **Conceptual ambiguity of GenAI definitions** | Inconsistent understanding of what GenAI is, and how it differs from broader AI or automation, can lead to unclear goals, unsuitable use cases, and misaligned governance expectations. | GenAI value-pathway and classification frameworks | Clarify GenAI scope by distinguishing between AI agents, content creation, knowledge querying, foundation models, and upstream infrastructure, and match each pathway to its required governance maturity. | Teng et al. (2025); Clear et al. (2025); Hughes et al. (2026) |
| **Skills shortages and AI literacy gaps** | Employees may lack the technical, governance, prompt engineering, and critical evaluation skills needed to use GenAI safely and effectively in IT service delivery. | Role-based AI upskilling | Provide role-specific GenAI training focused on safe usage, prompt design, output evaluation, data awareness, governance responsibilities, and decision support. | Hughes et al. (2026); Kasireddy & Sreenivasu (2025); Fernández & Uriarte (2026); Kemell et al. (2025) |
| **Resource constraints** | Organisations may lack sufficient budget, time, data infrastructure, technical capacity, or operational resources to pilot, scale, maintain, and govern GenAI responsibly. | Phased investment and cost governance | Use phased deployments, total-cost-of-ownership modelling, infrastructure capacity planning, and cost governance to prevent underestimation of scaling and maintenance costs. | Kasireddy & Sreenivasu (2025); Smith (2025); Pushpakumara & Ahsan (2025) |
| **Governance immaturity and unclear AI policies** | Organisations may lack GenAI-specific governance structures, internal controls, accountability mechanisms, and practical usage policies suitable for non-deterministic AI systems. | AI governance framework establishment | Establish GenAI governance bodies, internal controls, ethical review processes, accountability models, and layered assurance mechanisms across technical, operational, and executive levels. | Smith (2025); Taeihagh (2025); Neumann et al. (2026) |

| | | | | |
|---|---|---|---|---|
| **Leadership and strategic alignment gaps** | GenAI initiatives may remain fragmented when they lack senior sponsorship, strategic ownership, clear priorities, and integration into enterprise objectives. | Strategic alignment models | Integrate GenAI into enterprise strategy, OKRs, leadership agendas, investment priorities, and governance decision-making so pilots can move toward coordinated scaling. | Kasireddy & Sreenivasu (2025); Kumar et al. (2025); Hughes et al. (2026) |
| **Cultural resistance and change capacity limitations** | Employees and middle managers may resist GenAI adoption due to mistrust, fear of role displacement, workflow disruption, or weak organisational change capacity. | Structured change management | Align GenAI adoption with organisational culture through communication, engagement, cross-functional learning routines, change capacity development, and pilot success communication. | Hughes et al. (2026); Clear et al. (2025); Fernández & Uriarte (2026) |
| **Relative advantage and value clarity** | Business benefits may remain unclear beyond pilots, making it difficult to justify investment or connect GenAI outcomes to SLA improvement, service quality, risk reduction, or margin. | ROI and value measurement frameworks | Track financial, operational, and service-quality outcomes using ROI frameworks that connect GenAI adoption to measurable business value and client-facing performance indicators. | Kemell et al. (2025); Smith (2025); Hughes et al. (2026); Al-Khatib (2023) |
| **Human-AI collaboration and over-reliance risks** | Practitioners may become over-reliant on GenAI outputs, reducing independent judgement, critical evaluation, and motivation when returning to solo work. | Human-AI work design | Redesign workflows to clarify which tasks remain human-led and which can be AI-assisted; structure the sequencing of AI-supported and independent work to protect judgement and motivation. | Basha & Rodríguez-Pérez (2025); Wu et al. (2025); Clear et al. (2025) |
| **Regulatory uncertainty** | Evolving and fragmented AI regulations create uncertainty for organisations attempting to scale GenAI across jurisdictions and regulated client environments. | Compliance-by-design | Embed auditability, documentation, regulatory interpretation, and compliance requirements into GenAI systems and governance processes from initial deployment. | Cheung (2025); Taeihagh (2025); Hughes et al. (2026); Neumann et al. (2026) |
| **Unclear legal liability** | Accountability may be unclear when GenAI outputs cause harm, produce incorrect decisions, or involve client data, vendor platforms, or third-party models. | Contractual liability frameworks | Use contractual GenAI usage clauses, warranties, indemnities, liability caps, data-use agreements, and clear allocation of responsibility between providers, clients, and vendors. | Fernández & Uriarte (2026); Kemell et al. (2025); Taeihagh (2025) |

| | | | | |
|---|---|---|---|---|
| **Ethical concerns: bias, fairness, and accountability** | GenAI may create bias, fairness, transparency, and accountability concerns, especially as adoption scales into client-facing or decision-critical services. | Ethical leadership and responsible AI governance | Embed ethical leadership, transparency mechanisms, bias mitigation, accountability structures, and responsible AI governance into scaling decisions from the beginning. | Kumar et al. (2025); Kittipanya-Ngam et al. (2025); Fernández & Uriarte (2026) |
| **Customer and stakeholder pressure** | Clients and external stakeholders may demand AI-enabled service improvements before the organisation has sufficient governance, readiness, or assurance capacity. | Client co-design and ecosystem engagement | Engage clients and stakeholders in co-designing governance, data-use rules, assurance mechanisms, and service expectations before scaling GenAI into operational delivery. | Teng et al. (2025); Fernández & Uriarte (2026); Al-Khatib (2023) |
| **Competitive pressure** | Organisations may feel pressure to adopt GenAI quickly to avoid falling behind competitors, creating a risk of premature deployment before readiness conditions are met. | Readiness-based competitive positioning | Adopt GenAI strategically through maturity-based sequencing, readiness assessments, and disciplined pilot-to-scale decisions that preserve competitive positioning without sacrificing governance. | Al-Khatib (2023); Taeihagh (2025); Smith (2025); Cheung (2025) |

# Appendix B: Conceptual Model Variables, TOE Constructs, Barriers, Solutions, and Measurement Criteria

Table B.1: Conceptual Model Variables, TOE Constructs, Barriers, Solutions, and Measurement Criteria

| TOE Dimension | Construct | Barriers | Proposed Solutions | Survey and FDM Indicator | Key Literature |
|---|---|---|---|---|---|
| **Technology** | Model Reliability, Hallucination, Explainability, and Reproducibility | Model hallucinations; explainability limitations; monitoring and reproducibility challenges; technology immaturity | RAG and grounding techniques; human-in-the-loop validation; standardized evaluation pipelines; GenAI explainability frameworks; maturity-based adoption roadmaps | Perceived frequency of hallucination incidents; adequacy of output validation protocols; explainability sufficiency for client-facing use; reproducibility of production outputs | Pushpakumara & Ahsan (2025); Basha & Rodríguez-Pérez (2025); Neumann et al. (2026); Hughes et al. (2026); Voorneveld (2024); Taeihagh (2025) |
| **Technology** | Data Quality, Availability, and Privacy | Data quality and privacy; integration complexity (data pipeline dimension) | Enterprise data governance programmes; data quality pipelines; privacy-by-design; data labelling and cleansing initiatives; data sovereignty agreements | Adequacy of organisational data for GenAI use cases; sufficiency of data governance controls; privacy constraint severity on use case scope | Voorneveld (2024); Smith (2025); Pushpakumara & Ahsan (2025); Taeihagh (2025); Neumann et al. (2026) |
| **Technology** | Integration Complexity and Technical Debt | Integration complexity with legacy systems; technology immaturity (workflow integration dimension) | Modular API-first integration architecture; middleware orchestration; phased integration roadmaps; architecture readiness assessment | Severity of legacy system integration barriers; adequacy of API management capabilities; time and cost of GenAI workflow embedding | Neumann et al. (2026); Kasireddy & Sreenivasu (2025); Voorneveld (2024); Kemell et al. (2025) |
| **Technology** | Security Vulnerabilities and IP Risks | Security vulnerabilities (prompt injection, data leakage); unclear legal liability (IP dimension) | Security-by-design architecture; GenAI-specific security controls (prompt, data, model level); data anonymisation; vendor data sovereignty agreements | Perceived severity of prompt injection risk; adequacy of data leakage controls; IP protection sufficiency in GenAI contracts | Taeihagh (2025); Neumann et al. (2026); Kemell et al. (2025); Fernández & Uriarte (2026); Basha & Rodríguez-Pérez (2025) |
| **Technology** | AI Infrastructure Readiness and Scalability | Resource constraints (infrastructure dimension); technology immaturity (infrastructure dimension) | Cloud-native infrastructure investment; managed AI platform services; scalable containerised deployment; cost-efficient AI infrastructure; hybrid AI deployment models | Cloud architecture maturity for GenAI workloads; ability to scale beyond pilot environment; adequacy of compute and storage for production deployment | Kasireddy & Sreenivasu (2025); Smith (2025); Voorneveld (2024); Kemell et al. (2025) |
| **Technology** | Relative Advantage Clarity and Value Demonstration | Relative advantage and value clarity; conceptual ambiguity of GenAI definitions | ROI measurement frameworks; AI value tracking dashboards; use case definition per role; critical mindset development; technical classification frameworks | Clarity of GenAI value proposition in IT services context; ability to measure GenAI ROI; alignment between GenAI capabilities and service delivery KPIs | Teng et al. (2025); Kemell et al. (2025); Smith (2025); Hughes et al. (2026); Al-Khatib (2023); Simaremare & Edison (2024) |
| **Organisation** | Leadership Commitment and Strategic Alignment | Leadership and strategic alignment gaps | C-suite AI sponsorship; AI-integrated strategic planning; executive AI literacy programmes; strategic alignment models | Senior leadership visibility and active sponsorship of GenAI initiatives; alignment of GenAI roadmap with enterprise strategy | Kasireddy & Sreenivasu (2025); Kumar et al. (2025); Hughes et al. (2026); Smith (2025) |
| **Organisation** | Organisational Readiness and Change Capacity | Cultural resistance and change capacity limitations | Structured change management programmes; AI culture-building initiatives; employee AI readiness assessments; iterative pilot-to-scale frameworks | Organisational capacity to adapt workflows for GenAI; level of cultural resistance to AI adoption; adequacy of change management support during rollout | Hughes et al. (2026); Clear et al. (2025); Fernández & Uriarte (2026); Kemell et al. (2025) |
| **Organisation** | AI Skills, Literacy, and Capability Gaps | Skills shortages and AI literacy gaps | AI upskilling and reskilling programmes; workforce enablement strategy; AI Centre of Excellence; role-based AI training; strategic AI talent acquisition | Availability of AI-skilled staff for production deployment; adequacy of AI literacy training for delivery roles; quality of prompt engineering and output evaluation competencies | Hughes et al. (2026); Kasireddy & Sreenivasu (2025); Fernández & Uriarte (2026); Kemell et al. (2025); Clear et al. (2025) |

| TOE Dimension | Construct | Barriers | Proposed Solutions | Survey and FDM Indicator | Key Literature |
|---|---|---|---|---|---|
| **Organisation** | Governance Structures, Accountability, and AI Policy Maturity | Governance immaturity and unclear AI policies | AI governance framework adoption; enterprise AI policy development; risk classification and review processes; ethics review boards; accountability mapping | Clarity of AI policy and accountability structures; maturity of AI risk management processes; presence of formal approval and oversight mechanisms for GenAI deployment | Smith (2025); Taeihagh (2025); Neumann et al. (2026); Hughes et al. (2026) |
| **Organisation** | Resource Availability and Investment Capacity | Resource constraints (budget, time, IT capacity) | Dedicated GenAI investment budgets; portfolio prioritisation frameworks; phased investment roadmaps; build-partner-buy frameworks; minimum viable AI approach | Adequacy of financial resources for GenAI scaling; availability of dedicated AI investment budget; tolerance for iterative experimentation before ROI materialises | Kasireddy & Sreenivasu (2025); Smith (2025); Pushpakumara & Ahsan (2025); Fernández & Uriarte (2026) |
| **Organisation** | Human-AI Collaboration Design and Over-Reliance Risk | Human-AI collaboration and over-reliance risks | Use case definition per role; explicit human oversight frameworks; AI-assisted versus solo task recommendation strategies; cognitive ergonomics design for AI interfaces | Clarity of human versus AI task boundaries in service delivery; adequacy of over-reliance mitigation protocols; quality of work redesign for AI-integrated service roles | Wu et al. (2025); Basha & Rodríguez-Pérez (2025); Clear et al. (2025); Kemell et al. (2025); Neumann et al. (2026) |
| **Environment** | Regulatory Frameworks, Compliance, and Legal Liability | Regulatory uncertainty; unclear legal liability | Compliance-by-design governance integration; contractual liability allocation frameworks; regulatory monitoring and early engagement | Severity of regulatory compliance complexity on GenAI deployment decisions; adequacy of legal liability frameworks; impact of GDPR and EU AI Act on use case scope | Cheung (2025); Taeihagh (2025); Fernández & Uriarte (2026); Hughes et al. (2026); Neumann et al. (2026) |
| **Environment** | Ethical Concerns, Bias, and Legitimacy Expectations | Ethical concerns (bias, fairness, accountability) | Ethical leadership; ethical AI frameworks; standardised bias evaluation criteria; transparency and accountability mechanisms; ethics board establishment | Presence of ethical AI evaluation processes; adequacy of bias detection mechanisms; strength of ethical leadership in GenAI governance | Kumar et al. (2025); Kittipanya-Ngam et al. (2025); Fernández & Uriarte (2026); Taeihagh (2025) |
| **Environment** | Competitive Pressure and Adoption Urgency | Competitive pressure and premature adoption risk | Deliberate adoption sequencing strategies; competitive intelligence programmes; market positioning frameworks balancing adoption speed with execution readiness | Intensity of competitive pressure to adopt GenAI; risk of premature deployment due to competitive urgency; adequacy of strategic sequencing in GenAI roadmaps | Al-Khatib (2023); Taeihagh (2025); Smith (2025); Cheung (2025); Fernández & Uriarte (2026) |
| **Environment** | Customer and Stakeholder Pressure | Customer and stakeholder pressure | AI partnership strategies; client co-design of GenAI governance; AI-enabled service assurance frameworks; stakeholder communication plans | Intensity of client demands for GenAI-enabled service innovation; adequacy of stakeholder communication on AI capabilities and limits; client auditability requirements | Teng et al. (2025); Hughes et al. (2026); Taeihagh (2025); Fernández & Uriarte (2026); Al-Khatib (2023) |
| **Environment** | Vendor Ecosystems and AI Platform Maturity | Integration complexity (vendor dimension); resource constraints (vendor dependency dimension) | Strategic vendor assessment and selection; multi-vendor ecosystem diversification; vendor data sovereignty contractual controls; vendor partnership management | Availability of enterprise-ready, compliant GenAI platforms; vendor ecosystem maturity for IT services use cases; dependency risk severity from vendor lock-in, vendor policy changes, model deprecation, and data sovereignty limitations | Kemell et al. (2025); Neumann et al. (2026); Kasireddy & Sreenivasu (2025); Taeihagh (2025); Smith (2025) |

# Appendix C: Survey Questionnaire Participant Overview

Table C.1: Survey Questionnaire Participant Overview

| Participant | Industry | Primary Market | Role |
|---|---|---|---|
| **Participant 1** | System Integrator | Nordics | Strategy Development Specialist |
| **Participant 2** | Managed Service Provider | Global | AI Engineer |
| **Participant 3** | IT Consulting | EU | AI Architecture Head |
| **Participant 4** | Global IT consulting firm | US | Cloud Infrastructure Presales / AI Consulting |
| **Participant 5** | IT Consulting | Nordics | AI in IT Operations |
| **Participant 6** | Managed IT services organisation | Nordics | Product Owner / Service Desk Delivery Head |
| **Participant 7** | IT Consulting | Nordics | Workplace Services Owner |
| **Participant 8** | SaaS Service | Global | Strategy |
| **Participant 9** | Cloud and Platform Service Provider | EU | Strategy |
| **Participant 10** | Large Scale consulting firm; client-facing AI strategy | US | Technology Strategy and Transformation Consultant |
| **Participant 11** | Managed Service Provider | Nordics | Head of Customer Success |
| **Participant 12** | IT Consulting | Nordics | Head of Public Delivery |
| **Participant 13** | Managed IT services organisation | Nordics | Senior IT Architect / Delivery Lead |
| **Participant 14** | IT Consulting | Nordics | Automation and AI Expert |
| **Participant 15** | IT services firm; client facing delivery | Nordics | IT Programme / Delivery Manager |
| **Participant 16** | System Integrator | Global | Automation and AI Head EMEA |
| **Participant 17** | Managed Service Provider | EU | Global Enterprise Architect |
| **Participant 18** | IT Consulting | US | ServiceNow AI Product Manager |
| **Participant 19** | IT Consulting | Global | Solution Architect |
| **Participant 20** | IT Consulting | Nordics | Product Manager |

# Appendix D: Survey Instrument

**Survey Questionnaire: Bridging the GenAI Execution Gap in IT Service Organisations**
**MBA (Master of Business Administration) Thesis, Phase 1 Data Collection Instrument**
Theoretical Framework: Technology-Organisation-Environment (TOE)
Research Focus: Barriers and Solutions for Scaling Generative AI from Pilot to Enterprise-Level Operations

**Introduction and Informed Consent**
Thank you for participating in this research.
This survey is conducted as part of an MBA thesis at Blekinge Institute of Technology. It forms Phase 1 of a mixed-methods study investigating the barriers and solutions that determine whether Generative AI (GenAI) initiatives successfully transition from pilot experimentation to enterprise-scale operational deployment, a phenomenon referred to in this study as the *GenAI Execution Gap*.

**Key definitions:**

- *Generative AI (GenAI):* Systems that generate content (e.g., text, code) using large models (e.g., LLMs).
- *GenAI Execution Gap:* The difficulty of moving GenAI from pilot stages to sustainable, governed, value-producing operations.

Your responses will contribute to an empirically grounded understanding of the technological, organisational, and environmental factors that shape GenAI scaling in IT service organisations. The survey takes approximately *35–45 minutes* to complete.

*Confidentiality:* All responses are strictly anonymous and will be used solely for academic research purposes. No individual or organisation will be identifiable in the final thesis or any associated publications.
*Voluntary Participation:* Participation is entirely voluntary. You may withdraw at any time without consequence.

By proceeding to Section A, you confirm that you have read and understood this preamble and consent to participate.

**How to Complete the Survey**
The survey has six sections (A-F). Sections B, C, and D ask you to assess barriers and solutions across three dimensions, Technology, Organisation, and Environment, using two rating scales:

- Barrier Severity Scale (BSS): How much does each item hold back GenAI scaling in your organisation, or across the IT services sector more broadly?
  1 = Not a barrier | 2 = Minor barrier | 3 = Moderate barrier | 4 = Significant barrier | 5 = Critical barrier
- Solution Effectiveness Scale (SES): How effective do you think each proposed solution would be in addressing the corresponding barrier?
  1 = Not effective | 2 = Slightly effective | 3 = Moderately effective | 4 = Highly effective | 5 = Extremely effective

Respondents should base their answers on direct professional experience wherever possible. Where such experience is limited, responses may reflect informed professional judgement; however, purely speculative responses should be avoided.
**Section A: About You**
This section collects some background information so we can analyse responses by role, organisational context, and level of GenAI experience.
*A1. What is your current job role?* (Select one)

- C-Suite Executive (CEO (Chief Executive Officer), CTO (Chief Technology Officer), CIO (Chief Information Officer), CDO (Chief Data Officer), CAIO (Chief Artificial Intelligence Officer))
- VP (Vice President) / Director of Technology or Digital Transformation
- AI / GenAI Programme or Product Manager
- Enterprise Architect / Solution Architect
- Data Governance / Data Engineering Lead
- IT Service Delivery Manager / Operations Lead
- Software Engineer / AI/ML Engineer
- Risk, Compliance, or Legal Professional
- Other (please specify): _____

*A2. How many years of professional experience do you have in the IT services industry?* (Select one)

- Less than 3 years
- 3-5 years
- 6-10 years
- 11-15 years
- More than 15 years

*A3. How large is your organisation?* (Select one)

- Fewer than 100 employees
- 100-499
- 500-1,999
- 2,000-9,999
- 10,000 or more

*A4. What best describes your organisation's primary focus?* (Select all that apply)

- Managed IT Services
- IT Consulting and Advisory
- Software Development and Engineering
- Cloud Services and Infrastructure
- Cybersecurity Services
- Data Analytics and AI Services
- Enterprise Application Services (ERP, DOI, etc.)
- Business Process Outsourcing (BPO)
- Other (please specify): _____

A5. Which industries does your organisation primarily serve? (Select all that apply)

- Financial Services
- Healthcare / Life Sciences
- Public Sector / Government
- Retail / Consumer
- Energy / Utilities
- Manufacturing / Industrial
- Technology / Digital Native
- Telecommunications
- Other (please specify): _____

*A6. Where is your organisation primarily headquartered?* (Select one)

- Europe (EU / EEA(European Economic Area))
- United Kingdom
- North America
- Asia-Pacific
- Middle East and Africa
- Latin America
- Other (please specify): _____

*A7. What is the primary context in which your organisation applies GenAI?* (Select one)

- Primarily internal (e.g., productivity, internal delivery efficiency)
- Primarily client-facing (e.g., embedded in client services or deliverables)
- Both internal and client-facing use equally
- Not sure / not aware of how GenAI is applied

*A8. How would you describe your organisation's current stage of GenAI adoption?* (Select one)

- No active GenAI initiatives (exploring, or not yet engaged)
- Early exploration (proof-of-concept or sandbox testing only)
- Pilot phase (one or more pilots underway, not yet in production)
- Selective deployment (GenAI live in a limited number of production contexts)
- Enterprise-scale deployment (GenAI embedded across multiple service lines or business units)

*A9. How closely are you involved in your organisation's GenAI strategy, adoption, or deployment decisions?* (Select one)

- Directly responsible (primary decision-maker or programme owner)
- Closely involved (key contributor to decisions or delivery)
- Partially involved (consulted or kept informed)
- Limited involvement (aware but not directly engaged)
- Not involved / no exposure to GenAI activities

**Section B: Technology Barriers and Solutions**

This section covers the technology factors that affect an organisation's ability to move GenAI from pilot into full-scale operation. For each item, please use both the Barrier Severity Scale (BSS) and the Solution Effectiveness Scale (SES) as described above. Please base your answers on your direct professional experience wherever possible.

For barriers, you may also draw on your broader experience and observations of the IT services sector where relevant, particularly where you have not encountered the issue directly but have a clear understanding of its impact.

This section asks you to assess barriers and solutions across Technology dimension using two rating scales:

- Barrier Severity Scale (BSS): How much does each item hold back GenAI scaling in your organisation, or across the IT services sector more broadly?
  1 = Not a barrier | 2 = Minor barrier | 3 = Moderate barrier | 4 = Significant barrier | 5 = Critical barrier
- Solution Effectiveness Scale (SES): How effective do you think each proposed solution would be in addressing the corresponding barrier?
  1 = Not effective | 2 = Slightly effective | 3 = Moderately effective | 4 = Highly effective | 5 = Extremely effective

For solutions, please respond only where you have sufficient familiarity or understanding of the approach. If you are not familiar with a particular solution, please leave the item unanswered rather than providing an estimate.

**B1, Model Reliability, Hallucination Risk, Explainability, and Reproducibility**
*This construct covers challenges around the accuracy, transparency, and consistency of GenAI outputs in live production environments.*
*Barriers, Please rate how much each item holds back GenAI scaling in your organisation (BSS: 1-5):*
**Barriers (BSS: 1-5):**
B1.1 GenAI models produce incorrect or hallucinated outputs often enough to create unacceptable quality risks in client-facing work.
B1.2The inability to explain or audit how a GenAI model reached its output limits our confidence in using it for accountable, contractual service delivery.
B1.3 Inconsistent outputs across repeated runs make it difficult to monitor and assure quality at scale.
B1.4 Current GenAI model maturity does not consistently meet our reliability standards for production.
**Solutions (SES: 1-5):**
# Solution Item
B1.S1 Using Retrieval-Augmented Generation (RAG) and grounding techniques to reduce the rate of hallucinated outputs in live systems.
B1.S2 Requiring human review of GenAI outputs (human-in-the-loop validation) before they are shared with clients.
B1.S3 Establishing standardised evaluation pipelines and explainability frameworks to monitor output quality and auditability in production.
B1.S4 Adopting maturity-based deployment roadmaps that restrict production use of GenAI to use cases where reliability thresholds have been empirically validated. .

**B2, Data Quality, Availability, and Privacy**
*This construct covers the adequacy of organisational data assets and the privacy constraints that restrict what data can safely be used with GenAI systems.*
**Barriers (BSS: 1-5):**
B2.1 Poor data quality, incomplete, inconsistent, or unstructured data, makes it difficult to rely on GenAI outputs in our day-to-day workflows.
B2.2 Fragmented data governance across business units or client environments prevents us from building coherent data pipelines for GenAI at scale.
B2.3 Privacy obligations (e.g., GDPR, client data classification policies) restrict which data can be used with GenAI systems, limiting how widely we can deploy.
B2.4 A lack of resources for data labelling, curation, and cleansing slows deployment timelines well beyond the pilot stage.
**Solutions (SES: 1-5):**
B2.S1 Rolling out an enterprise data governance programme that sets clear, organisation-wide standards for data quality, classification, and access for GenAI use.
B2.S2 Applying privacy-by-design principles, including anonymisation and pseudonymisation, to broaden the range of data that can safely be used with GenAI.
B2.S3 Deploying automated data quality pipelines (profiling, labelling, cleansing) to keep data consistently fit for purpose in live GenAI operations.
B2.S4 Negotiating data sovereignty agreements with vendors and clients to establish clear boundaries on how data can be used in GenAI workloads.

**B3, Integration Complexity and Technical Debt**
*This construct covers the technical challenges of connecting GenAI with existing infrastructure, ITSM workflows, legacy applications, and multi-client environments.*
**Barriers (BSS: 1-5):**
B3.1 Legacy system architectures make it technically complex and costly to integrate GenAI into our existing service workflows.

B3.2 Accumulated technical debt significantly increases the time, cost, and risk of scaling GenAI from pilot to full deployment.
B3.3 Multi-client data requirements, such as tenant isolation and segregated environments, add significant complexity when scaling GenAI workflows.
B3.4 Weak API management capabilities make it difficult to embed GenAI reliably into standardised, repeatable service delivery processes.
**Solutions (SES: 1-5):**
B3.S1 Adopting a modular, API-first integration architecture that decouples GenAI components from legacy systems, reducing complexity and dependency.
B3.S2 Using middleware orchestration layers to manage handoffs between GenAI services and existing systems, without requiring full legacy replacement.
B3.S3 Developing phased integration roadmaps with explicit architecture readiness assessments before each stage of GenAI rollout.

**B4, Security Vulnerabilities and Intellectual Property Risks**
*This construct covers data leakage risks, prompt injection vulnerabilities, IP ownership uncertainty, and compliance obligations arising from GenAI use in client engagements.*
**Barriers (BSS: 1-5):**
B4.1 Prompt injection vulnerabilities create serious security risks when GenAI is deployed in contexts that handle sensitive client data.
B4.2 The risk of unintentional data leakage, such as client data exposed through model training or inference, is a significant obstacle to scaling GenAI.
B4.3 Unclear IP ownership of GenAI-generated outputs (e.g., code, documentation, service deliverables) creates legal uncertainty that gets in the way of wider adoption.
B4.4 Existing vendor contracts provide insufficient IP protection and data sovereignty guarantees for how our sector uses GenAI.
**Solutions (SES: 1-5):**
B4.S1 Embedding security-by-design principles that incorporate GenAI-specific controls, at the prompt, data, and model levels, from the outset of any deployment.
B4.S2 Applying data anonymisation and tokenisation to reduce the risk of sensitive client information being exposed through GenAI interactions.
B4.S3 Developing vendor agreements with specific IP clauses and data residency commitments tailored to the risks of GenAI-generated outputs.

**B5, AI Infrastructure Readiness and Scalability**
*This construct covers the compute infrastructure, cloud architecture maturity, and platform capabilities needed to scale GenAI workloads beyond controlled pilots.*
**Barriers (BSS: 1-5):**
B5.1 Our current cloud infrastructure and compute resources are not sufficient to support GenAI workloads beyond controlled pilot environments.
B5.2 The cost of scaling AI infrastructure (compute, storage, networking) to operational levels is prohibitive for our organisation.
B5.3 Gaps in cloud architecture maturity, such as limited containerisation or missing auto-scaling, prevent us from running GenAI reliably at enterprise scale.
B5.4 Relying on a single cloud or AI infrastructure provider creates scalability constraints and concentration risk for enterprise GenAI deployment.
**Solutions (SES: 1-5):**
B5.S1 Investing in cloud-native AI infrastructure, including auto-scaling, containerised deployment, and managed AI services, to scale GenAI workloads cost-effectively.
B5.S2 Adopting hybrid deployment models that let organisations balance on-premise and cloud-based GenAI infrastructure based on sensitivity and performance needs.
B5.S3 Using managed AI platform services from specialist providers to reduce the internal infrastructure management burden and speed up deployment readiness.

**B6, Relative Advantage Clarity and Value Demonstration**
*This construct covers the difficulty of making a clear, measurable business case for scaling GenAI beyond pilots, and the challenge of connecting GenAI outputs to KPIs, SLAs, and client value.*
**Barriers (BSS: 1-5):**
B6.1 We struggle to build a clear, quantifiable business case for scaling GenAI beyond pilots, the value proposition remains hard to pin down.
B6.2 It is difficult to connect GenAI outcomes, such as productivity gains or quality improvements, to billable value, SLA improvements, or risk reduction metrics.

B6.3 Confusion about what GenAI actually is, versus other AI or automation tools, creates mixed signals in investment decisions and stakeholder conversations.
B6.4 Without robust ROI measurement frameworks, it is hard to justify continued resource allocation once initial pilot excitement fades.
**Solutions (SES: 1-5):**
B6.S1 Developing ROI frameworks and value-tracking dashboards that link GenAI outputs directly to service KPIs, cost savings, and client outcomes.

B6.S2 Defining role-specific GenAI use cases with clear benefit hypotheses and success measures before committing to production investment.

B6.S3 Establishing shared GenAI definitions and classification frameworks that align expectations and investment logic across technical and business stakeholders.

**Section C: Organisation Barriers and Solutions**
This section covers internal organisational factors, including leadership, culture, skills, governance, and resources, that affect an organisation's ability to scale GenAI beyond pilots.
This section asks you to assess barriers and solutions across Organisation dimension using two rating scales:

- Barrier Severity Scale (BSS): How much does each item hold back GenAI scaling in your organisation, or across the IT services sector more broadly?
  1 = Not a barrier | 2 = Minor barrier | 3 = Moderate barrier | 4 = Significant barrier | 5 = Critical barrier
- Solution Effectiveness Scale (SES): How effective do you think each proposed solution would be in addressing the corresponding barrier?
  1 = Not effective | 2 = Slightly effective | 3 = Moderately effective | 4 = Highly effective | 5 = Extremely effective

Please base your answers on your direct professional experience wherever possible.

For barriers, you may also draw on your broader experience and observations of the IT services sector where relevant, particularly where you have not encountered the issue directly but have a clear understanding of its impact.

For solutions, please respond only where you have sufficient familiarity or understanding of the approach. If you are not familiar with a particular solution, please leave the item unanswered rather than providing an estimate.

**C1, Leadership Commitment and Strategic Alignment**
*This construct covers the degree to which senior leaders actively champion GenAI, allocate resources, and embed AI goals within the organisation's strategy.*
**Barriers (BSS: 1-5):**
C1.1 Weak C-suite sponsorship of GenAI means programmes tend to stall between a successful pilot and a broader organisational commitment.
C1.2 GenAI is not sufficiently embedded in the organisation's wider business strategy, leaving AI teams without a clear mandate or direction.
C1.3 Senior leaders lack enough AI literacy to make well-informed decisions about GenAI investment, risk, and governance.
C1.4 A focus on short-term financial performance limits the sustained investment
C1.5 Limited tolerance for iterative experimentation slows production scaling.
**Solutions (SES: 1-5):**
# Solution Item
C1.S1 Establishing a formal C-suite AI sponsorship model with a named executive accountable for GenAI scaling outcomes and resource allocation.
C1.S2 Integrating GenAI objectives explicitly into the organisation's strategic planning cycles, OKRs, and annual business roadmaps.
C1.S3 Delivering executive AI literacy programmes that give leaders the knowledge to evaluate GenAI risks and opportunities without relying solely on technical advisors.

**C2, Organisational Readiness and Change Capacity**
*This construct covers the structural, cultural, and process-level readiness of the organisation to manage the scale of change that enterprise GenAI deployment involves.*
**Barriers (BSS: 1-5):**
C2.1 Cultural resistance to AI, including fears about job displacement or distrust of AI outputs, limits our ability to embed GenAI in everyday operations.

C2.2 We lack the structured change management capability to redesign workflows, roles, and processes around production-grade GenAI.
C2.3 The breadth of GenAI-related change, affecting multiple teams, roles, and processes at once, exceeds our current change management capacity.
C2.4 Low AI readiness among staff and a weak internal AI culture reduce the organisation's willingness to adopt GenAI tools at scale.
**Solutions (SES: 1-5):**
C2.S1 Running structured change management programmes (e.g., Kotter, PROSCI ( Professional Science (change management methodology brand) ADKAR (Awareness, Desire, Knowledge, Ability, Reinforcement (change management model)) specifically tailored to GenAI-driven transformation.
C2.S2 Conducting employee AI readiness assessments and using the findings to design targeted culture-building and communication initiatives.
C2.S3 Using iterative pilot-to-scale frameworks that allow incremental GenAI integration, building organisational confidence and change capacity over time.

**C3, AI Skills, Literacy, and Capability Gaps**
*This construct covers the availability of AI technical skills and the broader AI literacy needed across the organisation to deploy, govern, and safely use GenAI in production.*
**Barriers (BSS: 1-5):**
C3.1 A shortage of AI engineers, data scientists, and MLOps (Machine Learning Operations) professionals limits our ability to deploy and maintain GenAI systems at scale.
C3.2 Weak prompt engineering skills and limited output evaluation competency among delivery staff create quality and reliability risks in live GenAI operations.
C3.3 Low AI literacy among non-technical staff, such as service managers, account leads, and consultants, limits the safe and effective use of GenAI tools in everyday work.
C3.4 Intense competition for AI talent makes it difficult to attract and retain the specialist skills needed to sustain an enterprise GenAI programme.
**Solutions (SES: 1-5):**
C3.S1 Launching comprehensive AI upskilling and reskilling programmes, with separate tracks for technical roles (engineers, data scientists) and operational roles (delivery staff, managers).
C3.S2 Setting up an AI Centre of Excellence (CoE) to centralise GenAI expertise, develop standards, and spread capability across the organisation.
C3.S3 Pursuing strategic AI talent acquisition through targeted hiring and partnerships with universities or AI vendors to close internal capability gaps.
C3.S4 Developing role-based AI training that equips each team with the specific GenAI skills and safety awareness relevant to their work.

**C4, Governance Structures, Accountability, and AI Policy Maturity**
*This construct covers the formal governance arrangements, AI policies, oversight bodies, risk protocols, and accountability frameworks, that govern GenAI use at enterprise scale.*
**Barriers (BSS: 1-5):**
C4.1 We lack a mature AI governance framework with clear policies, accountability structures, and risk management protocols for GenAI deployment decisions.
C4.2 Without clear AI policy, delivery teams end up using GenAI informally or in unsanctioned ways, creating inconsistent risk management and potential compliance exposure.
C4.3 Unclear decision rights and accountability gaps, who approves, monitors, and owns GenAI outputs, make it difficult to move from pilot to governed production deployment.
C4.4 AI risk classification and review processes are not developed enough to provide consistent oversight of GenAI deployment decisions across the organisation.
**Solutions (SES: 1-5):**
C4.S1 Adopting a formal AI governance framework (e.g., NIST AI RMF, ISO/IEC 42001, or a proprietary model) with explicit accountability mapping and risk classification.
C4.S2 Developing and enforcing enterprise-wide AI usage policies that define approved use cases, required safeguards, and approval processes for production deployment.
C4.S3 Setting up an ethics review board or AI oversight committee with cross-functional representation to govern high-risk GenAI deployment decisions.
C4.S4 Implementing a structured GenAI risk classification and review process that requires a documented risk assessment before any new production use case is approved.

**C5, Resource Availability and Investment Capacity**
*This construct covers the financial, human, and technological resources the organisation can commit to GenAI scaling, including appetite for iterative investment before returns materialise.*

**Barriers (BSS: 1-5):**
C5.1 Insufficient funding for GenAI scaling, beyond initial pilot budgets, prevents us from sustaining the investment required to reach full production deployment.
C5.2 Competing investment priorities and internal resource conflicts reduce the funding and capacity available for GenAI programmes.
C5.3 Low tolerance for the iterative experimentation and delayed returns that GenAI scaling involves makes it difficult to maintain investment commitment beyond early pilots.
C5.4 Key staff, architects, engineers, change managers, simply don't have enough time or bandwidth to dedicate to GenAI deployment alongside existing operational commitments.
**Solutions (SES: 1-5):**
C5.S1 Ring-fencing dedicated GenAI budgets, kept separate from general IT spend, to provide sustained, predictable funding for scaling programmes.
C5.S2 Using portfolio prioritisation frameworks and phased investment roadmaps to sequence GenAI investments by expected ROI, strategic value, and execution readiness.
C5.S3 Applying a minimum viable AI (MVA) approach, starting with the smallest viable deployment that can demonstrate tangible value and justify further investment.

**C6, Human-AI Collaboration Design and Over-Reliance Risk**
*This construct covers the challenge of designing effective human-AI working arrangements that preserve appropriate human judgement, especially in quality-sensitive delivery roles.*
**Barriers (BSS: 1-5):**
C6.1 Unclear boundaries between AI-assisted and fully human-performed tasks create quality, accountability, and liability risks in our delivery model.
C6.2 If staff rely too heavily on GenAI outputs without applying sufficient critical judgement, there is a real risk of eroding professional competence and service quality over time.
C6.3 Inadequate protocols for escalating from AI-assisted tasks to human decision-making leave gaps in oversight and accountability within live GenAI workflows.
C6.4 GenAI interface design does not adequately support safe, ergonomic collaboration between humans and AI, leading to misuse or inappropriate trust in automated outputs.
**Solutions (SES: 1-5):**
C6.S1 Defining explicit human oversight frameworks for each GenAI use case, specifying which tasks require human review, when escalation applies, and who has override authority.
C6.S2 Implementing task guidance strategies that help staff decide when to rely on GenAI, when to seek human review, and when to override AI suggestions.
C6.S3 Applying cognitive ergonomics principles to GenAI interface design, to reduce automation bias and encourage users to actively evaluate AI outputs rather than accept them passively.

**Section D: Environment Barriers and Solutions**
This section covers the external factors, regulatory, competitive, market, and stakeholder, that shape GenAI scaling decisions in IT service organisations. This section asks you to assess barriers and solutions across Environmental dimension using two rating scales:

- Barrier Severity Scale (BSS): How much does each item hold back GenAI scaling in your organisation, or across the IT services sector more broadly?
  1 = Not a barrier | 2 = Minor barrier | 3 = Moderate barrier | 4 = Significant barrier | 5 = Critical barrier
- Solution Effectiveness Scale (SES): How effective do you think each proposed solution would be in addressing the corresponding barrier?
  1 = Not effective | 2 = Slightly effective | 3 = Moderately effective | 4 = Highly effective | 5 = Extremely effective

Please base your answers on your direct professional experience wherever possible.

For barriers, you may also draw on your broader experience and observations of the IT services sector where relevant, particularly where you have not encountered the issue directly but have a clear understanding of its impact.

For solutions, please respond only where you have sufficient familiarity or understanding of the approach. If you are not familiar with a particular solution, please leave the item unanswered rather than providing an estimate.

**D1, Regulatory Frameworks, Compliance Requirements, and Legal Liability**
*This construct covers external legal and regulatory obligations (e.g., EU AI Act, GDPR, sector mandates) and the contractual liability uncertainty that comes with using GenAI in client-facing deployments.*
**Barriers (BSS: 1-5):**

D1.1 Uncertainty about current and emerging AI regulations, such as the EU AI Act, GDPR, or sector-specific mandates, makes it risky to commit to large-scale GenAI deployment.
D1.2 Ambiguity about who is legally responsible when GenAI outputs cause errors or harm in client engagements keeps many organisations in the relative safety of pilot phases.
D1.3 The burden of complying with multiple, potentially conflicting regulations across jurisdictions or client sectors is too high to manage effectively at scale.
D1.4 Regulatory audit and documentation requirements for AI systems currently exceed our capacity to demonstrate compliant GenAI usage in production.
**Solutions (SES: 1-5):**
D1.S1 Embedding compliance-by-design into GenAI deployment processes so that regulatory requirements are built into architecture and workflow from the start, not retrofitted afterwards.
D1.S2 Developing contractual liability frameworks, covering user-centric, balanced, and provider-centric liability models, for GenAI-enabled service contracts.
D1.S3 Establishing a regulatory monitoring programme that tracks AI legislation developments and engages proactively with regulators to reduce compliance uncertainty.

**D2, Ethical Concerns, Bias, and Legitimacy Expectations**
*This construct covers societal and stakeholder expectations around fairness, transparency, and accountability in AI decision-making, including the reputational risks of GenAI use.*
**Barriers (BSS: 1-5):**
D2.1 Concerns about bias, unfairness, or discriminatory outputs from GenAI systems create reputational and ethical risks that limit deployment in client-facing contexts.
D2.2 External expectations around AI transparency and accountability, from clients, regulators, and civil society, exceed what current GenAI systems can credibly demonstrate.
D2.3 The absence of standardised, auditable bias detection tools makes it difficult to demonstrate ethical compliance in live deployments.
D2.4 Insufficient ethical leadership within our organisation means GenAI deployment decisions are made without adequate ethical review or clear accountability.
**Solutions (SES: 1-5):**
D2.S1 Developing and applying an ethical AI framework that defines fairness criteria, transparency obligations, and accountability mechanisms for GenAI use in client-facing work.
D2.S2 Making standardised bias detection and evaluation a mandatory pre-deployment step for each new GenAI production use case.
D2.S3 Establishing an AI ethics board with board-level or senior leadership sponsorship to provide independent oversight of high-stakes GenAI deployment decisions.

**D3, Competitive Pressure and Adoption Urgency**
*This construct covers the competitive market dynamics that simultaneously accelerate GenAI adoption intent and create risks of deploying before the organisation is genuinely ready.*
**Barriers (BSS: 1-5):**
D3.1 Competitive pressure from peers who appear to be moving faster on GenAI creates urgency to deploy before we are adequately prepared.
D3.2 The need to showcase GenAI capabilities in client pitches and market positioning leads to deployment commitments that outpace our actual operational readiness.
D3.3 The pace of change in the GenAI market makes it hard to commit to stable, long-term deployment strategies without the risk of rapid technology obsolescence.
**Solutions (SES: 1-5):**
D3.S1 Adopting deliberate adoption sequencing strategies that explicitly balance competitive speed with execution quality and genuine readiness.
D3.S2 Establishing a competitive intelligence programme that tracks industry GenAI adoption patterns, supporting more calibrated, evidence-based timing decisions.
D3.S3 Building a market positioning narrative around responsible, reliable GenAI delivery, using quality and governance, rather than speed alone, as a competitive differentiator.

**D4, Customer and Stakeholder Pressure**
*This construct covers client demands for AI-enabled service innovation, auditability requirements, and broader market expectations that create both adoption pressure and governance obligations.*
**Barriers (BSS: 1-5):**
D4.1 Client demands for GenAI-enabled services push us to deploy capabilities before we have established the governance and quality controls needed for responsible production use.

D4.2 Clients' auditability and explainability requirements for AI-assisted services exceed what current GenAI systems, or our deployment practices, can currently provide.
D4.3 Divergent expectations about GenAI capabilities, between client business sponsors and their technical or legal teams, create contract management and delivery complexity.

**Solutions (SES: 1-5):**

D4.S1 Developing AI partnership strategies that involve clients in co-designing GenAI governance, acceptable use policies, and auditability frameworks from the start of any deployment.
D4.S2 Implementing AI-enabled service assurance frameworks that give clients verifiable evidence of GenAI output quality, governance compliance, and risk management.
D4.S3 Creating structured stakeholder communication plans that proactively manage expectations about GenAI capabilities, limitations, and deployment timelines, both with clients and internally.

**D5, Vendor Ecosystems and AI Platform Maturity**

*This construct covers the availability, reliability, compliance-readiness, and dependency risks of commercial GenAI platforms, foundation model providers, and implementation partners.*

**Barriers (BSS: 1-5):**

D5.1 Available commercial GenAI platforms are not yet enterprise-ready, lacking sufficient security controls, SLA guarantees, or compliance certifications, for use in our client-facing work.
D5.2 Heavy dependence on a single GenAI vendor or foundation model creates concentration risk and limits our ability to adapt to regulatory changes, vendor policy shifts, or model deprecation.
D5.3 GenAI vendor data sovereignty practices and contract terms do not adequately protect client data or meet the compliance requirements that apply to our delivery context.
D5.4 The rapid pace of change in the GenAI platform market makes it difficult to build stable, maintainable deployments on current vendor offerings.

**Solutions (SES: 1-5):**

D5.S1 Conducting structured vendor assessment and selection processes that evaluate enterprise readiness, compliance posture, data sovereignty controls, and long-term viability before signing production-scale contracts.
D5.S2 Pursuing a multi-vendor ecosystem strategy that diversifies GenAI platform dependencies and reduces the risk of over-reliance on any single provider.
D5.S3 Negotiating enhanced vendor agreements that include enterprise-grade SLAs, data sovereignty commitments, compliance certifications, and model stability guarantees.

**Section E: Overall Assessment of the GenAI Execution Gap**

This section asks for your overall view of the GenAI execution gap and the relative contribution of the three TOE dimensions to the challenge. Please respond based on your experience and professional judgement.

*E1. How significant is the gap between GenAI pilot experimentation and enterprise-scale deployment in your organisation or the sector you work in?*

- No meaningful gap (pilots transition smoothly to production)
- Minor gap (some friction, but manageable with existing processes)
- Moderate gap (notable barriers exist that require deliberate effort to address)
- Significant gap (substantial barriers that consistently stall GenAI scaling)
- Critical gap (most GenAI initiatives remain in pilot phase; enterprise-scale deployment is rare)
- None of the above

*E2. How would you rate the current state of solutions available to bridge the GenAI execution gap in IT services?*

- 1, Solutions are largely absent or underdeveloped
- 2, Some solutions exist but are not yet widely implemented
- 3, Solutions are available but unevenly adopted
- 4, Effective solutions are emerging and gaining traction
- 5, Mature, validated solutions are widely adopted and demonstrably effective
- None of the above

*E3. Which of the following best describes where your organisation expects to be in 12-24 months?* (Select one)

- Still primarily in pilot or experimentation phases
- Selective production deployment in 1-2 limited use cases
- Meaningful progress toward multi-use-case production deployment
- Enterprise-scale GenAI operations substantially in place
- Already at enterprise scale and actively optimizing
- None of the above

**E4. TOE Dimension Weight Allocation**

Please distribute *100 points* across the three TOE dimensions below to reflect how much each one contributes to the GenAI execution gap in your organisation or the sector. Allocate more points to dimensions you see as more consequential. Points must sum to exactly 100.

***Note:*** *This captures your holistic judgement about relative importance, separate from the individual barrier ratings above. If you believe all three dimensions contribute roughly equally, allocating around 33-34 points each is a valid and meaningful response.*

| TOE Dimension | What it covers (examples) | Points allocated (total = 100) |
|---|---|---|
| **Technology** | Model reliability, data quality, integration complexity, infrastructure readiness, security and IP risks, relative advantage clarity | ___ |
| **Organisation** | Leadership commitment, change capacity and culture, AI skills and literacy, governance maturity, resource availability, human-AI collaboration design | ___ |
| **Environment** | Regulatory and legal uncertainty, ethical expectations, competitive pressure, client and stakeholder demands, vendor ecosystem maturity | ___ |
| **Total** | | **= 100** |

**E5.** *Stage of Stalling, Where does your organisation most commonly stall when trying to scale GenAI from pilot to production?* (Select one)

- We have not yet moved beyond early exploration (e.g., experimentation or initial investigation stage)
- At the pilot stage itself (pilots don't produce usable or reliable results)
- At the pilot-to-production transition (results are promising, but scaling is consistently blocked)
- In early production (deployed in 1-2 cases but unable to extend further)
- During mid-scaling (multi-use-case deployment underway but plateauing or losing momentum)
- At governance or compliance review (technically ready but approval processes block release)
- We haven't experienced significant stalling, scaling is progressing broadly as planned
- *If your organisation has not yet moved beyond early exploration or pilot stages, please skip this question.*

**E6. Solutions Attempted and Feasibility Assessment**

For each solution below, please: (a) tick whether your organisation has actively tried or implemented it in the context of GenAI scaling, and (b) if yes, rate how feasible it would be to scale across the enterprise. Please choose between
1 = Extremely difficult | 2 = Difficult | 3 = Moderately feasible | 4 = Mostly Feasible | 5 = Highly feasible at Enterprise scale

Leave the feasibility rating blank if the solution has not been attempted.

- AI governance framework or enterprise AI policy development
- AI Centre of Excellence (CoE) or Centre of Competency (CoC)
- Enterprise-wide AI upskilling, reskilling, or AI literacy programme
- Structured change management programme for GenAI transformation
- RAG / grounding techniques to reduce hallucination in production outputs
- Human-in-the-loop (HITL) validation workflows for GenAI output review
- Compliance-by-design architecture to meet regulatory requirements
- ROI measurement framework or GenAI value-tracking dashboard
- Multi-vendor GenAI platform or diversified ecosystem strategy
- Formal C-suite AI sponsorship or named executive champion programme
- None of the above / we are at the early stages of exploring solutions
- Other (please specify): _____

**Section F: Open-Ended Questions**
This section invites you to share qualitative insights that extend beyond the structured items above. Your responses will provide rich contextual data that complements the quantitative survey results and informs the subsequent interview guide
**F3a. Missed Barrier**
Are there barriers to enterprise-scale GenAI deployment that you consider important but that weren't captured in this survey? If so, please describe the barrier briefly and rate its severity.
Barrier Severity Rating (BSS: 1–5)

Barrier Severity Scale (BSS): Rate the extent to which each item represents a barrier to scaling GenAI beyond pilots in your organisation or the IT services sector more broadly.
1 = Not a barrier at all | 2 = Minor barrier | 3 = Moderate barrier | 4 = Significant barrier | 5 = Critical barrier
*F3b. Any solutions we missed?*
Are there solutions to GenAI scaling challenges that you consider important but that weren't covered in this survey? If so, please describe the solution briefly and rate its effectiveness.
Solution Effectiveness Rating (SES: 1–5)
Solution Effectiveness Scale (SES): Rate how effective you believe each proposed solution is in addressing the corresponding barrier.
1 = Not effective | 2 = Slightly effective | 3 = Moderately effective | 4 = Highly effective | 5 = Extremely effective
*[Open text field, Describe the solution: recommended 30-100 words]*
Solution Effectiveness Rating (SES: 1-5): ___
*Thank You*
Thank you for your time and contribution to this research. Your professional insights are invaluable in developing an empirically grounded understanding of the GenAI execution gap in IT service organisations. If you have any questions about this research, please contact the principal researcher *Sara Mansouri at samn23@student.bth.se or Shefali Rattan at shrt23@student.bth.se* This survey is conducted in partial fulfilment of the requirements for the MBA degree at *Blekinge Institute of Technology*.

# Appendix E: Fuzzy Delphi Method analysis.

This appendix contains the tables referenced in Chapter 6. The tables are presented separately to preserve readability in the main chapter while maintaining the numerical audit trail of the Fuzzy Delphi Method analysis.

Table E.1: Survey/FDM Respondent Profile Summary

| Profile Dimension | Summary |
|---|---|
| **Number of valid survey/FDM respondents** | 20 |
| **Industry context** | IT services and IT-enabled organisations, including system integrators, managed service providers, IT consulting firms, cloud and platform service providers, and SaaS organisations |
| **Dominant professional roles** | AI/GenAI Programme or Product Manager: 10 (50%); Enterprise Architect / Solution Architect: 3 (15%); VP/Director of Technology or Digital Transformation: 2 (10%); Software Engineer / AI-ML Engineer: 2 (10%); C-Suite Executive (CEO, CTO, CIO, CDO, CAIO): 2 (10%); IT Service Delivery Manager / Operations Lead: 1 (5%) |
| **Professional experience** | More than 15 years: 10 (50%); 11–15 years: 6 (30%); 6–10 years: 3 (15%); 3–5 years: 1 (5%) |
| **Organisation size** | 2,000–9,999 employees: 9 (45%); 10,000 or more: 8 (40%); 100–499 employees: 2 (10%); Fewer than 100 employees: 1 (5%) |
| **Primary organisational focus** | Managed IT Services: 12 (60%); IT Consulting and Advisory: 11 (55%); Data Analytics and AI Services: 9 (45%); Cloud Services and Infrastructure: 7 (35%); Software Development and Engineering: 6 (30%); Cybersecurity Services: 6 (30%) |
| **Geographic headquarters** | Europe (EU/EEA): 10 (50%); North America: 5 (25%); Asia-Pacific: 4 (20%); United Kingdom: 1 (5%) |
| **GenAI application context** | Both internal and client-facing use equally: 16 (80%); Primarily internal: 2 (10%); Primarily client-facing: 2 (10%) |
| **Current GenAI adoption stage** | Enterprise-scale deployment: 11 (55%); Selective deployment in limited production contexts: 6 (30%); Pilot phase: 3 (15%) |
| **Involvement in GenAI decisions** | Closely involved (key contributor): 10 (50%); Directly responsible (primary decision-maker or programme owner): 8 (40%); Partially involved: 1 (5%); Limited involvement: 1 (5%) |

Table E.2: Item-Level FDM Results for All 122 Barrier and Solution Items

| Item | TOE | Construct | Type | Avg $\bar{d}$ | Decision | Avg Crisp | Global Rank | Interpretation |
|---|---|---|---|---|---|---|---|---|
| B1.1 | Technology | B1 | Barrier | 0.228 | REMOVE | 0.509 | 48 | Significant barrier |
| B1.2 | Technology | B1 | Barrier | 0.227 | REMOVE | 0.512 | 43 | Significant barrier |
| B1.3 | Technology | B1 | Barrier | 0.199 | KEEP | 0.525 | 38 | Significant barrier |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B1.4 | Technology | B1 | Barrier | 0.240 | REMOVE | 0.456 | 64 | Moderate barrier |
| B1.S1 | Technology | B1S | Solution | 0.144 | KEEP | 0.700 | 21 | Effective solution |
| B1.S2 | Technology | B1S | Solution | 0.104 | KEEP | 0.759 | 2 | Highly effective solution |
| B1.S3 | Technology | B1S | Solution | 0.120 | KEEP | 0.725 | 7 | Effective solution |
| B1.S4 | Technology | B1S | Solution | 0.110 | KEEP | 0.706 | 17 | Effective solution |
| B2.1 | Technology | B2 | Barrier | 0.231 | REMOVE | 0.681 | 2 | Significant barrier |
| B2.2 | Technology | B2 | Barrier | 0.162 | KEEP | 0.681 | 3 | Significant barrier |
| B2.3 | Technology | B2 | Barrier | 0.211 | REMOVE | 0.637 | 8 | Significant barrier |
| B2.4 | Technology | B2 | Barrier | 0.228 | REMOVE | 0.559 | 21 | Significant barrier |
| B2.S1 | Technology | B2S | Solution | 0.153 | KEEP | 0.647 | 47 | Effective solution |
| B2.S2 | Technology | B2S | Solution | 0.161 | KEEP | 0.697 | 22 | Effective solution |
| B2.S3 | Technology | B2S | Solution | 0.160 | KEEP | 0.622 | 55 | Effective solution |
| B2.S4 | Technology | B2S | Solution | 0.167 | KEEP | 0.697 | 23 | Effective solution |
| B3.1 | Technology | B3 | Barrier | 0.184 | KEEP | 0.644 | 7 | Significant barrier |
| B3.2 | Technology | B3 | Barrier | 0.194 | KEEP | 0.653 | 5 | Significant barrier |
| B3.3 | Technology | B3 | Barrier | 0.165 | KEEP | 0.553 | 25 | Significant barrier |
| B3.4 | Technology | B3 | Barrier | 0.187 | KEEP | 0.562 | 20 | Significant barrier |
| B3.S1 | Technology | B3S | Solution | 0.135 | KEEP | 0.703 | 19 | Effective solution |
| B3.S2 | Technology | B3S | Solution | 0.151 | KEEP | 0.656 | 46 | Effective solution |
| B3.S3 | Technology | B3S | Solution | 0.148 | KEEP | 0.691 | 28 | Effective solution |
| B4.1 | Technology | B4 | Barrier | 0.217 | REMOVE | 0.656 | 4 | Significant barrier |
| B4.2 | Technology | B4 | Barrier | 0.244 | REMOVE | 0.688 | 1 | Significant barrier |
| B4.3 | Technology | B4 | Barrier | 0.231 | REMOVE | 0.613 | 11 | Significant barrier |
| B4.4 | Technology | B4 | Barrier | 0.256 | REMOVE | 0.569 | 17 | Significant barrier |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B4.S1 | Technology | B4S | Solution | 0.146 | KEEP | 0.753 | 3 | Highly effective solution |
| B4.S2 | Technology | B4S | Solution | 0.190 | KEEP | 0.684 | 35 | Effective solution |
| B4.S3 | Technology | B4S | Solution | 0.219 | REMOVE | 0.647 | 48 | Effective solution |
| B5.1 | Technology | B5 | Barrier | 0.234 | REMOVE | 0.450 | 66 | Moderate barrier |
| B5.2 | Technology | B5 | Barrier | 0.258 | REMOVE | 0.566 | 18 | Significant barrier |
| B5.3 | Technology | B5 | Barrier | 0.269 | REMOVE | 0.481 | 59 | Moderate barrier |
| B5.4 | Technology | B5 | Barrier | 0.219 | REMOVE | 0.472 | 62 | Moderate barrier |
| B5.S1 | Technology | B5S | Solution | 0.094 | KEEP | 0.688 | 32 | Effective solution |
| B5.S2 | Technology | B5S | Solution | 0.141 | KEEP | 0.713 | 12 | Effective solution |
| B5.S3 | Technology | B5S | Solution | 0.118 | KEEP | 0.694 | 25 | Effective solution |
| B6.1 | Technology | B6 | Barrier | 0.205 | REMOVE | 0.516 | 40 | Significant barrier |
| B6.2 | Technology | B6 | Barrier | 0.260 | REMOVE | 0.478 | 61 | Moderate barrier |
| B6.3 | Technology | B6 | Barrier | 0.178 | KEEP | 0.406 | 67 | Moderate barrier |
| B6.4 | Technology | B6 | Barrier | 0.249 | REMOVE | 0.512 | 44 | Significant barrier |
| B6.S1 | Technology | B6S | Solution | 0.148 | KEEP | 0.691 | 29 | Effective solution |
| B6.S2 | Technology | B6S | Solution | 0.156 | KEEP | 0.709 | 13 | Effective solution |
| B6.S3 | Technology | B6S | Solution | 0.190 | KEEP | 0.694 | 26 | Effective solution |
| C1.1 | Organisation | C1 | Barrier | 0.251 | REMOVE | 0.606 | 13 | Significant barrier |
| C1.2 | Organisation | C1 | Barrier | 0.243 | REMOVE | 0.609 | 12 | Significant barrier |
| C1.3 | Organisation | C1 | Barrier | 0.229 | REMOVE | 0.581 | 14 | Significant barrier |
| C1.4 | Organisation | C1 | Barrier | 0.299 | REMOVE | 0.544 | 30 | Significant barrier |
| C1.5 | Organisation | C1 | Barrier | 0.247 | REMOVE | 0.559 | 22 | Significant barrier |
| C1.S1 | Organisation | C1S | Solution | 0.200 | KEEP | 0.625 | 52 | Effective solution |
| C1.S2 | Organisation | C1S | Solution | 0.164 | KEEP | 0.719 | 9 | Effective solution |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| C1.S3 | Organisation | C1S | Solution | 0.112 | KEEP | 0.716 | 11 | Effective solution |
| C2.1 | Organisation | C2 | Barrier | 0.184 | KEEP | 0.491 | 58 | Moderate barrier |
| C2.2 | Organisation | C2 | Barrier | 0.228 | REMOVE | 0.516 | 41 | Significant barrier |
| C2.3 | Organisation | C2 | Barrier | 0.172 | KEEP | 0.503 | 51 | Significant barrier |
| C2.4 | Organisation | C2 | Barrier | 0.197 | KEEP | 0.506 | 50 | Significant barrier |
| C2.S1 | Organisation | C2S | Solution | 0.166 | KEEP | 0.644 | 50 | Effective solution |
| C2.S2 | Organisation | C2S | Solution | 0.175 | KEEP | 0.684 | 36 | Effective solution |
| C2.S3 | Organisation | C2S | Solution | 0.115 | KEEP | 0.747 | 4 | Effective solution |
| C3.1 | Organisation | C3 | Barrier | 0.210 | REMOVE | 0.537 | 32 | Significant barrier |
| C3.2 | Organisation | C3 | Barrier | 0.184 | KEEP | 0.497 | 57 | Moderate barrier |
| C3.3 | Organisation | C3 | Barrier | 0.252 | REMOVE | 0.512 | 45 | Significant barrier |
| C3.4 | Organisation | C3 | Barrier | 0.240 | REMOVE | 0.500 | 55 | Significant barrier |
| C3.S1 | Organisation | C3S | Solution | 0.161 | KEEP | 0.709 | 14 | Effective solution |
| C3.S2 | Organisation | C3S | Solution | 0.164 | KEEP | 0.719 | 10 | Effective solution |
| C3.S3 | Organisation | C3S | Solution | 0.190 | KEEP | 0.694 | 27 | Effective solution |
| C3.S4 | Organisation | C3S | Solution | 0.159 | KEEP | 0.678 | 40 | Effective solution |
| C4.1 | Organisation | C4 | Barrier | 0.259 | REMOVE | 0.525 | 39 | Significant barrier |
| C4.2 | Organisation | C4 | Barrier | 0.249 | REMOVE | 0.512 | 46 | Significant barrier |
| C4.3 | Organisation | C4 | Barrier | 0.234 | REMOVE | 0.553 | 24 | Significant barrier |
| C4.4 | Organisation | C4 | Barrier | 0.239 | REMOVE | 0.503 | 52 | Significant barrier |
| C4.S1 | Organisation | C4S | Solution | 0.221 | REMOVE | 0.669 | 44 | Effective solution |
| C4.S2 | Organisation | C4S | Solution | 0.210 | REMOVE | 0.634 | 51 | Effective solution |
| C4.S3 | Organisation | C4S | Solution | 0.232 | REMOVE | 0.625 | 53 | Effective solution |
| C4.S4 | Organisation | C4S | Solution | 0.213 | REMOVE | 0.678 | 42 | Effective solution |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| C5.1 | Organisation | C5 | Barrier | 0.259 | REMOVE | 0.528 | 37 | Significant barrier |
| C5.2 | Organisation | C5 | Barrier | 0.269 | REMOVE | 0.512 | 47 | Significant barrier |
| C5.3 | Organisation | C5 | Barrier | 0.237 | REMOVE | 0.550 | 27 | Significant barrier |
| C5.4 | Organisation | C5 | Barrier | 0.278 | REMOVE | 0.581 | 15 | Significant barrier |
| C5.S1 | Organisation | C5S | Solution | 0.156 | KEEP | 0.709 | 15 | Effective solution |
| C5.S2 | Organisation | C5S | Solution | 0.161 | KEEP | 0.709 | 16 | Effective solution |
| C5.S3 | Organisation | C5S | Solution | 0.206 | REMOVE | 0.672 | 43 | Effective solution |
| C6.1 | Organisation | C6 | Barrier | 0.218 | REMOVE | 0.503 | 53 | Significant barrier |
| C6.2 | Organisation | C6 | Barrier | 0.276 | REMOVE | 0.566 | 19 | Significant barrier |
| C6.3 | Organisation | C6 | Barrier | 0.269 | REMOVE | 0.516 | 42 | Significant barrier |
| C6.4 | Organisation | C6 | Barrier | 0.260 | REMOVE | 0.456 | 65 | Moderate barrier |
| C6.S1 | Organisation | C6S | Solution | 0.198 | KEEP | 0.681 | 38 | Effective solution |
| C6.S2 | Organisation | C6S | Solution | 0.208 | REMOVE | 0.647 | 49 | Effective solution |
| C6.S3 | Organisation | C6S | Solution | 0.198 | KEEP | 0.681 | 39 | Effective solution |
| D1.1 | Environment | D1 | Barrier | 0.229 | REMOVE | 0.537 | 33 | Significant barrier |
| D1.2 | Environment | D1 | Barrier | 0.273 | REMOVE | 0.509 | 49 | Significant barrier |
| D1.3 | Environment | D1 | Barrier | 0.220 | REMOVE | 0.500 | 56 | Significant barrier |
| D1.4 | Environment | D1 | Barrier | 0.253 | REMOVE | 0.531 | 35 | Significant barrier |
| D1.S1 | Environment | D1S | Solution | 0.182 | KEEP | 0.684 | 37 | Effective solution |
| D1.S2 | Environment | D1S | Solution | 0.199 | KEEP | 0.688 | 33 | Effective solution |
| D1.S3 | Environment | D1S | Solution | 0.142 | KEEP | 0.691 | 30 | Effective solution |
| D2.1 | Environment | D2 | Barrier | 0.230 | REMOVE | 0.531 | 36 | Significant barrier |
| D2.2 | Environment | D2 | Barrier | 0.229 | REMOVE | 0.466 | 63 | Moderate barrier |
| D2.3 | Environment | D2 | Barrier | 0.265 | REMOVE | 0.559 | 23 | Significant barrier |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| D2.4 | Environment | D2 | Barrier | 0.283 | REMOVE | 0.481 | 60 | Moderate barrier |
| D2.S1 | Environment | D2S | Solution | 0.197 | KEEP | 0.703 | 20 | Effective solution |
| D2.S2 | Environment | D2S | Solution | 0.185 | KEEP | 0.706 | 18 | Effective solution |
| D2.S3 | Environment | D2S | Solution | 0.212 | REMOVE | 0.691 | 31 | Effective solution |
| D3.1 | Environment | D3 | Barrier | 0.237 | REMOVE | 0.550 | 28 | Significant barrier |
| D3.2 | Environment | D3 | Barrier | 0.228 | REMOVE | 0.537 | 34 | Significant barrier |
| D3.3 | Environment | D3 | Barrier | 0.251 | REMOVE | 0.619 | 10 | Significant barrier |
| D3.S1 | Environment | D3S | Solution | 0.213 | REMOVE | 0.625 | 54 | Effective solution |
| D3.S2 | Environment | D3S | Solution | 0.213 | REMOVE | 0.659 | 45 | Effective solution |
| D3.S3 | Environment | D3S | Solution | 0.166 | KEEP | 0.688 | 34 | Effective solution |
| D4.1 | Environment | D4 | Barrier | 0.216 | REMOVE | 0.503 | 54 | Significant barrier |
| D4.2 | Environment | D4 | Barrier | 0.259 | REMOVE | 0.547 | 29 | Significant barrier |
| D4.3 | Environment | D4 | Barrier | 0.226 | REMOVE | 0.553 | 26 | Significant barrier |
| D4.S1 | Environment | D4S | Solution | 0.136 | KEEP | 0.734 | 5 | Effective solution |
| D4.S2 | Environment | D4S | Solution | 0.154 | KEEP | 0.730 | 6 | Effective solution |
| D4.S3 | Environment | D4S | Solution | 0.228 | REMOVE | 0.678 | 41 | Effective solution |
| D5.1 | Environment | D5 | Barrier | 0.279 | REMOVE | 0.544 | 31 | Significant barrier |
| D5.2 | Environment | D5 | Barrier | 0.287 | REMOVE | 0.572 | 16 | Significant barrier |
| D5.3 | Environment | D5 | Barrier | 0.266 | REMOVE | 0.647 | 6 | Significant barrier |
| D5.4 | Environment | D5 | Barrier | 0.240 | REMOVE | 0.634 | 9 | Significant barrier |
| D5.S1 | Environment | D5S | Solution | 0.148 | KEEP | 0.722 | 8 | Effective solution |
| D5.S2 | Environment | D5S | Solution | 0.160 | KEEP | 0.772 | 1 | Highly effective solution |
| D5.S3 | Environment | D5S | Solution | 0.179 | KEEP | 0.697 | 24 | Effective solution |

Note: Global rank is the crisp-based position of each item across all 122 items combined. Separate barrier and solution rankings are provided in Tables E.6 and E.7. KEEP items appear in bold. Items classified as KEEP in Barriers have $\bar{d} \leq 0.20$; all KEEP items in Solutions also meet this criterion. The global rank column uses the original workbook ranking across all 122 items before barrier/solution separation.

Table E.3: Construct-Level Aggregation, Retained Barriers and Solutions

| Aggregate Code | TOE | Construct Name | Type | Items Proposed | Items Retained | Coverage | Mean $\bar{d}$ (retained) | Mean Crisp (retained) | Priority Rank (crisp) | Interpretation |
|---|---|---|---|---|---|---|---|---|---|---|
| **B2** | Technology | Data Quality, Availability and Privacy | Barrier | 4 | 1 | 25.0% | 0.162 | 0.681 | 1 | Significant barrier |
| **B3** | Technology | Integration Complexity and Technical Debt | Barrier | 4 | 4 | 100.0% | 0.183 | 0.603 | 2 | Significant barrier |
| **B1** | Technology | Model Reliability, Hallucination Risk and Explainability | Barrier | 4 | 1 | 25.0% | 0.199 | 0.525 | 3 | Significant barrier |
| **C2** | Organisation | Organisational Readiness and Change Capacity | Barrier | 4 | 3 | 75.0% | 0.184 | 0.500 | 4 | Significant barrier |
| **C3** | Organisation | AI Skills, Literacy and Capability Gaps | Barrier | 4 | 1 | 25.0% | 0.184 | 0.497 | 5 | Moderate barrier |
| **B6** | Technology | Relative Advantage Clarity and Value Demonstration | Barrier | 4 | 1 | 25.0% | 0.178 | 0.406 | 6 | Moderate barrier |
| **B4** | Technology | Security Vulnerabilities and IP Risks | Barrier | 4 | 0 | 0.0% | — | — | — | No retained items |
| **B5** | Technology | AI Infrastructure Readiness and Scalability | Barrier | 4 | 0 | 0.0% | — | — | — | No retained items |
| **C1** | Organisation | Leadership Commitment and Strategic Alignment | Barrier | 5 | 0 | 0.0% | — | — | — | No retained items |
| **C4** | Organisation | Governance Structures and AI Policy Maturity | Barrier | 4 | 0 | 0.0% | — | — | — | No retained items |
| **C5** | Organisation | Resource Availability and Investment Capacity | Barrier | 4 | 0 | 0.0% | — | — | — | No retained items |

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|---|---|---|---|---|---|---|---|---|---|---|
| **C6** | Organisation | Human-AI Collaboration Design and Over-Reliance Risk | Barrier | 4 | 0 | 0.0% | — | — | — | No retained items |
| **D1–D5** | Environment | All Environment Constructs | Barrier | 18 | 0 | 0.0% | — | — | — | No retained items, high cross-jurisdictional variability |
| **D4S** | Environment | Customer and Stakeholder Pressure | Solution | 3 | 2 | 66.7% | 0.145 | 0.732 | 1 | Effective solution |
| **D5S** | Environment | Vendor Ecosystems and AI Platform Maturity | Solution | 3 | 3 | 100.0% | 0.162 | 0.730 | 2 | Effective solution |
| **B1S** | Technology | Model Reliability, Hallucination Risk and Explainability | Solution | 4 | 4 | 100.0% | 0.119 | 0.722 | 3 | Effective solution |
| **B4S** | Technology | Security Vulnerabilities and IP Risks | Solution | 3 | 2 | 66.7% | 0.168 | 0.719 | 4 | Effective solution |
| **C5S** | Organisation | Resource Availability and Investment Capacity | Solution | 3 | 2 | 66.7% | 0.159 | 0.709 | 5 | Effective solution |
| **D2S** | Environment | Ethical Concerns, Bias and Legitimacy | Solution | 3 | 2 | 66.7% | 0.191 | 0.704 | 6 | Effective solution |
| **C3S** | Organisation | AI Skills, Literacy and Capability Gaps | Solution | 4 | 4 | 100.0% | 0.169 | 0.700 | 7 | Effective solution |
| **B5S** | Technology | AI Infrastructure Readiness and Scalability | Solution | 3 | 3 | 100.0% | 0.118 | 0.698 | 8 | Effective solution |
| **B6S** | Technology | Relative Advantage Clarity and Value Demonstration | Solution | 3 | 3 | 100.0% | 0.165 | 0.698 | 9 | Effective solution |
| **C2S** | Organisation | Organisational Readiness and Change Capacity | Solution | 3 | 3 | 100.0% | 0.152 | 0.692 | 10 | Effective solution |

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| **D1S** | Environment | Regulatory Frameworks, Compliance and Legal Liability | Solution | 3 | 3 | 100.0% | 0.174 | 0.688 | 11 | Effective solution |
| **D3S** | Environment | Competitive Pressure and Adoption Urgency | Solution | 3 | 1 | 33.3% | 0.166 | 0.688 | 11 | Effective solution |
| **C1S** | Organisation | Leadership Commitment and Strategic Alignment | Solution | 3 | 3 | 100.0% | 0.159 | 0.687 | 13 | Effective solution |
| **B3S** | Technology | Integration Complexity and Technical Debt | Solution | 3 | 3 | 100.0% | 0.145 | 0.683 | 14 | Effective solution |
| **C6S** | Organisation | Human-AI Collaboration Design and Over-Reliance Risk | Solution | 3 | 2 | 66.7% | 0.198 | 0.681 | 15 | Effective solution |
| **B2S** | Technology | Data Quality, Availability and Privacy | Solution | 4 | 4 | 100.0% | 0.160 | 0.666 | 16 | Effective solution |
| **C4S** | Organisation | Governance Structures and AI Policy Maturity | Solution | 4 | 0 | 0.0% | — | — | — | No retained items |

Table E.4: Full TOE-Level Aggregation by Category and Item Type, Retained Items Only

| **TOE Category** | **Item Type** | **Items Proposed** | **Items Retained** | **Consensus Coverage** | **Mean $\bar{d}$ (retained)** | **Mean Crisp (retained)** | **Priority Rank (crisp)** | **Interpretation** |
|---|---|---|---|---|---|---|---|---|
| **Technology** | Barrier | 24 | 7 | 29.2% | 0.181 | 0.575 | 1 (barriers) | Significant barrier |
| **Technology** | Solution | 20 | 19 | 95.0% | 0.144 | 0.696 | 2 (solutions) | Effective solution |
| **Organisation** | Barrier | 25 | 4 | 16.0% | 0.184 | 0.499 | 2 (barriers) | Moderate barrier |
| **Organisation** | Solution | 20 | 14 | 70.0% | 0.166 | 0.694 | 3 (solutions) | Effective solution |

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|---|---|---|---|---|---|---|---|---|
| **Environment** | Barrier | 18 | 0 | 0.0% | — | — | — | No retained items; high contextual variability |
| **Environment** | Solution | 15 | 11 | 73.3% | 0.168 | 0.710 | 1 (solutions) | Effective solution, highest mean crisp of all solution groups |

Table E.5: Full TOE-Level Descriptive Results, Retained Barriers and Solutions Combined

| TOE Category | Items Proposed | Items Retained | Consensus Coverage | Mean $\bar{d}$ (retained) | Mean Crisp (retained) | Descriptive Rank | Interpretation |
|---|---|---|---|---|---|---|---|
| **Technology** | 44 | 26 | 59.1% | 0.154 | 0.664 | 2 | Highest barrier retention; second-highest combined crisp |
| **Organisation** | 45 | 18 | 40.0% | 0.170 | 0.651 | 3 | Moderate retention; second-highest barrier mean crisp |
| **Environment** | 33 | 11 | 33.3% | 0.168 | 0.710 | 1 | All retained items are solutions; highest combined mean crisp |

Table E.6: Full Ranking of Highest-Priority Retained Barriers Based on Average Crisp Value

| Barrier Rank | Item | TOE | Construct | Barrier Statement | Avg $\bar{d}$ | Avg Crisp | Interpretation |
|---|---|---|---|---|---|---|---|
| **1** | B2.2 | Technology | B2, Data Quality, Availability and Privacy | Fragmented data governance across business units or client environments prevents coherent data pipelines for GenAI at scale. | 0.162 | 0.681 | Significant barrier |
| **2** | B3.2 | Technology | B3, Integration Complexity and Technical Debt | Accumulated technical debt significantly increases the time, cost, and risk of scaling GenAI from pilot to full deployment. | 0.194 | 0.653 | Significant barrier |
| **3** | B3.1 | Technology | B3, Integration Complexity and Technical Debt | Legacy system architectures make it technically complex and costly to integrate GenAI into existing service workflows. | 0.184 | 0.644 | Significant barrier |

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|---|---|---|---|---|---|---|---|
| **4** | B3.4 | Technology | B3, Integration Complexity and Technical Debt | Weak API management capabilities make it difficult to embed GenAI reliably into standardised, repeatable service delivery processes. | 0.187 | 0.562 | Significant barrier |
| **5** | B3.3 | Technology | B3, Integration Complexity and Technical Debt | Multi-client data requirements, including tenant isolation and segregated environments, add significant complexity when scaling GenAI workflows. | 0.165 | 0.553 | Significant barrier |
| **6** | B1.3 | Technology | B1, Model Reliability, Hallucination Risk and Explainability | Inconsistent outputs across repeated runs make it difficult to monitor and assure quality at scale. | 0.199 | 0.525 | Significant barrier |
| **7** | C2.4 | Organisation | C2, Organisational Readiness and Change Capacity | Low AI readiness among staff and a weak internal AI culture reduce the organisation's willingness to adopt GenAI tools at scale. | 0.197 | 0.506 | Significant barrier |
| **8** | C2.3 | Organisation | C2, Organisational Readiness and Change Capacity | The breadth of GenAI-related change exceeds the organisation's current change management capacity. | 0.172 | 0.503 | Significant barrier |
| **9** | C3.2 | Organisation | C3, AI Skills, Literacy and Capability Gaps | Weak prompt engineering skills and limited output evaluation competency among delivery staff create quality and reliability risks in live GenAI operations. | 0.184 | 0.497 | Moderate barrier |
| **10** | C2.1 | Organisation | C2, Organisational Readiness and Change Capacity | Cultural resistance to AI, including fears about job displacement or distrust of AI outputs, limits the ability to embed GenAI in everyday operations. | 0.184 | 0.491 | Moderate barrier |
| **11** | B6.3 | Technology | B6, Relative Advantage Clarity and Value Demonstration | Confusion about what GenAI is, compared with other AI or automation tools, creates mixed signals in investment decisions and stakeholder conversations. | 0.178 | 0.406 | Moderate barrier |

Table E.7: Full Ranking of Highest-Priority Retained Solutions Based on Average Crisp Value

| Solution Rank | Item | TOE | Construct | Solution Statement | Avg $\bar{d}$ | Avg Crisp | Interpretation |
|---|---|---|---|---|---|---|---|
| **1** | D5.S2 | Environment | D5S, Vendor Ecosystems and AI Platform Maturity | Pursuing a multi-vendor ecosystem strategy that diversifies GenAI platform dependencies and reduces the risk of over-reliance on any single provider. | 0.160 | 0.772 | Highly effective solution |
| **2** | B1.S2 | Technology | B1S, Model Reliability, Hallucination Risk and Explainability | Requiring human review of GenAI outputs (human-in-the-loop validation) before they are shared with clients or embedded in service deliverables. | 0.104 | 0.759 | Highly effective solution |
| **3** | B4.S1 | Technology | B4S, Security Vulnerabilities and IP Risks | Embedding security-by-design principles that incorporate GenAI-specific controls at the prompt, data, and model levels from the outset of any deployment. | 0.146 | 0.753 | Highly effective solution |
| **4** | C2.S3 | Organisation | C2S, Organisational Readiness and Change Capacity | Using iterative pilot-to-scale frameworks that allow incremental GenAI integration, building organisational confidence and change capacity over time. | 0.115 | 0.747 | Effective solution |
| **5** | D4.S1 | Environment | D4S, Customer and Stakeholder Pressure | Developing AI partnership strategies that involve clients in co-designing GenAI governance, acceptable use policies, and auditability frameworks from the outset. | 0.136 | 0.734 | Effective solution |
| **6** | D4.S2 | Environment | D4S, Customer and Stakeholder Pressure | Implementing AI-enabled service assurance frameworks that give clients verifiable evidence of GenAI output quality, governance compliance, and risk management. | 0.154 | 0.730 | Effective solution |
| **7** | B1.S3 | Technology | B1S, Model Reliability, Hallucination Risk and Explainability | Establishing standardised evaluation pipelines and explainability frameworks to monitor output quality | 0.120 | 0.725 | Effective solution |

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| | | | | and auditability in production environments. | | | |
| **8** | D5.S1 | Environment | D5S, Vendor Ecosystems and AI Platform Maturity | Conducting structured vendor assessment and selection processes that evaluate enterprise readiness, compliance posture, data sovereignty controls, and long-term viability before signing production-scale contracts. | 0.148 | 0.722 | Effective solution |
| **9** | C1.S2 | Organisation | C1S, Leadership Commitment and Strategic Alignment | Integrating GenAI objectives explicitly into the organisation's strategic planning cycles, OKRs, and annual business roadmaps. | 0.164 | 0.719 | Effective solution |
| **10** | C3.S2 | Organisation | C3S, AI Skills, Literacy and Capability Gaps | Setting up an AI Centre of Excellence (CoE) to centralise GenAI expertise, develop organisation-wide standards, and spread capability across business units. | 0.164 | 0.719 | Effective solution |

# Appendix F: Thematic Analysis

## Table F.1 Semi-Structured Interview Participant Overview

| Pseudonym | Role Type | Work Experience | Organisational Context | Geographic Market | Organisation Size | GenAI Adoption Stage (Self-Reported) |
|---|---|---|---|---|---|---|
| **A** | Enterprise Agentic AI Solution Architect | More than 15 years | Global IT services organisation | Asia / Global | 10,000 or more | MVP to selective production; 1-2 live deployments |
| **R2** | Cloud Infrastructure Pre-Sales / AI Consulting | More than 15 years | Global IT consulting firm | US | 10,000 or more | Early pilot and client readiness assessment |
| **R1** | Senior IT Architect / Delivery Lead | More than 15 years | Managed IT services organisation | Asia / Europe | 2,000–9,999 | Pilot; technical architecture scoping |
| **S** | IT Programme / Delivery Manager | More than 15 years | IT services firm; client-facing delivery | Asia/ Europe | 2,000–9,999 | Pilot; nascent internal GenAI initiatives |
| **D** | Product Owner / Service Desk Delivery Head | More than 15 years | Global managed IT services provider | Europe (Nordic) | 10,000 or more | Pilot, close to production rollout (ITSM and contact management) |
| **H** | Technology Strategy and Transformation Consultant | 6–10 years | Large Scale consulting firm; client-facing AI strategy | Asia / Global | 10,000 or more | Client AI readiness assessment; proposed use case scoping |

Note. The qualitative thematic analysis is based on 6 semi-structured interview participants selected from the broader 20-person survey/FDM respondent group reported in Appendix C and Appendix E.

## Table F.2 Coding Summary

| Theme Code | Theme Label | TOE | Contributing Participants | FDM Relationship |
|---|---|---|---|---|
| **T1** | The Data Foundation Paradox: Structural Fragmentation vs. Resolvable Complexity | Technology | A, R1, D, H, S | Validates B2.2 (top barrier); refines severity framing |
| **T2** | The Integration Bottleneck: Technical Debt as Architectural Reality | Technology | R1, D, A, R2 | Validates B3.1, B3.2, B3.3, B3.4 |
| **T3** | Output Reliability and the Human Oversight Imperative | Technology | A, D, H, R1, R2 | Validates B1.3; strongly supports B1.S2 |
| **T4** | The Governance Architecture Gap: Risk-Tiered Frameworks vs. One-Size Approaches | Technology/ Organisation | A, R2, H | Extends B6.3; emergent theme not fully in FDM taxonomy |
| **O1** | The Frozen Middle: Cultural Inertia and Middle Management Friction | Organisation | S, A, H, D, R1 | Validates C2.1, C2.4; explains dimension weight discrepancy |
| **O2** | Strategic Ambiguity and the ROI Credibility Gap | Organisation | A, D, R2, H, S | Validates B6.3; refines C1.S2; extends open-ended FDM findings |
| **O3** | Capability Infrastructure: The AI Council, CoE, and Upskilling Triad | Organisation | A, S, H, D, R2 | Strongly supports C3.S2, C1.S3; validates C3.2 |
| **E1** | Client-Side Readiness as the External Scaling Ceiling | Environment | D, H, R1, S | Explains Environmental barrier non-consensus; supports D4.S1, D4.S2 |
| **E2** | Regulatory Contextuality: Jurisdiction-Specific Pressure vs. Universal Solution Logic | Environment | D, H, A, R2 | Explains Environmental paradox; supports D1.S1, D1.S2, D1.S3 |

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|---|---|---|---|---|
| **X1** | The Governance-Technology Interaction: Security Gatekeeping as Amplifier | Cross-dimensional (E x T) | D, A, R2, H | Operationalises compliance gap (conceptual model) |
| **X2** | Iterative Scaling vs. Pilot Purgatory: The Governance Decision Gate | Cross-dimensional (O x T) | D, A, H, S, R2 | Validates C2.S3; challenges D5.S2 customisation logic |

Table F.3 Full Mixed-Method Triangulation Matrix

| **Qualitative Theme** | **FDM Finding** | **Triangulation Relationship** | **Analytical Implication** |
|---|---|---|---|
| **T1: Data Foundation Paradox** | B2.2 (crisp = 0.681, rank 1) | Validation with refinement | Barrier is real and severe, but character varies between structurally unavoidable (multi-client) and resolvable (data platform investment). Solution requires architectural commitment before GenAI deployment, not concurrent with it. |
| **T2: Integration Bottleneck** | B3.1-B3.4 (100% consensus) | Strong validation | 100% consensus confirmed. Qualitative evidence adds: a significant share of integration delay originates in environmental compliance processes (Theme X1), not technical infeasibility alone. |
| **T3: Output Reliability and Human Oversight** | B1.3 (crisp = 0.525); B1.S2 (crisp = 0.759) | Validation with contextual calibration | Output inconsistency confirmed as barrier; human-in-the-loop confirmed as top solution. Qualitative adds: risk-contextual design rather than universal mandation; threshold modelling as practical governance tool. |
| **T4: Governance Architecture Gap** | B6.3 (crisp = 0.406, rank 11); open-ended FDM contributions | Extension and elevation | FDM underweighted this barrier at item level. Qualitative evidence reveals it as a governance design problem (one-size assessment) with significant operational consequences beyond stakeholder communication. Risk-tiered AI placement frameworks emerge as an unranked but critical solution. |
| **O1: Frozen Middle** | C2.1 (crisp = 0.491); C2.4 (crisp = 0.506); holistic weight: Organisation = 38% | Explanatory contribution | Qualitative evidence explains why holistic Organisation weight (38%) exceeds item-level retention (16%). The frozen middle is a systemic, incentive-driven dynamic that cannot be captured by individual survey items but profoundly shapes Technology barrier severity. |
| **O2: ROI Credibility Gap** | B6.3 (crisp = 0.406); open-ended FDM (ROI ambiguity); C1.S2 (crisp = 0.719) | Validation and elevation | Open-ended FDM contributions identified ROI ambiguity as an important omitted barrier. Qualitative evidence confirms this as central to scaling failure and reveals it as both a prospective design requirement (KPIs before development) and a retrospective accountability mechanism. |
| **O3: AI Council, CoE, and Upskilling Triad** | C3.S2 (crisp = 0.719); C1.S3 (crisp = 0.716); C3.2 (crisp = 0.497) | Validation with structural elaboration | FDM solution items validated. Qualitative adds a sequenced architecture: AI Council (governance foundation) must precede AI CoE (delivery engine). Upskilling needs differ by context: displacement anxiety management vs. technical prompt engineering capability. |
| **E1: Client-Side Readiness** | D4.S1 (crisp = 0.734); D4.S2 (crisp = 0.730); Environmental barrier non-consensus | Explanatory and extension | Client-side readiness as an explicit external variable was not captured in the FDM's Environmental barrier taxonomy. Qualitative evidence identifies it as a primary scaling constraint in multi-client IT services, explaining why co-design solutions rank so highly: they address client readiness as well as governance. |
| **E2: Regulatory Contextuality** | Environmental paradox (0 barriers retained; | Explanatory validation | Qualitative evidence directly explains the Environmental paradox. Regulatory barriers are genuinely contextual; solution logic is genuinely |

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| | highest solution crisp = 0.710) | | universal. The paradox is a real finding, not a measurement artefact. |
| **X1: Security Gatekeeping as Amplifier** | B3.4 (crisp = 0.562); conceptual model compliance gap | Operationalisation | Compliance gap interaction effect operationalised: environmental security governance processes directly amplify Technology integration barriers. Integration solutions must be complemented by environmental approval process redesign. |
| **X2: Iterative Scaling vs. Pilot Purgatory** | C2.S3 (crisp = 0.747); C1.S2 (crisp = 0.719) | Validation with critical qualification | Iterative frameworks confirmed as effective but insufficient without explicit governance decision gates. ROI governance, standardised MVPs, and explicit pilot-to-production transition criteria are necessary complements to iterative scaling. |